MOBILE DJ / PRODUCT BLUEPRINT

DESIGN HANDOFF / VERSION 1.0

Mobile DJ planning dossier

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version 1.0 - Local first, ready for design

Nineteen coordinated specifications for a new Android mobile DJ app. Build the local phone version first; AWS is a later optional extension. The future AWS planning target is about US$35/month, excluding ParDay AI charges. No further owner answer is needed to start design. This baseline is a plan, not an implemented or certified app.

Area	Current baseline
Platform and audience	Android first; Galaxy S26 Ultra is the first test target. Serious mobile DJs primary; casual streaming DJs secondary.
Delivery sequence	Personal tool for practice/private gatherings. Phone-native first, with no required server; optional AWS expansion later.
Accessories and scope	USB audio interfaces and DJ splitters allowed. All selected professional modules retained; lighting and external DJ controllers excluded.
Visual direction	C - Crystal Current selected. Pale glass, titanium and restrained brass; A remains historical comparison.
DSP and stack	Compatible open source preferred; justified paid DSP acceptable. Kotlin/Compose plus C++/Oboe proposed.
ParDay	Preparation plus specifically armed bounded assistance and manual takeover. AI provider, spending limit and voice remain later choices.
AWS and source sharing	US$35/month future AWS planning target, excluding AI; not a hard cap or deployment approval. Optional GitHub sharing; no project licence or publication yet.

Start with Document 19, then 04, 03, 08 and 12. The pack retains 49 functional requirements, 24 shared quality gates, 90 specialist/deployment cases and 30 asset classes. AWS and home-lab service-specific variants are deferred; applicable local cases remain required.

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Readiness and agent handoff

Status: final planning and design baseline, version 1.0. Confirmed choices, proposed defaults and deferred engineering evidence are distinct. No unanswered owner question blocks design. No device benchmark, source adapter, AI integration, app implementation or deployment is claimed.

Owner and design agents: 19 -> 04 -> 03 -> 08 -> 12, then trace the complete selected scope through 02.

Architecture/audio agents: 02 -> 05 -> 06 -> 09 -> 13 -> 15; use work packages and local-first gates in 10.

Data/acquisition agents: 07 -> 14 -> 16 -> 11. Prove the on-device path first; preserve actual source, quality and publication/cancellation rules.

Future cloud/publication agents: 17 -> 18, only when that extension is requested. Use the deferred comparison and current-cost review.

Design now; evidence at the relevant gate

Use Mobile DJ as a working name and typed/touch ParDay as a design default. Capture the actual phone build and audio route before physical tests; evaluate DSP, stems, plugins and source paths before claiming support. Choose a paid AI configuration before real calls. AWS account, region and live quote can wait until expansion.

Companion pack

Nineteen editable specifications and PDFs, this dossier, decision and requirement ledgers, semantic schemas, readiness and deployment profiles, proposed C tokens, 30 asset work orders, two generated concepts with prompts, and the PDF renderer. Concept art shows materials, not measured native controls.

Scope and authority

The first two-deck slice is not the complete four-deck professional product. Preserve requested unofficial acquisition candidates and their independent technical/access/rights/distribution evidence. ParDay proposals need deterministic local authority. Prepared local sets must remain usable without AWS or AI.

Planning toolkit and revision

This pass uses Product Lens, Premium Product UI and independent scope, audio and acquisition/cloud reviews. Earlier Council and Learning Engine work remains in the dossier. Version 1.0 supersedes the AWS-first comparison sequence with local-first delivery; it does not claim a new full Council exercise or production validation.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 01 OF 19

Product vision and Product Lens review

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DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

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Version 1.0 | 29 October 2026 | Design handoff baseline - implementation and device validation pending

1. Product direction

Create a self-contained mobile DJ instrument that lets a person prepare music, start a set quickly, perform precise transitions, remix expressively, monitor the result, and recover predictably when the phone or a music source changes state. All performance controls must work through touch. External DJ-controller integration is excluded by the user's instruction.

This is a new project. No Mixline code, screenshots, interaction decisions, dependencies, project names, or earlier approvals are inherited. Android first is now user-confirmed. The product name remains open; ParDay is the confirmed name of the AI assistant. The user has confirmed steampunk and selected **C - Crystal Current** after reviewing the A/C concepts. The first test device is a **Samsung Galaxy S26 Ultra**, reported by the user as up to date; its exact installed build and audio route remain to be recorded.

The selected direction combines liquid glass and mechanical instrument craft through pale frosted glass, titanium and restrained brass detail. A is retained as a historical comparison, not a selected night theme. The product must remain useful when decorative glass and animation are reduced.

The owner has selected a personal tool and confirmed the delivery order: **a local Android version first, an optional AWS extension later**. Local first means the phone owns the library, preparation, performance state and audio engine, with no required LAN or cloud server. Network discovery, downloading, broadcasting and optional AI calls remain network-dependent features; local first does not mean that every feature works in airplane mode. The future AWS planning target is **about US$35 per month, excluding ParDay AI-provider charges**. It is a design target, not a current quote, hard billing cap or instruction to provision resources.

They are open to publishing the code on GitHub so other people can build and operate their own home-lab solutions; publication and the project licence are not yet selected or executed. Establish the on-device acquisition baseline first. Compare that measured baseline with owner-operated AWS when the extension is considered; the earlier immediate D-versus-AWS trial is no longer a dependency for the local build. No existing local host, AWS account, region or instance is selected. The complete target includes professional audio DJ capability, video mixing, karaoke, broadcasting and third-party plugin hosting adapted to Android; lighting is explicitly excluded. The early engineering milestone is a smaller vertical slice that proves the audio and touch architecture. Passing that milestone does not mean the full product is finished or authorize reducing the final scope.

2. Confirmed instructions, proposals, and unknowns

Classification	Item	Consequence
Confirmed	Start from scratch	Begin a fresh design and, when implementation is requested, a fresh codebase; no Mixline inheritance. A temporary working label is sufficient while the app name remains open.
Confirmed	Mobile, quick, all-in-one live DJing	Optimize time to first mix, touch operation, preparation, routing, and recovery.
Confirmed	Professional feature breadth	Maintain a requirements ledger; do not call a two-player demo a professional release.
Confirmed	No external DJ controllers	Exclude MIDI/HID mappings, controller certification, and motorized control. Treat DVS/timecode exclusion as a proposed interpretation to confirm.
Confirmed	C - Crystal Current; liquid glass plus steampunk	Refine the pale frosted-glass/titanium direction into measured native screens; retain A only as comparison history.
Confirmed	Planning, questions, at least eight guiding documents	Deliver a reviewable dossier; implementation is a later phase.
Confirmed	Android first; Galaxy S26 Ultra first	Record its actual build, variant and route before benchmarking; broader phone/tablet coverage remains to be specified.
Proposed	Two-deck default; four-deck advanced workspace	Keep the interface quick while retaining a professional destination.
Confirmed ambition; admission conditional	Any and all music sources	Preserve broad discovery and approved adapters; no universal permission or catalog availability is implied.
Confirmed	USB audio interfaces and dedicated DJ splitters allowed	Certify actual routes separately from excluded MIDI/HID controller integration.
Confirmed	Personal tool only	Personal installation, reliable operation and maintenance are the baseline; commercial pricing and public-store release are not current acceptance gates.
Optional future direction	Open-source GitHub release for others to home lab	Prepare reproducible code/docs; choose licences and publication contents before any public release.
Confirmed delivery sequence	Local Android first; AWS expansion later	No server setup, cloud comparison or AWS account is required to start the phone design/build. Establish on-device acquisition evidence first.

Classification	Item	Consequence
Retained later evaluation	Compare the local baseline with owner-operated acquisition before selecting an expansion runtime	Keep the common contract and comparison protocol; local-first sequencing does not establish a measured long-term runtime winner.
Confirmed future cloud target	AWS accepted; about US$35/month excluding ParDay AI charges	Region, account, runtime, actual quote and deployment remain later choices. No local host purchase is required; optional home-lab profiles remain portability references.
Confirmed	Serious mobile/live DJs primary; casual streaming DJs secondary	Preserve deep touch control while offering guided preparation and clear source access.
Confirmed	Personal practice and private gatherings initially	Source terms still apply; broadcasting is a distinct use requiring its own eligibility.
Confirmed	Paid DSP acceptable; compatible open source preferred	Compare quality, licences and total maintenance cost before a purchase decision.
Confirmed	ParDay assisted performance, option C	Provide preparation assistance plus explicitly armed bounded transitions/automix with immediate manual takeover.
Confirmed capability request; mechanism unresolved	In-app link-to-file acquisition, including unofficial routes	Evaluate own/self-hosted extraction and third-party integration without assuming a working or permitted API; see Document 16.
Unknown	Genres, development budget, time, team, exact device build/routes and provider access	Prevents a reliable completion date or capacity claim. These gaps do not block initial design; the future AWS target is already recorded separately.

3. Primary user hypotheses

The user selected serious mobile/live DJs as the primary audience and casual streaming DJs as the secondary audience. The operating context is personal practice and private gatherings, and personal-tool delivery is now confirmed. The audience definitions describe workflow depth and potential community users rather than a selected customer-acquisition strategy. Professional capability describes the depth and reliability expected; it does not imply that public or paid events were selected. Backup/travel use remains a plausible adjacent situation, not a newly approved primary audience.

These users require different promises. A performance DJ may accept preparation in exchange for reliable cueing. A casual user may reject the product if personal streaming playlists are unavailable. An emergency user may prioritize fast recovery over four-deck remixing. Resolve conflicts in favor of the confirmed primary audience while retaining the secondary audience in first-use and music-discovery testing. Do not simplify away required professional controls to imitate a streaming player.

Do not infer market demand from the existence of DJ apps. Current official product pages demonstrate that mobile DJing, stems, automated transitions, performance pads, effects, and music integrations already exist in the category. They establish a competitive baseline, not evidence that this new product will attract

paying customers. [P1-P4]

4. Core loop and product promise

Trigger: I want to practice, take over music at a gathering, perform a prepared set, or respond to a track request.

Action: Open a recent set or choose tracks, verify the output route, load a free deck, audition through cue where supported, match timing and key, transition, and shape the result with performance controls.

Outcome: Music stays under the DJ's control, the next track is ready, levels are visible, the session is recoverable, and the DJ can save useful work.

Return: Saved crates, reliable cue points, corrected beat grids, reusable sample banks, and set history reduce preparation next time. Optional rehearsal teaches useful gestures and timing without interrupting live operation.

A proposed positioning statement is: "A complete touch DJ instrument for people who want to perform directly from a phone, with fast preparation, expressive control, and clear live-session safeguards." This is a hypothesis to test. It is not a proven advantage.

5. What makes the product worth switching to

Liquid glass and steampunk can create recognition, but visual polish is copyable. The stronger opportunity is the combination of three measurable benefits:

1. **Fast readiness:** a recent local set opens, validates playable media, restores preparation data, and reaches a mix with few decisions.
2. **Confident touch control:** the user can change decks, manipulate the waveform, and browse the next track without moving the wrong audible control.
3. **Trust during performance:** source restrictions, output routing, recording eligibility, thermal limits, and recovery are understandable before they become a surprise.

Prototype these benefits against the participant's current workflow. A visual preference test alone cannot validate the product.

6. Product Lens assessment

Historical v0.1 assessment: idea-stage prosumer/consumer lens with professional usage requirements. Preserve the score below as the earlier commercial-evidence baseline; it is not a current personal-project launch gate. No implementation, usability result or provider agreement is established. The current success question is whether the owner can prepare and perform the intended set reliably, with a reproducible setup others could adopt if the code is later published.

Criterion	Weight	Score	Evidence and implication
Problem severity	3	?	The user wants the tool, but frequency, losses, and current workaround are unspecified. Interview intended DJs.
Target user and market	3	3	The v0.1 brief combined audiences; v0.2 now names serious mobile DJs primary and casual streaming DJs secondary. This clarifies scope but is not new demand evidence.
Differentiation	2	2	Professional parity plus a visual style is currently the main stated distinction; competitors already provide broad functionality. [P1-P4]

Criterion	Weight	Score	Evidence and implication
Core loop and value	1	2	A useful loop can be described, but delivery depends on unbuilt audio, library, routing, and source features.
Feasibility and effort	2	?	Native approaches exist, but team capability, SDK budget, target devices, and performance have not been established.
Business viability	2	?	No target price, acquisition channel, support model, or willingness-to-pay evidence.
Risk and assumptions	2	2	Device routing, music access, thermal load, and differentiation remain unresolved.
Evidence	1	2	Founder intent and competitor capability are secondhand signals; no research with this product's users.

Weighted score = (3x3 + 2x2 + 2x1 + 2x2 + 2x1) / (3+2+1+2+1) = 21/9 = **2.3/5**. Known criteria: 5/8. Weighted evidence coverage: 9/16 = **56.25%**. This score is not a probability of success.

Historical commercial verdict: rethink differentiation before a commercial parity race. The owner has since selected a personal tool, so willingness to pay, revenue and commercial differentiation do not block personal engineering. Continue the requested planning and bounded feasibility work. The low-coverage score does not reject a personal tool or the creative direction; it says the commercial reason to switch is not established. The 2.3/5 score is retained as the dated v0.1 evidence baseline. Audience selection is now resolved, but no interviews, willingness-to-pay tests or audio experiment have occurred. Recompute after those evidence-producing steps; do not present the old score as a fresh measured assessment. Do not raise scores merely because this dossier is more detailed than the original brief.

7. Ranked next work

The idea-stage ranking uses core-job centrality (0-3) plus learning value (0-2). Effort is unestimated until the team and SDK approach are known; rankings do not imply schedules.

Rank	Work package	Type	Centrality	Learning	Why it is early
1	Output, cue, and two-deck audio experiment	Blocker	3	2	Tests whether the target phone can support the basic performance promise.
2	Local import, playability, search, and preparation	Table stakes	3	2	Tests whether a DJ can get useful music into the app quickly.
3	Portrait and landscape touch workflow prototype	Differentiator hypothesis	3	2	Tests focus errors, control reach, and performance while browsing.
4	Safe deck loading, state recovery, and interruption handling	Blocker	3	2	Defines behavior when ordinary phone events occur.
5	Stretch, key lock, beat grid, and sync comparison	Table stakes	3	2	Makes an informed DSP choice before vendor coupling.
6	Cue, loop, pad, FX, and sampler vertical slice	Table stakes	2	1	Proves expressive controls on the same engine clock.

Rank	Work package	Type	Centrality	Learning	Why it is early
7	Prepared stems and four-deck load experiment	Expansion of core	2	1	Establishes credible advanced device tiers.

Video, karaoke, broadcasting, plugin hosting, prepared/live stems, four decks, sampler/sequencer, mic, recording and automix are selected for the complete target. Lighting is excluded. Delivery gates sequence and validate the selected scope; they do not authorize deleting it. Provider coverage remains an admission problem, and the user has not ranked services. ParDay and requested on-device link acquisition are local-version workstreams. The AWS comparison is a later expansion workstream, activated after a useful local baseline and a concrete expansion need; see Documents 13-18. Document 19 defines the immediate design handoff.

8. Proposed discovery and outcome targets

For optional community workflow research, recruit 8-12 intended users, including at least four who perform live. This study is not a requirement to install and validate the owner's personal build. This sample is for workflow discovery, not market sizing. Ask each to bring legally usable tracks and demonstrate their current preparation and next-track workflow. Include Caribbean/open-format material alongside electronic tracks when that matches the selected audience; do not assume every track has a fixed tempo or a clean downbeat.

Proposed gates: at least 8 of 10 participants complete local import and a first transition without help; median time from a prepared cold launch to first audible track under 15 seconds; no accidental replacement of an audible deck in 20 load tasks per participant; at least 6 of 10 prefer the touch workflow for one named situation after comparing it with their current app. Record failures and actual timings. These are targets for discussion, not achieved results.

If the owner later changes the objective to commercial distribution, revisit pricing and demand using real purchase intent or a paid pilot. For the current personal tool, evaluate the owner's usable music collection, required sources and actual preparation workflow directly. Optional open-source adoption is a separate question from paid demand.

9. Learning Engine application

Use the Learning Engine to make the planning decisions understandable and, later, to design optional rehearsal. Teach the distinctions between deck sync and automatic transitions, key lock and key shift, master and headphone cue, and streaming playback and permission to transform music.

For rehearsal, progress from demonstration to guided attempt to independent practice. Suggested exercises: set a cue, align an eight-bar phrase, recover a misaligned beat grid, move an EQ without changing the wrong deck, and identify cue-route leakage. Track assisted versus independent attempts only if the user opts in. A completed tutorial is not proof of DJ competence. Live performance must never require a lesson, quiz, badge, account, or tutorial overlay.

10. Owner and acceptance

Owner: product lead. Consult: working DJs, audio architect, UX lead, music integration lead. The audience, first platform, operating context and audio-accessory allowance are recorded in Document 12. Crystal Current, the first Galaxy S26 Ultra target, personal-tool delivery, local-first sequencing and the future US$35/month AWS target are confirmed. Numerical success criteria, exact installed hardware/software/audio-route profile, development and AI budgets, licence/publication choices and the eventual expansion comparison outcome remain open or proposed. They are assigned to the phase that needs them. No additional owner answer is required to start the design in Document 19. This readiness judgment does not mean the technical or commercial hypotheses have passed.

Sources checked 29 October 2026

- [P1] Algoriddim product/platform overview: https://www.algoriddim.com/apps

- [P2] Serato DJ Pro feature overview: https://serato.com/dj/pro

- [P3] VirtualDJ feature overview: https://virtualdj.com/products/virtualdj/features.html

- [P4] Native Instruments Traktor product overview: https://www.native-instruments.com/products/traktor-pro

These sources support the competitive baseline only. Proposed priorities, research targets, positioning, and scores are this plan's judgments.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 02 OF 19

Professional product requirements

A new touch-first performance instrument.
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DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline - implementation and device validation pending **Product:** New mobile DJ application; no Mixline reuse **Platform direction:** Android first; Galaxy S26 Ultra is the confirmed first test device (current software reported, not inspected) **Owners:** Product lead, audio lead, mobile lead, design lead and QA lead

1. Requirement contract

The intended product is a complete mobile performance instrument: quick to start, deep enough for professional preparation and live creativity, and dependable during a set. The user confirmed that external-controller integration is excluded. External MIDI/HID DJ controllers and their mapping therefore do not belong in this baseline. DVS/timecode control is proposed as part of that exclusion and should be confirmed. USB audio interfaces and dedicated DJ splitters are explicitly allowed. Microphone functionality is selected; exact supported microphones, speaker routes and external displays still require a hardware matrix.

Every “shall” below expresses a proposed technical acceptance contract for the selected feature scope. This handoff is ready for design; implementation and test evidence are subsequent work. Two decks are the proposed default visible workspace; four-deck performance remains a selected advanced capability. Video mixing, karaoke, broadcasting and third-party plugin hosting are selected requirements. Lighting is excluded. Specific mobile formats, compatible plugins and device tiers remain to be proven, and a missing dependency must be reported rather than silently deleting a selected feature.

The owner confirmed a local Android version first and AWS expansion later. No required LAN/cloud server, AWS account or cloud trial belongs in the local-version dependency chain. The future AWS target is about US$35/month excluding ParDay AI-provider charges; it is a planning target, not spending authorization or a hard cap. Online source operations, broadcasting and optional AI remain independently network-dependent. Preserve the requested local link-to-file feature and report source-specific feasibility gaps rather than silently replacing it with a mandatory cloud service.

Professional capability is measured through completed workflows, accurate timing, audible results, usable controls and recovery. A control that animates without driving the engine does not satisfy a requirement. An engine function without a usable and accessible control does not satisfy the corresponding product workflow.

2. Gate vocabulary

G0 establishes the decision baseline. G1 proves audio feasibility. G2 is a working vertical slice. G3 is core performance beta. G4 completes advanced performance. G5 covers source/provider and ParDay integrations. Cloud-specific G5 cases apply only when the later optional AWS extension is evaluated. G6 establishes release candidacy. G7 is the personal installed release. G8 is maintenance and optional community readiness. G3 is not the full requested final product. The user has selected all discussed professional modules except lighting for the complete target. G7 cannot be described as that complete product with selected modules missing, but it does not wait for an AWS extension. Exact provider coverage, compatible plugin formats and certified device combinations remain unresolved implementation constraints.

In the matrix, a gate is the earliest evidence checkpoint. All selected release requirements must also pass regression at G6. QA identifiers are stable test-case references for the quality plan; tests must assert observable behavior rather than repeat implementation code.

3. Functional requirement matrix

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-001	Core local playback shall work offline without an app account.	QA-FR-001: fresh guest session imports and mixes two authorized local tracks in airplane mode.	Mobile + Audio / G2
FR-002	Import shall use user-selected files/folders and retain durable access or make a verified local copy.	QA-FR-002: restart, moved file and revoked access produce correct available/missing states without lost metadata.	Data / G2
FR-003	Supported formats shall be enumerated by codec/container/bit depth/sample rate, initially evaluating MP3, AAC/M4A, WAV, AIFF, FLAC, Ogg/Vorbis and Opus.	QA-FR-003: approved corpus decodes correctly; malformed/unsupported files fail safely. Final format set follows G1 decoder/license review.	Audio + Security / G3
FR-004	Search shall filter title, artist, album, BPM range, key, tags, availability and played state.	QA-FR-004: exact fixtures return expected results; search remains usable during playback on the declared library-size tier.	Data + Design / G3
FR-005	Crates/playlists, folders, smart rules, ratings, colors, comments, favorites and history shall support set preparation.	QA-FR-005: reorder, rename, duplicate detection and saved filter behavior survive restart.	Data / G3
FR-006	BPM, downbeat, key, waveform and loudness analysis shall run outside the real-time callback and cache versioned results.	QA-FR-006: analysis progress/error is visible; playback does not depend on completed optional analysis.	DSP + Data / G3
FR-007	Variable-tempo beatgrids and phrase/energy hints shall distinguish detected results from user corrections.	QA-FR-007: live-drumming/tempo-change fixtures retain meaningful bar positions; low-confidence results are not presented as certain.	DSP / G4
FR-008	Users shall correct downbeats, beat positions, local tempo, half/double BPM and grid alignment, then lock edits.	QA-FR-008: loops and sync follow corrected maps; reanalysis cannot overwrite locked edits silently.	DSP + Design / G3-G4
FR-009	Overview and detailed waveforms shall be derived from media, with cue/loop/grid overlays and explicit zoom.	QA-FR-009: transient positions match reference audio; overview seeking and display clocks stay consistent.	Audio + UI / G3
FR-010	Each deck shall provide load, play/pause, stop, temporary cue, nudge, position and elapsed/remaining time.	QA-FR-010: independent state and multi-touch transport actions produce documented audible results.	Audio + UI / G2
FR-011	Each track shall support at least eight named/color-coded hot cues plus saved memory markers.	QA-FR-011: cues persist, trigger at intended musical positions and distinguish set/delete/trigger actions.	Audio + Data / G3
FR-012	The engine shall support four deck identities; four simultaneous decks shall be enabled only for a certified workload tier.	QA-FR-012: switching focus never changes transport or assignment; four-deck workload passes performance gates.	Audio + UI / G4

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-013	Touch jog/scratch shall support vinyl mode, nudge, directional movement and defined release behavior.	QA-FR-013: scratch/release tests measure audio response; pointer cancellation cannot leave a deck held.	Audio + UI / G3
FR-014	Waveform seek/scrub shall be a first-class action distinct from jog scratching.	QA-FR-014: precise seek, coarse navigation and accidental-gesture protection work with track running or paused.	Audio + Design / G3
FR-015	Tempo, pitch bend, key lock and independent key shift shall expose ranges and reset controls.	QA-FR-015: reference playback verifies pitch/time behavior and smooth parameter changes across the approved range.	DSP / G3
FR-016	Manual matching and optional sync shall coexist with visible leader/follower and phase state.	QA-FR-016: sync off returns predictable manual control; invalid grids cannot silently force a bad alignment.	Audio + DSP / G3
FR-017	Quantization shall support selectable musical resolution with explicit enabled state.	QA-FR-017: cue, loop and pad commands meet the documented event scheduling rule; unquantized commands are not delayed to a beat.	Audio / G3
FR-018	Manual/auto loops, loop in/out, reloop, loop roll, beat jump and saved loops shall be supported.	QA-FR-018: boundaries, negative jumps, track edges and changing-tempo maps are exercised without invalid reads.	Audio + DSP / G3-G4
FR-019	Slip and reverse shall preserve or replace timeline position exactly as the chosen mode specifies.	QA-FR-019: scratch/loop/cue during slip returns to the calculated musical position; release is click-controlled.	Audio / G4
FR-020	Mixer shall include trim, channel faders, EQ/isolator, filter, assignable crossfader sides/bypass and curve/cut-in.	QA-FR-020: signal-path tests verify gains, bypass, center behavior, resets and audible smoothness.	DSP + UI / G3
FR-021	Channel/master metering, peak hold, clipping indication and master protection shall reflect the actual signal.	QA-FR-021: calibrated fixtures exercise overload; meter location and limiter latency are documented.	DSP / G3
FR-022	Independent cue/PFL, cue/master blend and headphone gain shall work only on valid output routes.	QA-FR-022: isolated cue never leaks to main; split-mono mode clearly trades stereo for independence.	Audio + Platform / G3
FR-023	Route setup shall identify main/cue channels, sample rate and unsupported combinations.	QA-FR-023: connect/disconnect/reconfigure on supported hardware yields truthful routes and safe fallback.	Platform / G3
FR-024	FX shall offer essential echo, delay, reverb and filter plus advanced modulation/distortion, assignment, wet/dry and tempo/free rate.	QA-FR-024: impulse/reference tests verify routing, bypass and tails; no timing depends on UI frame rate.	DSP / G3-G4
FR-025	FX XY control and macros shall show parameter mapping, hold/latch mode and reset behavior.	QA-FR-025: two-axis gestures, multi-touch and cancellation map predictably; reset cannot leave hidden wet gain.	UI + DSP / G4

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-026	Performance pads shall have explicit cue, loop, roll, jump, slicer, sampler, pitch-play and stem modes where supported.	QA-FR-026: changing bank/mode cannot trigger a hidden action; active mode and deck remain legible.	UI + Audio / G4
FR-027	Sampler/sequencer shall support at least 16 slots, banks, trigger modes, editable step patterns, per-step velocity/mute, length/resolution, swing and quantized start/stop/change.	QA-FR-027: approved voices and saved banks/patterns survive restart; repeated patterns, tempo changes, bar-boundary edits, mute/stop and takeover follow engine time.	Audio + Data / G4
FR-028	Prepared stems shall support aligned vocals/drums/bass/other with mute/solo/gain and readiness status.	QA-FR-028: sum/alignment reference, cache integrity and unavailable-stem fallback are verified on authorized media.	DSP + Data / G4
FR-029	Live separation shall be conditional on source permission, model license and measured device capacity.	QA-FR-029: refused activation is understandable; overload does not disrupt already playing full-mix audio.	DSP + Platform / G4
FR-030	Microphone/aux input shall expose permission, route, gain, mute, monitoring and talkover/ducking.	QA-FR-030: feedback-aware defaults and supported input/output latency are verified on actual hardware.	Audio + Platform / G4
FR-031	Mix recording shall provide format choice, timer, storage estimate, markers, safe finalization and partial recovery.	QA-FR-031: low storage/process interruption recover valid segments; playback remains independent of writer failure.	Audio + Data / G3
FR-032	Queue/automix shall show next readiness, transition settings, reorder, skip, cancel and immediate manual takeover.	QA-FR-032: unavailable next track is never treated as ready; takeover causes no surprise transport restart.	Audio + Product / G4
FR-033	Preview and track preparation shall remain available while the current set plays.	QA-FR-033: browsing/search/editing never obscures required transport, output and fader access.	Design + UI / G3
FR-034	Session state shall persist deck assignments, cues, relevant controls, queue and history at defined checkpoints.	QA-FR-034: recovery presents a truthful saved state and requires deliberate audible resume after a crash.	Data + Audio / G3
FR-035	Live protection shall prevent unintended audible-deck replacement and destructive reset.	QA-FR-035: load protection, live lock, quick fade and escape actions are tested under realistic touch errors.	Design + Audio / G3
FR-036	Source capability resolution shall gate analysis, caching, DSP, stems, recording, export and broadcast independently.	QA-FR-036: unknown/expired permission fails closed; provider restrictions are enforced below the UI.	Integrations + Security / G5
FR-037	Background playback, audio focus and interruption behavior shall be explicit per operating system.	QA-FR-037: calls, notifications, lock screen, route loss and system termination follow approved state transitions.	Platform / G3
FR-038	Day/night, high contrast, reduced transparency/motion and color-independent identities shall be supported.	QA-FR-038: critical controls and labels remain usable in each mode; screen-reader/precise-value alternatives exist.	Design + Accessibility / G3

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-039	Portrait/landscape and compact/expanded layouts shall preserve engine state and pointer safety.	QA-FR-039: rotate during playback/gesture without audio reset, cue leakage or unintended fader movement.	UI + Platform / G3
FR-040	Accounts shall be optional for the offline core; cloud functions shall disclose the data they require.	QA-FR-040: expired app login cannot disable authorized local performance unexpectedly.	Product + Security / G3
FR-041	Backup/export shall include portable metadata, version information and explicit source references.	QA-FR-041: round-trip restoration restores user work; restricted media is not copied as a side effect.	Data / G4
FR-042	Diagnostics shall show meaningful audio-route, storage, battery/thermal and source readiness states.	QA-FR-042: simulated faults produce actionable messages; telemetry omits track contents/tokens by default.	Platform + Security / G3
FR-043	Bundled audio, fonts, icons, textures and models shall have documented rights and source versions.	QA-FR-043: every shipped asset resolves to an approved manifest entry and required attribution.	Assets + Release / G6
FR-044	Provider loss shall not invalidate the local engine, user-owned library or metadata unnecessarily.	QA-FR-044: provider unavailable/revoked while local decks play leaves those local workflows operational.	Integrations / G5

4. Selected specialist and assistant requirements

FR-045 - Video mixing: selected. Support authorized video decks, visual crossfades, linked/unlinked audio-video behavior and visual effects, with external display on named supported routes. Document 15 defines the codec, audio-clock, seek, visual fallback, thermal and output contracts. Owner: Audio/Video + Platform. QA-FR-045 and the AV cases in Document 15. First feasibility G1; integrated evidence G4; release regression G6.

FR-046 - Karaoke/broadcast: both selected, with separate submodule acceptance. Karaoke includes singer rotation, authorized timed lyrics/media, mic integration and supported output. Broadcasting includes eligible mix encoding, explicit destinations/start/stop, protected credentials, visible connection state and independent failure handling; camera/visual overlays require their own resource profile. Document 15 defines these contracts. Personal/private practice does not grant internet broadcast rights. Owner: Audio/Video + Integrations. QA-FR-046 plus AV cases; G4/G5, regression G6.

FR-047 - Third-party plugin hosting: selected; lighting is explicitly excluded from this requirement and the product. Support at least one genuinely third-party, Android-compatible effect format/path after feasibility, distribution and licence review. Built-in effects alone do not satisfy hosting, and desktop plugin binaries are not presumed compatible. Versioned ABI, latency reporting, resource budgets, failure isolation, preset recall and allowed installation path are required. Document 15 provides the feasibility and acceptance plan. Owner: Architecture + Audio + Security. QA-FR-047 plus AV cases; first feasibility G1, integration G4, release G6.

FR-048 - ParDay assistant: selected. Provide a provider-neutral assistant for eligible local-library search, crate preparation, readiness diagnosis, transition proposals and optional rehearsal and selected armed assisted performance. The local validator enforces source permissions, user authority, fresh deck/session state and bounded actions; a model never drives the audio callback. Option C is confirmed: preparation assistance plus explicitly armed bounded transitions/automix and immediate manual takeover. Before a concrete plan is armed, live behavior remains read/propose. Owner: AI + Product + Security + Audio.

QA-FR-048, AI-001 through AI-024 in Document 13; deterministic harness G2, cloud integration G5, release G6.

FR-049 - Music discovery and acquisition: selected in-app link-to-file acquisition, including the requested unofficial Spotify/YouTube downloader capability. Provide Find a playable version across catalogs, local media and acquisition adapters. Evaluate an own/self-hosted extractor and third-party integration as concrete candidates; Document 16 separates technical verification, provider access, content rights and distribution review. An official API is not a universal technical prerequisite. Preserve recording/version identity, actual media origin, provenance and operation-specific permissions. A URL/search result or streaming offline cache is not treated as an exportable music file. No named adapter is declared working or release-cleared before evidence exists. Owner: Integrations + Data + Security. QA-FR-049, SRC-01 through SRC-12 in Document 14, and DLA-01 through DLA-20 in Document 16; local import G2, admitted provider workflows G5, release G6.

5. Cross-cutting acceptance rules

Each test result records device/OS, audio route, sample rate/buffer, workload, media corpus, thermal condition and software versions. Performance targets live in the audio/quality documents; a single fast demonstration does not certify a platform. Differences between devices and source providers must be visible in the capability matrix.

Library editing, artwork decoding, analysis, stem generation, network work and recording writes must not block the real-time audio callback. UI animation follows the audio clock; it never becomes the clock. Operations entering the engine include stable deck identity and ordering so rapid gestures cannot affect the wrong deck after focus changes.

A full release review traces every selected FR to implementation, test evidence, known limitations and user documentation. A feature is complete only when success, loading, unavailable, error, cancellation and recovery states are implemented. Nothing is declared finished because an AI-generated screenshot shows the control.

6. Research basis

Official feature research informs category coverage; the requirements above are original proposals. Platform parity and partner permissions must be checked independently.

- Algoriddim, djay Pro capabilities.
- Serato, DJ feature comparison.
- AlphaTheta, rekordbox mobile and mobile FAQ.
- Native Instruments, Traktor Pro.
- VirtualDJ, professional feature categories.

Remaining implementation and later-phase decisions: capture the S26 Ultra's exact installed build/variant and audio route; qualify broader devices, source coverage and compatible formats/plugins; choose a ParDay provider and separate AI spending limit before paid model use; resolve optional voice and source-publication/licence details when needed. Team, development budget and schedule remain uncommitted. AWS account, region, service and current quote are deferred to expansion; the planning target of about US$35/month excluding AI is already confirmed. Establish the on-device acquisition baseline now and evaluate AWS later before choosing an expansion runtime. C Crystal Current, Galaxy S26 Ultra first, ParDay option C, personal-tool delivery and local-first sequencing are confirmed. No local host is selected or required. Future open-source sharing remains optional. Previously answered questions must not be asked again or reset to proposed status. No unanswered item blocks the first design work package in Document 19; feature implementation and device capacity remain unverified.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 03 OF 19

Mobile workflows and interaction contracts

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version 1.0 · 29 October 2026 · Design handoff baseline - implementation and performance remain unverified

Android first, with **C - Crystal Current** selected and Galaxy S26 Ultra as the initial test device. Its current software is owner-reported; exact build and route are uninspected. This new product reuses no Mixline design or code. USB audio interfaces and DJ splitters are allowed; external DJ controllers are excluded. Screen names and numerical usability/performance targets are proposals.

1. Experience contract

The user must be able to prepare, start and sustain a live set on the phone. Two visible decks are the proposed default workspace; four decks and a sampler remain selected advanced capabilities. Professional scope is preserved through deliberate focus and navigation rather than progressively smaller controls. Changing screens, selecting a deck or opening the library must not change musical state.

Local Android comes first; AWS is a later optional extension. First use opens local import and route setup, without server pairing, an AWS account or assistant subscription. The phone owns library, preparation, performance state and audio. Discovery/downloads, broadcasting and optional AI need their own connectivity; a prepared local set does not. Add AWS settings only when implemented. Its future target is about US$35/month excluding ParDay AI charges; cost settings stay outside the live surface.

Separate **Prepare**, **Perform** and optional **Rehearse** contexts. Prepare supports detailed beatgrid/cue work. Perform protects the controls needed during a set. Rehearse provides optional explanations and practice. Users can enter Perform without completing a tutorial.

The requested finished scope includes four decks, prepared and live stems, sampler/sequencer, recording, automix, microphone, video/karaoke, broadcast and plugin hosting. Lighting is excluded. These modules need dedicated views and compatibility gates; their presence in the plan does not establish universal device, plugin or source support. An intermediate two-deck build is a delivery milestone, not a redefinition of the requested finished product.

Persistent performance shell

On compact portrait phones the shell contains:

- A labelled status selector for every enabled deck: A/B, or A-D. Each shows playing/paused/empty/loading state, an identifiable shortened track name, and a level/activity cue. Colour supplements the deck letter.
- Play and Cue for the currently focused pair, with at least 48dp touch targets.
- The crossfader, its assignments and current position.
- Master meters, recording status, the current output route, and a 48dp Master button opening its control panel.

The remaining region hosts the active workspace or library. Four-deck portrait does **not** promise eight full-sized Play/Cue buttons plus every mixer control at once. A nonfocused deck is one selector tap from its transport. Selection only changes the view. If a held gesture would disappear, defer the focus change until release; never redirect that finger to another deck.

The browser docks within the remaining space. The search keyboard may reduce result rows, but cannot cover all live controls. In an extreme small-height layout, use a compact search workflow rather than overlapping targets. These rules need validation against chosen physical devices before support is promised.

2. Layouts and screen inventory

Portrait uses a two-deck overview and a larger focused control area. Landscape gives the active pair adjacent workspaces and a central mixer, with waveform lanes readable together. It is a different composition, not a rotated portrait stack. Tablets can expose four channels and the browser together after usable target sizes are verified. Foldables use current window dimensions and keep critical controls clear of hinges; changing posture preserves engine state.

Screen family	Core purpose	Required special states
Launch and resume	Import music, use clearly labelled demo media, or restore preparation/session	Empty library, missing files, interrupted session
Import and sources	Grant scoped access, select files/folders, connect permitted providers	Denied permission, DRM/unsupported format, expired access, partial indexing
Library, crates and playlists	Search/filter, organize, queue, audition where supported, choose Load A/B/C/D	Offline, duplicates, no results, analysis pending, unavailable source
Track preparation	Edit beatgrid, BPM/key, cues, loops, gain, tags and phrase markers	Uncertain analysis, manual override, undo, unsaved changes
Readiness and routing	Identify master/cue outputs and validate local media/storage	No separate cue route, mono tradeoff, missing track, low storage
Performance and deck detail	Transport, waveform, pitch, jog, sync, slip/reverse, beatjump	Near end, held scratch, pending quantized command, protected seek
Mixer	Gain, EQ, filter, faders, assignments, headphone mix and meters	Clipping, limiter action, muted channel, cue selection
Pads, loops and sampler	Cue/loop banks, samples, choke groups, sync and gain	Empty slot, edit mode, playing loop, missing sample, stop samples
FX and stems	Effect chain/XY controls, stem levels and preparation	Source restriction, processing, unavailable effect/model, active latch
Queue and automix	Reorder, prepare transitions, inspect next item, manual takeover	Locked current item, committed transition, missing next track
Recording and history	Record eligible audio, show duration/storage, export and recover	Restricted source, storage full, interrupted recording, recoverable fragments
ParDay assistant	Search permitted context, propose preparation and transitions, explain readiness and rehearse	Offline/local-only, thinking, proposal ready, expired proposal, unavailable provider, cost limit
Video and karaoke	Select eligible video/captions, preview the output, manage synchronization and display routing	Missing video/captions, unsupported format, unavailable output, late decoding
Broadcast and microphone	Review destinations and routing, monitor levels, deliberately start/stop transmission	Private monitoring, disconnected destination, muted input, feedback risk, restricted source
Plugin rack	Inspect compatible effects/instruments, parameters, presets, cost and bypass	Unsupported plugin format, failed validation, unavailable plugin, unsafe load or state restore

Screen family	Core purpose	Required special states
Settings, help and recovery	Search settings, save preferences, inspect licenses, restore safely	Draft/preview/save failure, route change, entitlement offline, interrupted operation

Routine account, subscription, announcement and permission prompts must not obstruct an ongoing performance. Request permissions before the affected operation starts. A mid-session source revocation or system interruption follows the actual provider/platform rule, with a clear explanation; uninterrupted access is not promised after authorization is lost.

3. End-to-end journeys

First useful mix: open → choose demo or import → select an available track → load Deck A → inspect output → play → load Deck B → cue/align → crossfade. If a track fails, keep the other deck running and return selection to the failed deck. Do not mistake a completed tutorial for a completed mix.

Prepared set: import → analyze → inspect uncertain grids → set cues/loops → build crate/queue → confirm offline availability and rights → route check → Perform → record when eligible → save session log and export. Analysis can be pending without blocking all library browsing.

Live browsing: open the docked library → search/filter → select → choose destination → protect an audible occupied deck from accidental replacement → show load status for that deck → confirm engine readiness. A tap on a row selects; it does not load immediately.

Discover and download: search, paste or share a supported link → resolve the requested recording and actual media source → select version/format → download through an enabled on-device adapter → validate → publish to the phone library → prepare → load deliberately. Include the requested unofficial extractor candidates alongside direct-file and provider routes. Show actual source, progress and cancellation; unavailable capabilities have specific reasons. A catalog match is not a ready file. Do not substitute server setup when an on-device adapter fails; expose the source-specific gap while preserving available local import. The same job contract can add an explicitly configured AWS execution path later. Documents 14 and 16 define source evidence, transfer and preparation.

ParDay-assisted preparation: a request to suggest a crate, transition or preparation edit produces a bounded proposal with named tracks/actions and permitted context; preview, accept selected changes, then verify actual results. A direct request to create a reversible crate can authorize saving it immediately under the existing task scope, with visible outcome and Undo. Audible performance still requires a specifically armed plan. Declining or dismissing a suggestion leaves music and preparation unchanged. A cloud timeout leaves the current set running.

Failure recovery: identify the affected capability → preserve unaffected audio → show a concise actionable state → attempt only a safe recovery → reconcile actual engine/source status → restore the relevant view. UI timeout does not prove an operation failed; do not send duplicate commands blindly.

4. Touch and musical behavior

Every control must define its engine command, units, legal range, pending/acknowledged state and cancellation path. Musical scheduling belongs to the engine. Visual animation is never the timing source.

Control	Required behavior	Alternative / recovery
Full-track overview	Navigation surface. Drag previews a time/beat target. Default Seek Lock on audible decks requires an explicit Seek action to commit. Stopped/cued decks may seek directly according to quantize settings.	Time/beat entry and Beatjump actions; cancelling preview leaves playback unchanged. Advanced jump-to-touch is a deliberate preference.
Detailed waveform	Audio-derived multiresolution waveform with engine-aligned playhead. Propose Nudge while playing and Scrub while paused; final mapping requires DJ testing. Scratch is explicitly enabled. Pinch changes zoom only.	Zoom buttons, beatjump and accessible position controls; no decorative repeating bars.
Jog wheel	Distinguishable centre scratch and rim nudge zones if large enough; otherwise an explicit Scratch/Nudge mode. Slip shows a continuation marker.	Waveform scratch and labelled controls remain available; pointer cancellation releases a held scratch safely.
Dial	Relative vertical drag starts from the existing value; no jump on first contact. Visual rotation follows value. Units remain visible.	Enlarged inspector, fine adjustment, numeric entry and accessible increment/decrement; explicit Reset.
Fader	Default pickup on the thumb/current value; no abrupt far-away jump from an incidental tap. Crossfader curve, reverse and assignment are inspectable.	Accessible set/increment actions; optional advanced jump-to-touch. Detents cannot resist intentional fast cuts.
Cue / Play	Proposed cue behavior: while paused, holding Cue auditions from the cue point and release returns; Play during hold commits playback. While playing, Cue returns and pauses. Final semantics must match the engine contract.	Accessible activation maps to one equivalent action; no duplicate pointer-down plus click command.
Hot cues / loop pads	Cue numbers and bank are always visible. Quantized commands display pending execution. Editing/deleting requires explicit edit state.	Labelled edit actions and undo for preparation changes; no destructive long-press shortcut during normal performance.
XY pad	Names both axes, units, wet/dry and Momentary/Latch mode. Starting-point policy is explicit. Momentary release returns according to the preset envelope; Latch remains clearly active.	Independent sliders, Engage and Bypass. Pointer cancellation follows safe momentary release.
Queue reorder	Drag by a handle; current item and committed next transition have defined protection.	Move Up/Down or position action; actual engine acknowledgement determines whether a late change took effect.
Deck selector	Changes visible focus only; deck ID is never inferred from current colour alone.	Stable names and semantics; held gestures retain their starting deck ID and block a conflicting view replacement.

All simultaneous touches bind independently to stable controls. A parent scroll region cannot steal a performance gesture. Every held action specifies pointer-up, pointer-cancel, background, route-loss and rotation behavior. Haptics and visual feedback run outside the audio callback.

5. Accessibility and common UI states

Android guidance recommends at least 48dp interactive targets. Compose can expand targets beyond visible bounds, so inspect overlaps rather than assuming a small icon is safe because its framework target grew. See Android Compose accessibility guidance, checked 29 October 2026. Provide meaningful semantics for custom dials, pads, meters and waveforms. Screen-reader announcements must report meaningful state changes without narrating every frame or peak.

Use text, position and shape alongside deck colour. Expose essential drag operations through accessible alternatives. Support font scaling, reduced motion, contrast modes and hardware keyboard/switch actions where applicable. Evaluate changes with TalkBack and on physical devices; an automated scan alone is insufficient.

Async states preserve usable controls. Library artwork may show a correctly sized skeleton; a playing deck never becomes a skeleton because analysis is refreshing. Progress percentages come from actual work units. Unknown progress is labelled; a stalled job has retry/cancel/continue-without-analysis paths.

Settings use explicit Save, a visible preview/draft distinction and scoped Reset. Performance controls apply immediately and are not settings drafts. Native search/filter state survives navigation and restoration; web URL persistence rules are adapted to local saved-view identifiers, without putting personal music paths in share links.

6. ParDay interaction contract

ParDay names the assistant; the app name remains open. Use a compact labelled button with a stable hit region. Its panel preserves jog wheels, pads, meters and transport. In landscape it may replace a secondary library or inspector; portrait retains the performance shell and a clear dismiss action. Opening ParDay leaves deck focus, recording, automation and routing unchanged.

Prepare may use a full-height conversation and structured proposal cards. The owner selected ParDay option C: preparation assistance plus explicitly armed assisted performance. Perform stays read/propose until a specific bounded plan is armed; the selected mode is not standing permission for arbitrary audible actions. Each actionable proposal states the affected deck or track, source eligibility, intended musical effect, timing, and whether it can be undone. Separate **Preview**, **Apply to preparation**, and **Arm transition** actions. A generic conversational acknowledgement cannot authorize an unbounded sequence of audible actions. The app validates the proposal against current deck identity, media revision, authorization and engine state when the user acts; a stale proposal becomes reviewable again instead of silently targeting the new track.

States are Ready, Local only, Thinking, Proposal ready, Needs review, Applied, Rejected and Unavailable. Saved results or engine acknowledgements determine success. Preserve useful content during loading and label unknown progress. Cancel stops future work and identifies already-applied changes. Retries cannot duplicate cues, crate entries or transport actions.

Voice is optional and explicitly started. Explain where the audio goes before first cloud voice use; microphone permission for ParDay is separate in meaning from sending a microphone to the master or a broadcast. Spoken replies are off in Perform by default and never injected into the master merely because chat is open. Text and accessible actions remain complete alternatives. No always-listening requirement is introduced.

Suggested quick actions are task-specific: “Find a compatible next track,” “Explain this route,” “Draft cues,” “Build a 30-minute crate,” and “Practice this transition.” They use actual eligible content and state. User-requested catalogue lookup follows source provenance and provider permissions; private playlists, lyrics or audio are not silently copied to a cloud model. Offline, available local controls, analysis and saved guidance continue; ParDay states which cloud-dependent capability is unavailable. Document 13 defines the tool and authorization contracts. The requested paste-link acquisition flow, including unofficial

extraction candidates, belongs in a dedicated source/download panel with clear progress, cancellation, exact-version review and import status; Document 16 owns feasibility and adapter behavior. It must not obstruct transport or be executed in the audio callback.

7. Extended professional workspaces

Video/karaoke, broadcasting and the plugin rack are first-class destinations in the requested finished scope, reached through a labelled workspace switcher when the phone cannot show every destination simultaneously. Lighting receives no controls or placeholder. These destinations keep the same persistent audio controls and route status; they never turn an unrelated swipe into a live transport command.

The video preview distinguishes **Preview on this device** from **Program output**. Caption offset, visibility and language remain explicit. Video frames may drop under a declared visual fallback; audio is not slowed to let the preview catch up. A separate display must be verified before presenting an output as live. Use original/authorized media fixtures for planning and validation.

Broadcasting always presents destination, selected buses and actual connection state before the user starts. A recording indicator is not a broadcast indicator. Show whether microphone, samples and permitted music are included. Stop has an immediate local effect on outgoing capture according to the implementation contract, with remote termination state reported honestly. Do not silently publish a rehearsal or start transmission from an AI suggestion.

Plugin hosting exposes only validated formats and compatible plugins; “everything” cannot mean desktop binaries become Android-compatible automatically. Loading, preset changes and unsafe plugin operations occur at qualified times and follow the engine's cost and failure model. Parameter inspection and bypass have accessible alternatives. ParDay may explain or propose a rack, but cannot install code or insert an effect into audible audio on its own.

Sequencer: Pads includes a dedicated step-pattern view with explicit sample/track assignment, per-step velocity/mute, length, resolution and swing. A visible pattern revision and next-bar activation marker distinguish a queued edit from the sounding pattern. Start, quantized stop, immediate stop and pattern change have different labeled actions. Pattern editing does not steal touch focus from the active deck, and the stopped/offline editor remains usable without ParDay. See FR-027 and AE-06.

8. Rehearsal and verification

Learning Engine contributes optional contextual explanation and practice that moves from demonstration to independent transfer. No guidance appears unsolicited during Perform. Users can skip rehearsal. Completing an assisted exercise is not evidence of independent skill; learning claims require later performance on a different track.

Test a two-track transition, browsing/loading during playback, cue/loop/FX use, recording failure, rotation, pointer cancellation and route removal. Include long titles, empty/error/offline states, four-deck focus changes and a software keyboard. Proposed broader-usability study: four of five participants complete the transition independently, with **zero wrong-deck or destructive-seek actions**. It is optional; personal acceptance requires the owner's independent workflow completion plus the functional/audio gates.

Test late replies after deck reload, offline use, cancellation, duplicate Apply, unarmed audible actions, text scaling and dismissal during a held gesture. When catalogue discovery has no eligible media, show the actual next action; metadata is not a loaded deck. Exercise video, broadcast and plugin failures under the intended workload.

Functional and visual acceptance remain separate. Current status: **NOT VERIFIED - planning only**. Failure in target size or access is resolved by changing composition, not by claiming a tiny control is professional density.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 04 OF 19

Art direction and design system

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.

DIRECTION C SELECTED / PROPOSED ROLE COLORS

Pale glass	Control base	Brass	Deck A ink	Deck B ink
#DCE6EA	#F1F5F6	#8B6B39	#00697A	#A53E18

DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Crystal Current selected

Generated material studies. The same performance task is shown, but exact control placement differs. These are not validated native layouts or audio screenshots.

Selected C - Crystal Current



Historical comparison A - Precision Atelier



C is selected. A is retained as historical comparison. Native geometry, labels and contrast still require implementation validation.

Version 1.0 · 29 October 2026 · C Crystal Current selected - design handoff baseline; native validation pending

New product, with no Mixline reuse. The user confirmed Android first and steampunk, reviewed **A - Precision Atelier** and **C - Crystal Current**, and selected **C**. Crystal Current is the approved art direction for the next design pass on the Galaxy S26 Ultra. A remains a historical comparison, not an approved alternative/night theme. Generated concept imagery does not verify a functional screen. ParDay is the assistant's name; the app's product name remains open.

1. Design brief

Primary user: a serious mobile DJ controlling a live set directly on a phone, including prepared mixing and fast spontaneous transitions. **Secondary user:** a casual streaming DJ who needs clear source availability and a quick route to a first mix. Initial intended use is personal practice and private gatherings. **First useful outcome:** two available tracks loaded, understood and mixed. **Hard moments:** dark or bright environments, one-handed browsing while music plays, small targets, long titles, unknown audio routes, interruptions and heat. **Return value:** preserved preparation, dependable recovery and faster repeat use.

Selected concept: a precision DJ instrument expressed through C pale frosted glass, titanium and subtle brass. The owner selected this material direction; precise colors, geometry, typography and render cost remain to be validated. Its critical information stays clear while material detail gives controls a tactile, crafted identity. Branding is allowed to be expressive; audio state remains literal.

Premium Product UI informs coherent composition, functional/visual acceptance, asset provenance and state completeness. The native-mobile brief overrides its default desktop/web-first sequence. Glass/brass is primarily identity and perceived craft; performance benefits are hypotheses to test, not established facts.

The first design target is the local Android app on the Galaxy S26 Ultra, with no required server or cloud account. Build the design around local library preparation and performance, then expose optional online source/ParDay capabilities through their own truthful states. AWS is a later extension with a roughly US$35/month planning target excluding AI, not an initial screen or onboarding requirement. Document 19 is the entry point for the first connected design deliverable.

2. Comparison record and selected direction

Direction	Composition, type and materials	Signature interaction	Tradeoff
A - Precision Atelier	Graphite instrument bed, smoked glass panels, champagne-brass rims, engraved index marks, clean industrial sans with stable numeric readouts	A value ring and sparse haptic detent clarify a dial's musical position	Most direct match for restrained steampunk; dark glass needs careful brightness and contrast tuning
C - Crystal Current - SELECTED	Pale mineral glass, frosted panes, brushed titanium, dark-ink readouts, restrained brass joints/index marks	The same value ring and focus behavior rendered with lighter material and edge treatment	More futuristic and airy; pale decoration needs strong opaque numerical/control contrast

The original comparison brief called for the **same geometry, controls, state and track content** in A and C. The user has now chosen C. Its next native layout must normalize the generated control differences rather than treating accidental render omissions as requirements. Both omit decorative spinning gears and animated steam: neither represents musical state. Genuine jog motion, meter movement and pending-beat

indication remain useful. The previous Engine Room option is retained only in version history and is not part of the requested render comparison.

Neither option is merely a dark/light toggle. A emphasizes warm mechanical craft; C emphasizes frosted optical depth and precise pale metal. C now leads and still requires separately tuned low-light, daylight and opaque high-contrast treatments. Derive those from Crystal Current; the selection does not authorize A as a separate night theme.

3. Selected direction and proposed tokens

These numerical values remain proposals. **C is selected**; use the C palette table for implementation experiments. The first palette is retained only to document A, while the shared spacing, targets, type and focus rules remain applicable. Contrast and legibility must be measured on the actual composed native screen.

Historical A palette / shared token	Candidate value / rule
Stage / opaque control surface / raised surface	`#0C1015` / `#171D26` / `#222B36`
Primary / secondary text	`#F3F5F6` / `#BAC2CC`
Structural brass	`#B9955D`; material cue, not a warning
Deck A / B / C / D	`#53D7E8` / `#FF955A` / `#B6A0FF` / `#82DEAB`; always paired with letters and state labels
Warning / critical	`#FFD75E` / `#FF667A`; always paired with explicit meaning
Spacing	4dp base rhythm; 8/12/16/24dp group gaps
Targets	48dp minimum on Android; aim for 56dp frequent transport/pads
UI type	Barlow candidate, system sans fallback
Numeric type	IBM Plex Mono candidate or tabular figures in the UI font
Type sizes	Start at 12/14/16/20/28sp; the smallest size is secondary detail, never a license to make essential live labels unreadable
Numerals	Stable width for BPM, time, loop length and meter units
Corner vocabulary	Small radii on instrument controls; softer bounded panel corners; never identical rounded cards for every element
Focus	Explicit high-contrast outline/shape independent of deck colour

Selected C palette	Proposed value / rule
Stage / opaque control / raised surface	`#DCE6EA` / `#F1F5F6` / `#FFFFFF`
Primary / secondary text	`#15252E` / `#455C68`
Structural titanium / restrained brass	`#98A9B1` / `#8B6B39`; no low-contrast metal-as-text shortcut

Selected C palette	Proposed value / rule
Deck A / B light-surface foreground	`#00697A` / `#A53E18`; preserve cyan/orange identities with clear letters and shapes
Deck C / D light-surface foreground	`#6541A5` / `#246A45`; distinguish label colour from luminous waveform accents
Warning / critical on pale surfaces	`#805600` / `#A01939`, with icons and written state

The machine-readable set records `selected_direction: "C"`, uses C for its top-level palette and retains A only as an unselected historical direction. Selecting a material family does not approve every proposed color value. A generated highlight or altered shade does not silently replace a token or establish a contrast pass.

Night and daylight themes are independently tuned. A high-contrast Stage option uses opaque surfaces, stronger separators and reduced decorative detail. Respect system/in-app reduced-motion settings. Persistent transport locations do not move with changing values, badges or labels.

4. Composition and component families

The performance screen is an instrument, not a dashboard of unrelated cards. Waveform, musical position and control hierarchy lead. Avoid giant greetings or artwork that displaces controls. Small album art identifies tracks but is never the primary state signal.

Build canonical components for deck summary, waveform lane, cue marker, gain/EQ/filter dial, pitch rail, channel/crossfader, transport button, performance pad, XY pad, meter, source capability chip and library row. Every component documents states, semantic labels, touch region, units, geometry, engine binding and performance cost. Procedural vector/Canvas geometry is preferred for controls over a bitmap for each state.

Add a compact ParDay access button and structured suggestion card family: context summary, proposal, timing, affected deck/track, Preview, Apply, Dismiss, expired and applied states. Status text carries meaning; an animated orb is optional decoration outside Perform and never a success indicator. The assistant's visual identity shares the instrument's materials but remains subordinate to track and route state. A catalogue search result and an eligible playable file have visibly distinct actions.

Glass panels have a stable opaque contrast backing under text, a restrained edge highlight and optional local optical treatment. Do not refract labels or blur active values. Brass is concentrated on rims, index marks and control grips. Its mechanical detail explains affordances only if it remains recognizable at the actual size.

Six shared UI contracts, adapted to native

Family	Required native application
Badges	Count actionable items only. Hide known zero; display 1-99 then 99+; unknown stays unknown. Stable geometry, exact accessible count and explicit clearing rules. No badge on every deck.
Navigation	Phone uses labelled workspaces/docked browser. Tablet may use a collapsible resizable library pane if useful. Preserve current location and safe return; no desktop sidebar requirement is imposed on a phone.
Loading	Region-sized skeletons only for genuinely asynchronous data. Playing controls stay active. Real progress/unknown state, timeout and recovery; no fake percentages.
Dates	Relevant only to history/export filters: typed/picked parity, clear range and timezone semantics. Date-only data is not silently converted through UTC. Do not add calendar widgets to performance.
Settings	Searchable categories, explicit Save, dirty-state protection, preview/discard, scoped Reset and clear consequences. Live controls remain immediate.
Filters	Removable chips above results, real/unknown counts, useful zero-results recovery, native state restoration. Sensitive paths stay out of public links and analytics.

5. Motion and haptics

Motion has four purposes: immediate input feedback, spatial continuity, musical visualization and optional presentation. Suggested initial timings are 90-160ms for state changes and 140-220ms for workspace transitions. These are tunable proposals, not standards. Animation must be interruptible and input remains usable before it completes.

Engine-driven waveforms and meters retain their meaning under reduced motion; remove ornamental travel, refraction and ambient loops. No looping steam, parallax or brand sequence runs behind a live set. Use fixed geometry and tabular numerals to avoid jumping controls.

Haptics distinguish accepted transport/pad actions, real detents, bank confirmation and rejected actions. They are optional and sparse. Android recommends clear action-oriented feedback and capability-aware fallback; rich effects are not supported uniformly.[1] Do not vibrate on every beat by default, use vibration as an assumed accurate metronome, or inject UI clicks into the master audio. Record device tests for fatigue, actuator quality and possible microphone pickup.

6. Rendering tiers and budget proposals

Tier	Rendering policy	Required behavior
Essential	Opaque/alpha fills, vector bevels, cached highlights, no dynamic blur	Complete qualified controls and state; same semantic hierarchy
Balanced - default	Bounded glass, restrained static metal detail, profiled local effects	Same musical behavior and hit targets
Enhanced	Optional local RuntimeShader effects and richer optics where measured safe	Graceful fallback without restarting audio or changing parameter values

Android's AGSL RuntimeShader support begins with Android 13.[2] It is a graphics capability gate. The audio eligibility and minimum OS decision must consider the equivalent fallback, not depend on the most decorative tier.

Initial proposed budget: roughly 8 MiB of packaged non-audio visual assets, separately budgeted fonts, and no mandatory live 3D scene or video background. Sustain a 60Hz baseline on the declared minimum device under its worst qualified audio workload; proposed visual benchmark is no more than 1% janky frames over five minutes of intense interaction. The exact metric and test fixture must be fixed before the gate is approved. Audio underrun/deadline gates remain separate. No benchmark result is claimed.

7. Comparable render brief

Use image generation for material and art-direction exploration, then rebuild approved controls as exact native components. Generated screens do not prove feasible typography, interaction, licensing or performance.

Shared composition and content

Use a straight-on landscape phone performance screen, approximately 20:9, without perspective distortion. Keep the same layout in both concepts: thin route/recording/ParDay status area; two readable stacked waveform lanes; Deck A on the left and Deck B on the right; track name, BPM and key; two shallow jog rings; a compact central mixer; clearly separated Play/Cue, Sync and four numbered hot-cue pads per visible deck; a lower central crossfader; and labelled access to Library, Pads, FX and Stems. The four visible pads are one bank of the planned eight hot cues; the render is not a reduction of that requirement.

Use fictional fixture tracks **AURORA DRIVE** and **COPPER SKY**, each at **124.0 BPM**, key **8A**; cyan Deck A and orange Deck B. Small text reads **USB Master + Cue**, **REC**, and **ParDay**. Do not invent a product brand or real performer artwork. Show a modest **Concept** label. The waveform is illustrative within the concept render and cannot be reused as real audio-derived data. The mockup does not establish actual 48dp hit bounds or native text fit; those remain build requirements.

Prompt A - Precision Atelier: “Apply the shared landscape two-deck composition in smoked liquid glass over a deep graphite instrument bed. Champagne-brass machined jog rims, restrained knurling, precise index marks and subtle bevels convey steampunk craftsmanship. Use crisp off-white numerals over opaque dark backing; cyan/orange deck states remain explicit. Keep labels undistorted, transport prominent and controls separated. No steam, exposed gears, pipes, antique parchment, external controllers or floating mascot. This is an original art-direction concept for a mobile DJ app.”

Prompt C - Crystal Current: “Apply exactly the same landscape two-deck composition and fixture content in pale frosted mineral glass, brushed titanium and subtle champagne-brass mechanical joints/index details. Use dark-ink labels over stable pale backing, clear cyan/orange deck states, restrained edge light and optically convincing glass depth. Keep typography, control locations, visible functions and framing comparable with A. No refracted labels, floating panels obscuring controls, ornamental steam, gears, pipes or external controllers. This is an original art-direction concept for a mobile DJ app.”

Portrait preparation and performance layouts are required next-design deliverables within selected C, alongside landscape; they do not require another art-direction vote or a second A/C comparison. Use consistent crate/track fixtures and ParDay proposals across orientations. Final production screens must recompose for real phone dimensions, safe areas, text scale and measured touch regions. The earlier approximately 20:9 render canvas is a concept-art framing choice, not a verified Galaxy S26 Ultra logical viewport.

Generated comparison review

Two concept images were generated and visually inspected on 29 October 2026. A shows dark smoked surfaces with champagne-brass rims and pale readouts. C shows pale frosted surfaces, titanium and dark-ink readouts. Both depict the same fixture tracks, two-deck task, cyan/orange identity and compact ParDay access. These images establish the material comparison from which the user selected C.

The images are **not pixel-matched layouts**: the jog/pad ordering differs, A repeats a Gain label and omits visible tempo faders that appear in C. Neither shows the full EQ/control requirement set at once. Their illustrated waveforms and meter values are not computed audio data. Evaluate colour, material, atmosphere and hierarchy; do not choose based on these accidental feature differences or copy them into the interaction contract. The chosen concept still needs an exact native composition with consistent control access, calibrated meters, corrected labels and measured hit targets.

8. Approval and verification record

Inspect native portrait/landscape/tablet/foldable renders with realistic names and all important states. Validate contrast over the worst underlying content, target overlaps, typography at scaled sizes, night/daylight, colour-independent meaning, reduced motion and loading/error treatments.

The owner explicitly selected **C - Crystal Current**. Retain pale frosted glass, titanium, dark-ink labels and restrained brass; do not silently switch to A or a combined A/C product. Compare the selected aesthetic against a flatter functional baseline on the Galaxy S26 Ultra. Record its actual software build, density, display configuration and text scale before testing. If task errors, wrong-deck actions or rendering costs increase, simplify materials before removing useful controls. **Functional status: NOT VERIFIED. Native visual status: NOT VERIFIED. Concept images: GENERATED AND VISUALLY REVIEWED. Art-direction selection: C CONFIRMED.** A polished mockup cannot close either implementation gate on its own.

Sources

[1] Android Developers, Haptics design principles. [2] Android Developers, AGSL. Both accessed 29 October 2026. Font acquisition and licensing checks are specified in Document 08.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 05 OF 19

Tech Stack and Architecture

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; implementation and validation pending **Product basis:** New product; no Mixline code, assets, architecture or approval is inherited. **Platform decision:** Android first; Samsung Galaxy S26 Ultra is the first test device, software reported up to date. **Accountable agents:** Architecture/Orchestrator; Audio Engine; Android Platform. **Related documents:** 06 Audio Engine and DSP; 09 Quality and Device Validation; product requirements and data/content specifications.

Purpose and architectural decision

Build the local Android version first, with a persistent phone-native audio engine, a touch-focused product interface, a durable local library and no required application server. AWS is a later optional extension. Local first does not mean Internet sources can be downloaded offline or that every ParDay function has an on-device language model. The live mix must remain independent of navigation, screen animation, network requests and analytics. Two music decks form the integration foundation; the intended full product grows to four music decks plus a separate sampler. Prepared stems belong to the product baseline; separation during performance is a selected additional capability that must pass device and quality gates.

The proposed stack is Kotlin and Jetpack Compose for Android, a portable C++ DSP core, and Oboe for Android audio I/O. Google recommends Oboe for high-performance native audio. Oboe alone does not decode music files or supply the musical operations of a DJ engine. Those responsibilities require explicit components. [S1, S2]

This decision is **provisionally ready** at the boundary level. The DSP provider, minimum device class and exact cue routes are **experiment-first**. Android first, the Galaxy S26 Ultra first device and permission to use USB audio interfaces/splitters are confirmed. Record the actual variant, OS/build and accessory route before using it as evidence; the phone is not yet certified. No performance result in this dossier has been measured. Budgets and limits are proposed requirements for validation.

Stack and ownership

Layer	Proposed technology	Owner and boundary
App shell and performance UI	Kotlin, Compose, native navigation and accessibility	Product UI agent; sends commands and renders snapshots
Waveforms and tactile controls	Custom Canvas drawing; measured shaders/blur where useful	Product UI agent; never owns musical time
Audio domain	Portable C++ with narrow JNI/C boundary	Audio Engine agent; owns scheduling, DSP and mix state
Android output	Oboe using the available native backend	Android Platform agent; opens, verifies and recovers streams
Decoder and DJ DSP	Adapter-selected commercial or modular components	Audio Engine agent; no vendor API in UI/data models
System playback lifecycle	mediaPlayback foreground service and Media3 session adapter	Android Platform agent; owns focus, notifications and safe resume
Local metadata	Room/SQLite plus versioned cache manifests	Library/Data agent; transactions and migrations outside callback
Analysis and selected prepared/live stems	Cancellable worker pipeline; chosen native/ML runtime	Analysis agent; queued/preparation work yields; audible live inference remains protected
Deferred owner-operated services	Later AWS acquisition comparison; optional sync and ParDay gateway evaluated separately	Services/AI agent; not a local-design or local-release prerequisite
ParDay	Local orchestration/validator, source-filtered context, replaceable OpenAI/Anthropic adapters	AI agent; structured proposals only, with ordinary engine authority
Video/karaoke/broadcast /plugins	Separate media, network and host modules	Specialists; Document 15 resource/distribution gates
iOS expansion	Swift/SwiftUI shell, shared C++ DSP, Apple audio adapter	iOS agent if commissioned; separate routing certification

Flutter and React Native remain viable shells around a native engine if the team can prove equivalent touch, rendering and lifecycle behavior. Their ordinary media plugins do not satisfy the complete engine contract. A web prototype can test layouts; it is not sufficient evidence for the production audio route. JUCE is a credible alternative for an experienced C++ team, but its current upstream framework licence is AGPLv3/commercial and must be evaluated at the selected version. [S3]

Personal deployment and optional owner-operated services

Personal-tool delivery and local-first delivery are confirmed. Implement and evaluate the on-device acquisition arm D first, preserving common fixtures and measurements for a later comparison. The local app must not require a home-lab machine, AWS account, application login or gateway merely to import, prepare and perform with its local library. Actual source extraction still needs technical and access/content evidence; the local-first decision is not a successful extractor result.

The owner accepts AWS as a later optional server extension and selected approximately US$35 per month as its planning target, excluding ParDay AI-provider charges. This target is neither a guaranteed bill nor

deployment/spending authorization. The earlier D-versus-AWS comparison remains the method for evaluating a future expansion, not a prerequisite to local design, development or personal release. No comparative winner has been measured. Document 16 and contracts/acquisition_comparison_protocol.json define the staged evaluation; Document 17 preserves optional home-lab/publication portability, and Document 18 specifies the deferred AWS trial. Account, region, service/instance and deployment remain unset/unexecuted.

Candidate	Comparison purpose	Boundary to preserve
On-device acquisition - first	Implement and measure native/package dependencies, preparation time, battery, heat and maintenance	Acquisition is outside protected audio; downloaded media remains locally usable afterward
Owner-operated AWS acquisition - deferred	Later measure resolver/processing behavior, phone offload, actual transfer mode, server resources, retention, cost and setup effort	Not required for local delivery; if enabled later, outage blocks only dependent acquisition tasks
Optional local home-lab profile - deferred	Preserve LAN/remote portability for a later owner or community deployment	Not a prerequisite for local delivery or the future AWS comparison; no local host has been selected
Hybrid split - deferred, if supported	Later evaluate server resolution with phone transfer or processing when source/session behavior permits it	Do not assume server-issued media URLs work from a phone; test the exact transfer mode

Acquisition execution, ParDay's gateway location, language-model inference location and optional metadata sync are independent decisions. An AWS or home-lab acquisition server does not determine where ChatGPT/Claude requests are processed, and an on-device extractor does not imply an on-device language model. A private gateway can hold provider credentials and enforce limits, while cloud providers still receive the permitted request content sent to them. A local language model is a separate quality/resource experiment. Do not add billing, multi-tenant accounts or a managed database as prerequisites for this personal app.

The service implementation language, persistence and packaging follow the chosen worker's needs; a small Ktor service or another well-supported runtime can be evaluated without prescribing a large managed platform. Document 18 and contracts/aws_deployment_profile.json define the deferred bounded AWS experiment; contracts/aws_cost_scenarios.json keeps researched prices separate from workload assumptions and actual spending. These are planning artifacts, not deployed resources. Preserve versioned contracts and replaceable workers. First record local evidence; when the cloud comparison is commissioned, use the same applicable outage, deletion and privacy tests for both candidates, record actual operating cost, and keep credentials out of distributable client binaries and repository examples. Optional future cloud-sync/commercial services require a separate scope decision.

Module and dependency boundaries

Use modules equivalent to `app`, `feature-performance`, `feature-library`, `feature-preparation`, `feature-recordings`, `core-domain`, `core-data`, `platform-audio`, `engine-api`, `engine-core`, `dsp-adapters`, `analysis` and `test-harness`. Names are provisional; dependency direction is mandatory. Feature modules depend on stable domain contracts, not each other’s implementation. The C++ core must not depend on Android UI or database APIs.

The engine service owns one session across Activity recreation and orientation changes. A Media3 custom `Player` adapter can represent that native engine to a media session; Android documents `SimpleBasePlayer` for custom players. Advertise only safe whole-session system commands. Do not map an unsolicited system “next” command to destructive replacement of the currently audible deck. [S4]

The data layer resolves a media reference and its permission/capability record. Decode workers prepare PCM cache pages. The engine accepts a ready handle and generation identifier. UI state reports the actual command outcome. Backend entitlements are evaluated before starting a feature; an already-authorized local session must not be interrupted by a routine network outage. Third-party streaming entitlements and revocations remain governed by their provider agreement and require their own behavior specification.

Contracts agents must implement together

Contract	Required fields/behavior	Failure behavior
`EngineCommand`	Sequence, engine instance, origin, deck generation/stream epoch, payload; scheduled automation has engine-issued ID/revision	Explicit rejected/stale/queue-full result; atomic batch admission and cancellation acknowledgement
`EngineSnapshot`	Engine instance, stream epoch, deck/manual/master revisions, automation state, configuration, source positions, meters, route, capabilities, warnings	Stale snapshot labelled; audio never waits for UI
`PreparedMedia`	Stable source ID, validated format, duration, PCM/cache handle, capability record, generation	Unsupported/unavailable media cannot replace a playing deck
`DspProvider`	Prepare off-thread; bounded render; reported latency; timestamped parameter application; reset/seek; supported ranges	Provider failure quarantined; no silent parameter change
`RouteDescriptor`	Device identity, actual rate/channels/buffer, master/cue mapping, verified profile	Unsupported mapping rejected before playback
`AnalysisResult`	Source fingerprint, algorithm/model version, confidence, manual overrides and cache location	Stale analysis invalidated without deleting user edits
`RecordingSession`	Permission decision, source eligibility, master tap, format, storage state, recovery journal	Disk failure ends recording while protecting master audio

The DSP contract must expose source-to-engine timeline mapping and algorithmic latency. A vendor whose latency changes without a usable notification/compensation mechanism cannot be combined safely with sample-scheduled cues or aligned stems. Do not assume that two libraries share the same clock simply because both expose a BPM value. Provider selection stops if repeated tempo changes create drift, unsupported blocking/allocation occurs in rendering, or the agreed warm-soak profile fails after reasonable documented tuning.

Thread and lifecycle boundaries

The audio callback performs bounded DSP and mixing only. Control, decoding, model inference, cache writes, database work and service callbacks run outside it. Use preallocated buffers and bounded queues. Critical commands are acknowledged; high-rate continuous parameters may coalesce. SDK initialization, file opening and licence checks occur before a provider becomes active.

Use an engine frame counter with monotonic host-time mapping, not display time, as the scheduling basis. A route reopen creates a new stream epoch. UI interpolation may make motion smooth, but it cannot alter the authoritative playhead. Commands carrying a prior deck generation must not affect a replacement track.

Android focus, calls and background behavior are platform constraints. The app must start the appropriate playback service from a user action and handle interruption states. Current Android 17 documentation applies additional background-audio rules, including foreground-service requirements; native Oboe

playback is not exempt. Test the supported OS versions explicitly. [S5]

Platform and output matrix

Target	Working proposal	Evidence still required
Android phone	First implementation on the selected S26 Ultra; wider support only when commissioned	Exact variant/build/route for the personal target; evidence for each additional advertised profile
Android tablet/foldable	Responsive layout with same engine	Resize, hinge, rotation and multitouch tests
iPhone/iPad	Portable engine boundary proposed; expansion uncommitted	Separate session/routing and performance validation
Emulator	Functional, contract and UI checks	Cannot certify physical latency or USB cueing
Bluetooth	Preparation and tolerant playback route	Not certified as precise live scratch/cue route

Independent stereo master and stereo cue require four output channels or a proven platform-specific combination. A dedicated DJ splitter supports mono master and mono cue on one stereo pair; it does not preserve stereo. The user explicitly allows USB audio interfaces and DJ splitters. This does not add MIDI/HID controller support, and an interface is certified by its actual negotiated channels rather than by its category name.

One verified multichannel USB stream is preferable to unrelated devices whose clocks and delays differ. Query the negotiated configuration rather than trusting the requested one. If unplugging invalidates mapping, mute/pause safely and reconfigure off-thread; never leak headphone cue to the audience. See QA-06 and QA-10.

Build-versus-buy experiment and delivery gates

Evaluate a compatible open-source modular implementation first, then compare a commercial integrated SDK on the same two-deck harness and corpus. Paid DSP is acceptable if justified, but open source is the user preference; neither preference waives quality, distribution or licence obligations. Superpowered publishes separate evaluation/commercial licence routes; no numerical commercial quote is established here. Rubber Band offers GPL and commercial terms, and proprietary distribution requires the applicable licence. [S6, S7]

Compare QA-03 latency, QA-04 callback headroom, QA-05 timing, QA-08 listening quality and QA-02 maximum load. Include seek/reverse, loop joins, key changes, startup buffers and licence availability offline. Pin the selected version, build flags, licence text, supported ABIs and reproducible fixtures in an architecture decision record. Avoid declaring a “free” path until all code, models and dependencies have been checked.

Before finalizing production rendering budgets, pass two-deck engine/lifecycle integration. Initial layout and component design can proceed from Document 19 in parallel. Before advanced feature certification, pass the aggressive profile with four decks, sampler, recording and the intended glass treatment. Suspend optional analysis and precomputation work and reduce visual cost first under pressure. Live inference supplying audible stems is part of the protected workload; it cannot be suspended without changing the sound. Do not change active tempo, stem alignment, limiter latency or output routing mid-set. If the profile remains unstable, it is not certified.

Canonical checkpoints are G0 Decision baseline, G1 Audio feasibility, G2 Working vertical slice, G3 Core performance beta, G4 Advanced performance, G5 Source/provider and ParDay integrations, G6 Release candidate, G7 Personal installed release and G8 Maintenance/community readiness. G1 resolves DSP/routing experiments; G2 proves the full command-to-sound path; G3 certifies P0; G4 certifies extended profiles. G5 cannot weaken G3/G4 behavior. Its local source and selected assistant requirements remain in scope; cloud-specific G5 cases apply only when the later optional extension is evaluated. AWS setup/comparison does not gate G6 or G7 for the local version. G6 verifies the actual distribution package and evidence set. G7 verifies the owner's signed installation, upgrades, backups and recovery. Optional GitHub publication or public-store distribution is separate from that personal completion gate. G8 monitors regressions and revisits device profiles.

Native release packaging must include 16 KB page-size compatibility checks for the core and every prebuilt SDK; Android documents both alignment and runtime requirements. Verify the final packaged APK/AAB, not only locally linked libraries. [S8]

ParDay and specialist integration boundaries

Document 13 owns the assistant contract. The user selected preparation assistance plus explicitly armed assisted performance; this does not give the model unrestricted engine authority. Add modules for assistant orchestration, model adapters, source-provenance filtering and deterministic action validation. Keep provider credentials in a controlled service for a distributed app. A consumer chat subscription is not a credential or authorization to automate a chat website; API billing/credits must be checked for the selected account and provider. All provider responses are untrusted proposals.

An accepted proposal is revalidated against current session, stream epoch, deck generation, manual-control revision, media/analysis/preparation revision, source grant and user authority before ordinary engine commands are issued. A stale proposal expires. Assistant timeouts, model changes and provider outages cannot change transport or mute a local deck. Local conversational inference yields before active audio and live stem inference. No raw audio, microphone capture or whole-library upload is a default.

Scheduled automation carries an engine-owned execution identity/revision checked immediately before each event executes, including events already queued. Manual takeover, Stop Automation, relevant load/route changes and grant invalidation revoke the affected execution revision. A model cannot mint or renew it. Batch admission is atomic: capacity or validation failure rejects the whole native batch, and later cancellation invalidates remaining events with an acknowledgement. A load-then-arm workflow waits for load acknowledgement and creates a fresh proposal. See the 0.2-draft engine contract and Document 13 for exact transaction semantics.

Document 16 owns requested in-app link acquisition, including unofficial extraction candidates. Keep the adapter/worker pipeline outside playback and separate technical success from provider access, content rights and distribution clearance. Evaluate and implement on-device extraction first. A later D-versus-AWS experiment can justify an optional extension; it is not a condition for starting or finishing the local version. A verified vendor API remains a candidate, not the selected mechanism. An official API is not a universal technical gate.

Document 15 owns video, karaoke, broadcast and real third-party plugin hosting. These add measured resource profiles and distribution constraints rather than weakening the audio clock or callback rules. A network encoder may fail independently. A plugin whose scheduling or latency behavior cannot satisfy the graph contract cannot be certified.

Completion criteria and unresolved decisions

Architecture completion means contracts compile in a minimal vertical slice, replacement/route/error cases are demonstrated, and QA-01 through QA-07, QA-10 and QA-17 have evidence against selected hardware. It does not mean the interface merely resembles a professional mixer.

No owner answer is required before interface design can begin from Document 19. Use the proposed stack and explicit capability/error states while implementation proves DSP, routes, extraction and specialist formats. Capture S26 Ultra variant/build/accessory evidence at G1; select actual DSP/model providers and their spending limits before paid integration. App naming, voice, wider device support and optional source-publication details can follow their relevant work packages.

Local first, AWS later and the approximately US$35/month future AWS planning target are confirmed. AWS account/region/runtime/access choices wait until a concrete expansion proposal; they do not block local work. Android first, selected C, ParDay armed assistance, allowed audio accessories and the full selected professional scope remain fixed. Record material architecture or scope changes in the decision ledger rather than presenting a default or a failed experiment as an owner decision.

Sources

Checked 29 October 2026. Source facts support constraints; recommendations and targets are this plan’s proposals.

- **S1:** Google, Low latency audio.
- **S2:** Google, Oboe FAQ.
- **S3:** JUCE, Upstream licence.
- **S4:** Android, The Player interface.
- **S5:** Android, Background audio hardening.
- **S6:** Superpowered, Pricing and licence categories.
- **S7:** Breakfast Quay, Rubber Band Library.
- **S8:** Android, Support 16 KB page sizes.
- **S9:** Supabase, Official documentation.
- **S10:** Firebase, Official documentation.
- **S11:** Ktor, Server project documentation.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 06 OF 19

Audio Engine and DSP

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; implementation and validation pending **Product basis:** New product; no Mixline reuse. **Platform decision:** Android first is user-confirmed. **Accountable agents:** Audio Engine; Analysis/Stems; Android Platform. **Dependencies:** 05 Tech Stack and Architecture; 09 Quality and Device Validation; content-rights requirements.

Scope and audio authority

Implement a phone-native, sample-scheduled engine for two foundational music decks, expanding to the selected four music decks plus an independent sampler. The local version comes first and requires no AWS or home-lab server; future service work cannot become part of its render graph. Include manual mixing, cueing, scratching, tempo/pitch, key lock, sync, grid correction, loops, beat jump, slip, reverse, EQ/filter, FX, stems, monitoring and eligible recording. All numerical limits and budgets below are proposed targets; no engine or device has been tested in this planning session.

The audio engine is the authority for what is audible, when commands execute and what the interface displays. The owner selected ParDay armed assisted performance; preparation plus bounded transitions/automix are required capabilities, with explicit arming and immediate manual takeover. Galaxy S26 Ultra is the first test device, not a measured performance guarantee. ParDay never supplies the render clock or performs conversational inference in the callback. It produces proposals; only accepted, fresh, capability-checked actions become ordinary generation/epoch-bound engine commands. Preparation metadata may be undone, but audio already heard cannot be reversed by an Undo label. Optional assistant workloads yield before active audio and audible live-stem inference. Video follows the audio clock; broadcast encoders consume bounded off-thread taps. See Documents 13 and 15. Waveforms must represent real analyzed samples and actual playhead positions. A visual animation or timer must never stand in for working audio behavior. Each deck exposes an explicit state: empty, preparing, ready, playing, paused, seeking, ended or error, with separate flags for loop, slip, reverse and sync.

AE-01 - Decode, cache and supported media

Separate decoding from rendering. Background workers decode validated source media into bounded PCM pages with source-frame metadata. Oboe handles PCM I/O rather than compressed-file extraction/decoding. Android’s format table provides a platform baseline, but the product must publish a tested codec/container matrix, including actual bit depth, sample rate, channel handling, VBR and seeking behavior. [S1, S2]

Propose MP3, AAC/M4A, WAV and FLAC as initial required formats; evaluate AIFF, ALAC, Ogg Vorbis and Opus for the full product. Do not claim support solely from filename extensions. Unsupported DRM or providers without permitted PCM access remain ineligible for engine operations. A corrupted file must produce a clear error without replacing a working deck.

Cache policies must anticipate reverse and scratching around the playhead. Preload active cue/loop regions. Long seeks outside cache are asynchronous and visible; arbitrary compressed-file seeking cannot be promised instantaneous. Each load increments a generation identifier so a late decoder completion cannot insert an old track into a new deck. Acceptance: QA-07, QA-12 and QA-15.

AE-02 - Timing, tempo and musical operations

Use an output frame clock with monotonic host-time mapping and a stream epoch. Keep source position separate from output position because tempo, reverse, scratching and loops transform that relationship. The beatgrid maps musical beats to source frames and includes downbeat, tempo segments and manual corrections. Analysis confidence must be visible when unreliable; manual overrides take precedence over fresh automated analysis until intentionally reset.

Quantization schedules to engine frames using the selected subdivision. Sync uses the designated master deck/clock and its grid; loading a track must not unexpectedly change the master. Define separate

tempo-sync and phase-sync behaviors. Pitch bend is temporary; the tempo fader sets the persistent rate. Key lock changes time while preserving nominal pitch; key shift changes pitch independently within the validated range. Disable incompatible combinations explicitly instead of silently changing behavior.

Loop in/out points, half/double, roll, beat jump and saved loops require exact source positions. Loop jumps use controlled de-clicking. Slip maintains a shadow position so releasing scratch/reverse/roll returns to the expected musical point. Document whether a specific effect advances that shadow position; all UI labels must match. Acceptance: QA-05, QA-07 and QA-08.

AE-03 - DSP provider contract

The selected provider must support these operations or disclose the unsupported capability before selection:

Operation	Required contract
`prepare(config)`	Off-thread allocation; sample rate, maximum block, channels, quality and ranges
`latencyFrames()`	Exact current algorithmic latency and notification of permitted changes
`process(n, inputs, outputs)`	Bounded render into caller-owned buffers; no unexpected allocation/blocking
`applyParameters(frame, values)`	Defined sample/frame granularity, smoothing and safe range validation
`seek/reset(generation, sourceFrame)`	Defined flush, warm-up and timeline-reset behavior
`mapTimeline()`	Source-to-output mapping, including stretch/reverse and latency compensation
`capabilities()`	Supported combinations and restrictions, not merely individual features

Do not select a backend that cannot reconcile its internal time with the engine clock. Compensation is tested at varying tempo/key settings, not inferred from a fixed nominal delay. A provider that passes steady playback but drifts after parameter changes fails the relevant capability.

Evaluate integrated commercial DSP against modular libraries. Superpowered documents separate rate/pitch controls, quality options and mobile workload caveats at large rate changes. Rubber Band provides time/pitch processing under GPL/commercial routes. Their availability does not establish product quality; use a shared blind-listening and device benchmark. [S3, S4]

AE-04 - Mix graph and gain behavior

The proposed deck chain is source read/scratch → aligned prepared/live stem mix when active → time/pitch processing → trim → EQ/filter → assigned channel FX → channel fader → crossfader assignment → master sum. The sampler and eligible microphone paths enter documented buses. FX send/return placement and wet/dry behavior must be specified per effect so bypass does not change level unexpectedly.

Selected microphone input requires explicit permission and route selection, input gain/mute, a clear active indicator and a documented talkover/duck policy. Keep it muted until enabled. Verify input/output clock behavior, latency compensation and whether platform processing alters the signal. Do not promise feedback prevention or professional microphone quality from every built-in phone mic. A recording-only mode and live-monitor mode need distinct states; permission denial must leave music playback usable.

Define the headphone cue tap before channel fader/crossfader. Provide cue selection, cue level and cue/master blend. Preview/metronome routes must never leak into master. Master processing includes

headroom, optional master FX, limiter and metering. Meter pre/post points must be named. Record after the agreed master FX/limiter point and before physical headphone routing; hardware volume should not alter the recording.

Use floating-point internal processing and measured headroom. Apply parameter smoothing to gains, EQ/filter changes, cuts and routing transitions where musically appropriate. Specify linear/equal-power/cut crossfader curves with endpoint behavior and channel assignments. A limiter prevents output overflow within its validated operating conditions; it is not a substitute for useful trim and clipping indicators. Acceptance: QA-06, QA-08, QA-13 and QA-14.

AE-05 - Real-time execution and concurrency

No file/network I/O, database calls, memory allocation/free, mutex waits, sleeps, model inference or unbounded one-time calculations may occur in the device callback. Google specifically identifies blocking and bursty callback work as a dropout risk. [S5]

Preallocate buffers, pools and command capacity. Critical play/cue/load commands need sequence acknowledgements; continuous knob values may coalesce. Define queue saturation errors and stale-generation rejection. Control-side state never races directly with render-side mutable state. Route changes, provider initialization and recording finalization occur on control/worker threads.

Send bounded snapshots to the UI containing actual state, meters, position, stream epoch, underruns and capability status. Visual refresh cannot delay render. External library calls must be checked for real-time safety under the exact operation used. Acceptance: QA-04 and QA-07.

Scheduled automation uses an engine-owned automation ID/revision and engine instance guard, in addition to deck generation and stream epoch. The bounded event execution path checks these guards immediately before applying an event. Manual takeover invalidates affected pending automation before the manual command takes effect, so an already queued fader event cannot later reassert an old value. Stop/cancel, route/load invalidation and capability changes revoke the applicable revision; acknowledge cancellation and the last applied state honestly. Admit each native automation batch atomically or reject it entirely; never leave a partially scheduled transition hidden behind a success message. Trusted control-side code supplies this authority, never a model proposal. Cancellation has reserved bounded control capacity so a full normal queue cannot drop it; its acknowledgement names the effective frame after which affected events cannot execute. At a shared control boundary, invalidation precedes affected due events and manual values apply with ordinary smoothing. The plan's own accepted parameter writes do not invalidate its manual-control revision. Restart discards execution tokens and queued events; fresh arming is required.

AE-06 - Stems and sampler

Prepared four-part stems - vocals, drums, bass and other - are the baseline design target for eligible sources. Each stem set records source fingerprint, model and version, aligned sample count, channel format, preparation settings and integrity checks. Controls must distinguish unavailable, preparing, ready and failed. Full-mix fallback keeps the original source and user cue metadata intact.

Start with stem summing before one deck stretch stage unless individual stem DSP requires otherwise. If stems are processed independently, compensate all paths consistently and test reconstruction/phase alignment. Separation artifacts are a quality concern; do not describe outputs as exact original studio stems.

Live separation is conditional on a real mobile export and thermal/quality benchmark. Demucs is a research candidate whose upstream repository is archived; this is not a supported turnkey mobile product. Check exact code, weights and dependency licences separately. ExecuTorch and LiteRT are mobile-runtime candidates; model conversion and supported operations still need validation. [S6, S7, S8]

The sampler is separate from the four music decks. Specify one-shot/loop/gate modes, quantized trigger, velocity/level behavior, choke groups and voice-stealing order. Proposed certified profiles start with eight simultaneous voices and evaluate sixteen in the extended profile. Reserve voice capacity so a new pad does not interrupt a music deck. Rights and provenance travel with sample packs. Acceptance: QA-09, QA-13 and QA-18.

The selected sequencer adds editable step patterns, per-step velocity and mute, pattern banks, length/resolution, swing and quantized start/stop/pattern changes. Proposed initial grid: 16 steps per bar with selectable 1-4-bar patterns and explicit straight/triplet resolution where supported; final limits remain configurable acceptance decisions. Store event times in musical units, then schedule against the authoritative beat/engine clock. Edits during playback become a new pattern revision at the selected next bar boundary, with immediate mute/stop as distinct controls. A tempo or leader change remaps future events without double-triggering or altering already sounded events. Preserve sample and pattern identity through bank changes; voice limits and choke groups still apply. Validate repeated playback, swing bounds, tempo changes, stop/cancel and manual takeover under QA-05, QA-07 and QA-13.

AE-07 - Output, recording and interruptions

Support explicit master-only stereo, split mono master/cue, and verified four-channel USB profiles. A standard duplicated headphone splitter does not produce independent cue audio. AAudio streams attach to a device and require actual negotiated configuration checks; endpoint routing is not automatically supplied by the API. [S9]

Prefer a single multichannel device clock for stereo master plus stereo cue. Bluetooth remains a convenience/preparation route because Oboe documentation lists it among routes that may not provide low-latency streams. Do not market universal accurate scratching over wireless audio. [S1]

Recorder disk writes consume a bounded ring off-thread. Full disk, permission loss or encoder failure stops recording visibly without blocking live audio. Use recoverable chunks/journals where practical, and an explicit incomplete-recording status. Recording eligibility is checked per content source, including mixed-source sets.

Audio focus loss, calls, route removal and process death are distinct events. Obey system focus policy. On unsafe route loss, pause/mute rather than rerouting cue/master blindly. A process crash stops audio; restore session metadata at next launch without claiming uninterrupted playback or automatically blasting sound. Acceptance: QA-10, QA-11 and QA-18.

Default route-loss state transitions

These proposed defaults apply to the first implementation and must be tested under QA-10. Never equate a UI-selected device with a verified audio-channel map.

Event / current topology	Immediate state	Recovery condition
Master device disappears, or channel topology becomes unknown	Mute output, pause deck transports safely, invalidate the old route and stream epoch; preserve session state	Reopen off-thread, verify channels and show route; explicit user resume before audible output
Cue endpoint alone disappears while a separately verified master path remains unchanged	Disable and mute cue, keep master only if unchanged master mapping is positively verified; show alert	Verify cue mapping before enabling it; never send cue material to master as fallback

Event / current topology	Immediate state	Recovery condition
Whole USB multichannel interface disconnects	Treat as lost master; pause/mute and retain state	No automatic phone-speaker fallback; verify the entire topology and require resume
New device appears or an old device reconnects	Keep existing safe state; inspect capabilities off-thread	Accept only a validated mapping; a device name match alone is insufficient
Sample rate/channel count changes without user request	Mark route invalid, mute/pause, establish a new epoch	Rebuild format/latency mapping off-thread; resume deliberately
Focus is lost or a call interrupts	Follow the OS focus rule and transition to the documented interrupted state	Default to deliberate resume after disruptive focus loss; do not promise continued audio against OS policy

If platform behavior reroutes audio before the app observes the event, that is a concrete certification risk. Capture actual outputs and exclude unsupported route combinations rather than claiming software prevents every transient leak.

AE-08 - Thermal policy and completion gate

When resources tighten, suspend optional analysis and queued or precomputation stem jobs and reduce decorative rendering first. Inference supplying currently audible stems is protected active audio work; do not suspend it or silently replace its outputs with a full mix. P3 must be certified for that sustained workload. Never silently change active tempo, key, stem alignment, limiter latency, cue topology or audio route. Deny new expensive feature activation when the certified profile cannot support it. If the already-active graph cannot survive the defined maximum-load test after non-audio reductions, the profile fails certification.

Reduced visual effects must preserve control locations, hit areas and current gestures. Musical operation cannot change because the glass surface was simplified.

Engine completion requires QA-01 through QA-11 plus applicable QA-13/14/18 results, a pinned dependency/licence manifest, reproducible test corpus and documented unsupported combinations. Unit tests alone cannot certify touch response, thermal stability, output routing or subjective audio quality. Agents must return evidence and remaining limitations, not a visual demo as proof.

Sources

Checked 29 October 2026. Engineering targets and graph design are proposals.

- **S1:** Google, Oboe FAQ.
- **S2:** Android, Supported media formats.
- **S3:** Superpowered, TimeStretching API.
- **S4:** Breakfast Quay, Rubber Band Library.
- **S5:** Google, Low latency audio.
- **S6:** Meta Research, Demucs repository.
- **S7:** PyTorch, ExecuTorch on Android.
- **S8:** Google, LiteRT overview.
- **S9:** Android NDK, AAudio.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 07 OF 19

Library, data and music-source architecture

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; implementation and device validation remain unperformed **Product:** New mobile DJ application; no Mixline reuse **Platform direction:** Android first confirmed; implementation stack remains subject to G1 **Owners:** Data lead, integrations lead, security lead and audio lead

1. Data principles

The user has confirmed serious mobile DJs as the primary audience, casual streaming DJs as secondary, and private practice/personal gatherings as the initial use context. The desired catalog scope is “any and all.” Treat this as a broad integration ambition, not a claim that every service permits every operation or is approved for this app. Document 14 specifies discovery and acquisition; Document 16 adds the explicitly requested unofficial link-to-file subsystem. Document 13 specifies ParDay, now selected for preparation plus explicitly armed assisted performance. The delivery objective is now a personal tool with optional future open-source sharing. Neither that choice nor self-hosting overrides a source contract or independently settles the rights for a particular gathering. The latest owner decision makes the first version local: the Android app, library, preparation and live engine run on the phone without a required server. Implement and validate the on-device acquisition path first. Retain the Document 16 comparison for a later optional AWS extension; no comparative runtime winner is claimed. Local first does not make Internet downloads or optional external AI available offline.

The local library is the durable record of the performer's preparation. It must remain useful without an app account, network connection or provider partnership. Cloud sync, if selected, supplements this record. This design supports FR-001-008, FR-031, FR-034, FR-036 and FR-040-044.

Keep user edits separate from machine-derived analysis. Cue points, corrected grids and playlists are user work; waveforms and detected key are recomputable caches. Losing either is undesirable, but their retention, conflict and migration policies differ. Imported media, user preparation and provider entitlements must not be conflated into one track object.

Use a transactional local database. Room/SQLite is the proposed Android implementation if the stack decision confirms Kotlin. Media bytes and large derived artifacts live outside database rows, with manifests and integrity checks. Real-time audio reads prepared immutable state, not database queries. Database, indexing, network and file operations execute off the audio thread.

2. Entity and identity contract

Entity	Required fields and invariants
Track	Stable app UUID, display metadata, tags, rating, explicit/clean status where known and creation/update metadata. A title or ISRC is not a unique binary-media identity.
MediaAsset	UUID, track reference, local/provider source kind, durable URI/reference, content hash when available, byte size, container/codec, sample rate/channels, duration frames, rendition/version and availability state.
ProviderReference	Provider ID, provider track ID, market/territory, rendition ID and account association. Provider IDs never collide with local UUIDs.
AnalysisVersion	Media identity/hash, algorithm/model version, parameters, completion/error status, detected BPM/key/grid/loudness and artifact manifest.
UserPreparation	Media revision, cue markers, saved loops, corrected tempo map, locked fields and edit revisions. Never overwrite this with automatic analysis.
Playlist/Entry	Playlist UUID/type/rule, entry UUID, track/media reference, stable ordering key and optional notes. Same track may occur multiple times intentionally.
Session/History	Session UUID, deck identities, media assignments, queue, routing configuration reference, checkpoint version and performance event/played records.
Recording	UUID, segment manifest, codec/sample format, sample count, creation time, markers, finalization state, recoverability and rights state.
CapabilityGrant	Provider/track/rendition/account context, decision values, evidence/contract version, validation time, expiry and restriction reason.
AssetLicense	Asset ID, creator/source, license snapshot, receipt/permission reference, allowed uses, attribution and file hash.
DiscoveryCandidate	Source-tagged catalog reference, displayed title/artist/version, timestamp, permitted retention, and playable-acquisition options. Discovery is not a media file.
ProvenanceLabel	Field-level origin, provider restrictions, allowed processing purposes, expiry/deletion rules and derived-from references. Survives joins and exports.
AcquisitionJob	Trusted request/revision, selected actual-source candidate and format, method/adapter version, separate technical/access/content/distribution evidence, transfer lease, cancellation revision, validation and publication journal. Document 16 owns the job schema.

A file rename or move should not create a new musical identity if content can be matched confidently. Conversely, a different edit, remaster or encoding with changed timing must not silently reuse sample-position cues. Content hashes help detect exact duplicates; partial fingerprints are candidate matches, not proof. Compute expensive fingerprints asynchronously and ask before merging ambiguous identities.

Cues and loops store an explicit media-time basis: source-frame position and source sample rate, plus optional musical beat position and grid revision. Tempo maps are ordered beat-to-source-time anchors. Half/double-BPM edits change the musical interpretation without silently moving the underlying cue audio. Every migration documents how old positions map into the current basis.

3. Import and availability workflow

User selects files or a folder through platform-supported access. Validate the selection, inspect headers within resource limits, stage metadata, and obtain durable URI permission or copy verified bytes into app-managed storage. A cloud document provider can require download; report that state before promising offline use. Import must not assume a transient URI remains readable after restart.

Availability states are explicit: discovered, importing, locally ready, stream ready, analysis pending, permission missing, file missing, expired, unsupported and failed. “Analyzed” and “playable” are separate. Basic playback may begin before optional analysis completes when the decoder and route are ready. Track load shall fail safely if a file changes underneath it.

Offer a no-login demonstration crate made from commissioned or otherwise redistribution-cleared music. The practical first-use path is demo → file/folder import → background analysis → ready crate → output check → mix. This makes the app immediately explorable while preserving a real local collection workflow.

Library search uses a local index. Search, tags and smart-crate rules cannot require a network round trip. Keep a readable missing-file repair flow; preserve the user's cues and playlist references while the media is unavailable. Test restart, permission revocation, removable-storage loss, duplicate imports and partial-copy failure.

4. Provider integration admission

Existing DJ partners demonstrate feasibility; they do not grant our app equivalent access. Ordinary catalog/playback APIs are not evidence of raw PCM access, simultaneous DJ playback or permission to manipulate recordings. For negotiated SDK/streaming integrations, admit the source only after written terms, SDK access, supported territories, commercial use, security obligations and feature restrictions are documented.

Source	Verified research finding	Planning treatment
Local authorized files	Unencrypted readable files can use the app's supported decoder/DSP path; technical support does not establish public-performance or redistribution rights.	Primary independent performance path.
Spotify	Selected DJ apps support online personal mixing; the ordinary Developer Policy prohibits DJ overlap, voice-control assistants and ingesting Spotify Content into AI. [1,2]	Partner negotiation for Spotify playback; provenance restrictions also apply to ParDay. A Spotify-link acquisition candidate is evaluated separately in Document 16 and must identify the actual audio source; no approved cache is relabelled as a local export.
Apple Music	Apple lists named DJ partners; MusicKit describes catalog/library/playback access. Ordinary media terms limit personal/noncommercial use. [3,4,5]	Separate partnership/usage review; do not infer custom-DSP access.
Tidal DJ	Official documentation states stem and offline support vary by partner. [6]	Obtain precise mobile SDK and contract capability matrix.
Beatport	Qualifying plans advertise offline storage and stems in supported DJ integrations. [7]	Candidate for partnership exploration; tier/track/territory restrictions remain explicit.
SoundCloud	Existing integrations differ: supported rekordbox configurations offer DJ-plan offline storage, while djay documents live streaming. [8,9]	No provider-wide offline assumption; validate intended mobile integration.

Source	Verified research finding	Planning treatment
YouTube	Policies restrict separation/modification and hidden/background playback; multiple automatic players are also restricted. [10,11]	Ordinary player/API access does not establish raw-audio permission. The requested extraction adapter is a separate feasibility and access review in Document 16; its result must pass validated local-media publication.
Artist/store downloads	Bandcamp exposes downloadable purchase formats; SoundCloud has uploader-enabled web downloads for eligible tracks. [12,13]	Link to the authorized acquisition flow, then import the resulting local file; do not scrape purchase sessions or bypass disabled downloads.
Creator-owned YouTube upload	YouTube documents Studio downloads and Google Takeout for one's uploaded videos. [14]	User exports through the official channel, then imports an eligible local original/export. Ownership of the upload alone does not establish rights to every third-party element.

Recording and stem permissions differ independently. djay documents recording restrictions across streaming providers and no Neural Mix for Apple Music/Spotify; this is evidence that source-dependent controls are necessary, not the final grant for this product. [9] A “DJ” subscription does not by itself establish the right to perform commercially in public. The user's intended territory, venue and use must be reviewed alongside the provider agreement.

5. Capability resolver

Permission values are **allowed / denied / unknown**. Unknown fails closed for the affected operation. Do not convert missing fields to true or interpret “playable” as “recordable.”

```
CapabilityContext
  sourceId, mediaRevision, providerAccountId, territory, accountTier
  contractVersion, checkedAt, expiresAt, offlineAuthorizationUntil

OperationGrant
  readPcm, seek, analyze, cacheAnalysis, changeTempo, changeKey
  applyEffects, separateStems, cacheStems, cacheAudio
  record, broadcast, exportAudio, publicPerformance
  displayMetadata, lookup, exportMetadata, aiInference, aiEmbedding
  aiTraining, voiceControl, downloadOriginal, extractLocalAudio
  each: allowed | denied | unknown, reason, evidenceReference

TechnicalLimits
  concurrentDecks, seekGranularity, tempoRange, cacheQuota
  supportedFormats, routeRequirements, certificationTier
```

Effective capability is the intersection of source grant, track/rendition, account, territory, authorization time, engine implementation and certified device capacity. A future commercial app entitlement, if one is selected, would be another layer and could not override source restrictions; product billing is not required for the personal local version. Resolve at load and again before sensitive operations or entitlement changes. Cache grants only within their validated lifetime.

For imported local media, do not invent a provider restriction or require users to upload ownership documents. The app can expose technically supported local operations while explaining that users are responsible for the rights applicable to their use. The public-performance field records known provider restrictions or explicit user-managed context; it is not an automated copyright adjudicator. Unknown provider grants still fail closed for provider-controlled operations.

The UI explains unavailable operations before commitment. The engine/encoder also enforces them. When a user is recording and selects an ineligible source, provide a clear choice before transition. Finalize the

existing recording before restricted audio enters it. If rights change during performance, preserve valid audience playback where permitted while stopping prohibited recording/export. Mark partial recordings accurately; never imply that the complete set was captured.

The integration layer must not expose protected provider caches through an unrestricted fallback file URL. Preserve metadata/attribution, cache, account and expiry requirements. The owner now explicitly requires in-app link acquisition, including unofficial extraction candidates. Documents 14 and 16 therefore include maintained extractor and documented third-party conversion adapters alongside direct-file transfer and approved streaming. Technical support, service access, content use and distribution evidence are separate dimensions. Ordinary API access is not claimed to authorize extraction, and unofficial classification alone is not a technical rejection. Keep unverified source support visible as an unresolved requirement.

Retain requested catalog identity and actual media origin independently. A Spotify reference resolved to another platform does not become a Spotify audio master, and its metadata restrictions are not erased by importing the matching file. Apply provenance rules to the particular fields/artifacts, without incorrectly assigning every restriction to independently measured local analysis. Acquire, validate and publish before the engine can load the result; source failures never replace a live deck.

5A. ParDay data boundary

ParDay must receive only data whose provenance permits the intended processing. Spotify's developer terms expressly cover AI ingestion, not merely model training; the ordinary integration therefore cannot send Spotify-returned audio, artwork, metadata or derived context into a language model without a separate applicable authorization. [2,15] A user's app-level consent is necessary for optional cloud processing but cannot override the source grant. Do not relabel provider metadata as “user data” merely because it appeared in a playlist, was copied to a local table or was selected on screen.

The gateway distinguishes user-authored requests, user-managed local metadata, licensed app knowledge and provider-controlled fields. Each tool request passes through a deterministic policy resolver before execution; its response passes through an outbound-data filter before any model receives it. Provider display cards may be rendered outside model context where the applicable agreement allows that display. This separation is an architectural control, not a claim that the combined AI/provider use case has automatically been approved.

A safe first integration uses ParDay to interpret a user's independently supplied mood, duration and local-library preferences, while a reviewed deterministic catalog connector performs eligible lookups. It returns source-labelled cards to the UI; it does not feed restricted catalog results back into ParDay. Keep provider-dependent voice playback disabled unless explicitly admitted. The local engine and user-managed local library remain useful when a catalog or AI gateway is unavailable.

A combined track record may contain local user tags and restricted provider artwork. Resolve at field and operation level rather than granting the whole record the least restrictive origin. Logs, analytics, embeddings, vector indexes, conversation summaries and crash reports are also destinations; deleting a visible chat message does not remove an embedding unless deletion is implemented there. Store provenance with derived objects so revocation can find and invalidate affected data.

6. Cache, storage and recovery

Separate media, analysis, waveform, artwork, prepared-stem and provider-cache budgets. Pin every currently playing or staged deck's required artifacts against eviction. Avoid a design that loads an entire long track into RAM. The audio subsystem defines bounded decode/read-ahead windows and communicates readiness without blocking.

Cache manifests include media hash/revision, algorithm version, format, completion marker and checksum. Write to a staging file, validate and atomically publish. A partially generated waveform or stem set must not

appear ready. User-authored metadata is never reclaimed as disposable cache.

Before recording, reserve a configurable safety margin and estimate duration from the actual selected format. Stream writes to recoverable segments and maintain a journal. Storage exhaustion stops/finalizes recording while playback continues. A crash can interrupt audio; recovery must offer valid completed segments and the last consistent state rather than claiming uninterrupted service.

Provider offline caches are encrypted or protected as their agreements require, and expire according to their own authorization. Local user files do not inherit those expirations. Reauthentication and cleanup must never delete unrelated owned audio or user preparation.

7. Migrations, backups and optional sync

Every schema has a numbered migration, rollback/recovery strategy and representative old-version fixtures. Run migrations outside an active performance. Preserve a backup/checkpoint before destructive transformation, check free space, and either complete atomically or retain the previous valid state. A failed migration displays an actionable recovery option.

Portable backup includes versioned metadata and media references; media copies are opt-in and rights-aware. OAuth tokens, provider secrets and restricted caches are excluded. Restore validates paths and rejects archive traversal, duplicate IDs and excessive sizes.

If sync is selected, use record revisions and tombstones. Preserve conflicts in user-edited cues/grids for deliberate resolution; do not silently choose a timestamp winner. Upload only consented data, keep local operation functional during conflict, and pin the session's snapshot while performing. Cloud analysis of owned audio requires a separate opt-in and retention policy.

8. Acceptance and unresolved decisions

Required test sets cover 10,000 proposed library entries, duplicated files, changed renditions, malformed tags, long paths, Unicode, interrupted imports, missing permissions, migrations, offline grants, account deletion and full storage. The 10,000-entry figure is a proposed design workload, to be ratified against devices and user need; it is not a measured capacity claim.

Confirmed direction: Android first; primary serious mobile DJ audience; secondary casual streaming audience; private/personal initial use; broad source coverage desired; USB interfaces and splitters allowed. No particular streaming partnership is approved. Implementation decisions include ordering of provider investigations, import reference versus managed-copy behavior, metadata portability formats and measured storage/device limits. An app account is not required for the local baseline. Cloud storage/sync and its account purpose are deferred optional decisions, not prerequisites for design. The user has not reduced the requested final scope: any necessary staged release or unavailable source must be explicitly identified, not silently substituted.

In addition to the existing library cases, test a restricted catalog field joined to a local track, a request to send provider artwork to a model, revoked metadata retention, an unavailable official download, a local video whose audio cannot be decoded, and a stale acquisition job returning after the deck changed. Every case must preserve existing cues and valid playback. Document 14 source cases, Document 16 DLA-01 through DLA-20, and Document 13 ParDay cases supply detailed acceptance criteria.

Later AWS extension data placement

The first release has no required server. Owner-operated AWS remains the accepted later server candidate, with no local host selected and an approximate USD 35 monthly planning target excluding ParDay AI-provider charges. Keep phone preparation and validated local media authoritative. Record server staging, artifact lifetime and actual transfer legs separately; a cloud-resolved URL is not assumed usable from the phone. The deferred AWS source-path feasibility and comparison are defined in Documents 16 and 18; their service-specific cases do not block local design or release. Source rights, provider restrictions, actual media provenance and ParDay context rules remain distinct from hosting location. No account, bucket, worker or media transfer has been created.

Sources

1. Spotify DJ integration support.
2. Spotify Developer Policy.
3. DJ with Apple Music.
4. MusicKit.
5. Apple Media Services terms, applicable territory to be confirmed.
6. Tidal DJ support.
7. Beatport Streaming.
8. rekordbox FAQ, desktop example, not mobile parity.
9. djay streaming capabilities.
10. YouTube Developer Policies.
11. YouTube Required Minimum Functionality.
12. Bandcamp purchase download formats.
13. SoundCloud uploader-enabled downloads.
14. YouTube own-upload exports.
15. Spotify Developer Terms, including AI ingestion, stream-ripping and conditional-download restrictions.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 08 OF 19

Asset creation, sourcing and provenance plan

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version 1.0 · 29 October 2026 · Design handoff baseline - production acquisition pending

This is a new product with no Mixline asset reuse. Steampunk and Android first are confirmed. The user requested comparable concept renders for A - Precision Atelier and C - Crystal Current, and named the AI assistant ParDay. Every production asset below is **proposed or to create**, unless a later manifest explicitly records acquisition, rights verification and integration. A generated concept is a review artifact, not a production asset or a completed acquisition of fonts, sample packs or stock.

Local Android delivery comes first; AWS expansion is later. Prioritize the reusable native control kit, source/download states, ParDay proposal states and authorized local demo/validation media. No cloud deployment asset or account setup is required for design. The roughly US$35/month future AWS target excludes ParDay AI charges and does not fund or authorize font, stock, audio or other asset procurement.

1. Asset strategy

Create original control geometry and material recipes for the app's distinctive instrument. Source coherent general-purpose icons and fonts from verified upstreams. Produce or commission demo music and samples with rights that explicitly cover application bundling and intended user output. Build test signals reproducibly. Use generated raster imagery for material exploration, illustration or marketing where appropriate; exact controls, waveforms, numerical data and diagrams require deterministic production assets.

The selected material direction is **C - Crystal Current: pale frosted glass, titanium and subtle brass**. Retain a clear instrument structure. A remains comparison history and is not selected as a night theme. The app must remain complete using its Essential visual tier. Asset failure cannot remove transport, meters, deck identity or critical status. User album artwork is cached appropriately and never substitutes for a track's textual identity. Prefer compatible open-source assets when they meet quality, maintenance and redistribution needs; paid options remain permitted after comparison, without spending being authorized by this plan.

2. Production asset matrix

Quantities and budgets are initial planning assumptions, not acquired inventories or fixed commercial commitments.

ID	Asset / initial quantity	Creation or sourcing route	Deliverable / initial budget	Acceptance and rights requirements
AS-01	Wordmark, app mark, 2-3 directions	Original vector design after name selection	SVG/source, monochrome, small-size and Android adaptive icon layers	Originality review, editable masters, author/provenance; legible without complex texture
AS-02	UI font	Barlow candidate from official author repository [1]	Selected TTF/OTF weights and license; part of 1-2 MiB initial font budget	Verify downloaded license/files; check I/l/1, O/0, punctuation, long titles and required languages
AS-03	Numeric font	IBM Plex Mono candidate from IBM upstream [2], or tabular UI font	Required files only, license and fallback metrics	Acquisition/license verification pending; readable time/BPM at actual size
AS-04	General icons, about 60-90	One coherent Lucide subset [3] or selected native symbol family	SVG masters and native vector conversions	Preserve ISC/MIT notices where applicable; consistent stroke/grid; accessible names

ID	Asset / initial quantity	Creation or sourcing route	Deliverable / initial budget	Acceptance and rights requirements
AS-05	DJ symbols, about 25-40	Original vectors matching selected icons	Slip, beatjump, sync, phase, cue/memory, stem, bus, tempo-bend symbols	DJ comprehension review; no copied competitor iconography or embedded brand logos
AS-06	Dial family	Original procedural native geometry	Gain/EQ/filter/pitch controls, states, size variants, unit/value rules	Real interactive components; legibility at 48/64/80dp; no baked text or bitmap per value
AS-07	Faders and transport	Original native component family	Crossfader/channel rail, thumb, play/cue/stop, as signed/disabled/pressed states	Distinct hit bounds, captured pointers, clear active deck, accessible alternatives
AS-08	Pads and XY surface	Original components	Empty/loaded/held/pending/playing/edit/error states, bank badges	Engine acknowledgement reflected honestly; labels and units; momentary/latch visibility
AS-09	Brass and titanium material recipes	Procedural formulas; original small texture if needed	C recipes plus optional 256-512px tiling noise/roughness maps; retain A only as reference	No unlicensed photo crops; no seams; bounded GPU/memory cost
AS-10	Glass materials, 3 tiers	Original native recipes and optional shader	Enhanced source, Balanced recipe, opaque Essential fallback	No refractive text; worst-content contrast; visual tier cannot change musical functionality
AS-11	Waveforms	Derived from actual audio analysis	Versioned multiresolution peak format and renderer fixtures	Correct duration/channel/peak mapping; never a stock bar animation
AS-12	Beat/phrase/cue markers	Original vector geometry	Active, pending, edited, uncertain and disabled styles	Shape/label plus colour; visible analysis uncertainty; stable scale
AS-13	Meters	Native procedural rendering	Master/cue/channel scales and clipping/limiter states	Documented units and ballistics from DSP contract; not decorative gauges
AS-14	FX and stem symbols	Original concise vector family	Effect identifiers, axis cues, vocal/drum/bass/other controls	Parameter names still written; restricted/unready states cannot resemble ready controls
AS-15	Album-art placeholders, 8-12 patterns	Original deterministic procedural/vector patterns	Small reusable patterns or seed-based render	Stable identity, no fake performer images, no invented real album covers
AS-16	Empty/error illustrations, about 6	Original vector or generated raster exploration	SVG where exact; optimized WebP/PNG where illustrative	Essential instructions remain text; provenance; content-appropriate alternative description

ID	Asset / initial quantity	Creation or sourcing route	Deliverable / initial budget	Acceptance and rights requirements
AS-17	Haptic profiles	Native platform events and capability fallback	Mapping table and physical-device test record	Optional, restrained, system preference respected; no buzzy routine fallback
AS-18	Motion recipes	Authored design-system parameters	Trigger/state/timing/interruption/reduced-motion specifications	Does not gate input or assert audio completion; costly decoration has static equivalent
AS-19	Factory one-shots, about 24	Original synthesis/recording or commission	WAV masters, playback versions, metadata and license	Explicit app redistribution and agreed commercial mix rights; no borrowed vocals
AS-20	Factory loops, about 12	Original production or expressly licensed commission	BPM/key/bar metadata, masters, loop seam ground truth	Typical sample-pack purchase may not permit app redistribution; verify exact terms
AS-21	Demo tracks, about 6-8	Original commission/production	Full mix, authored stems, cue/beatgrid reference and rights declaration	Contrasting tempos/styles; no named-performer imitation; track/stem start alignment checked
AS-22	Analytical audio fixtures	Deterministic synthesis and owned recordings	Impulses, sweeps, silence, tempo ramps, long tracks, varied sample rates and malformed controlled inputs	Reproducible generator, seed, hashes and sample-indexed ground truth
AS-23	Output-route diagrams	Exact original vectors	Single stereo, split mono and supported multichannel arrangements	Only qualified routes depicted; channel/mono tradeoffs explicitly labelled
AS-24	Help and rehearsal content	Original short text and actual product captures	Glossary, gesture cards, independent-practice tasks, recovery guides	Audio/product review; no mandatory onboarding or claims of mastery from assisted completion
AS-25	Optional store and promotional assets	Actual approved app captures plus original composition	Editable masters, selected-channel exports and short clips only if pursued	Conditional on future promotion/store distribution; no personal-release gate; depict shipped behavior and record media rights
AS-26	Optional 3D hero scene	Original modeled or generated concept, then validated	Editable scene, optimized stills; GLB only if a justified use exists	Marketing optional; no live audio surface dependency on a heavy 3D model
AS-27	ParDay assistant identity and states	Original compact mark, word treatment and UI icon family	Editable vector/source; small-button, proposal, offline, pending, applied and error states; optional nonlive illustration	ParDay is the assistant, not a final app name; no copied ChatGPT/Claude marks or claim of provider endorsement; no obstructive live mascot

ID	Asset / initial quantity	Creation or sourcing route	Deliverable / initial budget	Acceptance and rights requirements
AS-28	A/C art-direction comparison	Two comparable generated raster concepts using Document 04's common brief	Same landscape framing, controls, fictional tracks and state; separate material variants and saved prompts	Concept provenance and actual output recorded after generation; no claim of native implementation, real waveform correctness, source permission or performance
AS-29	Video/karaoke validation media	Original or expressly authorized video, timed captions and audio	Short clips, caption timing ground truth, clean program-output and preview examples	Recording/composition/video/caption rights checked separately; no ripped catalogue video, borrowed lyrics or unsupported display depicted as qualified
AS-30	Link acquisition UI states and icons	Original native components/vectors using selected C tokens	Local-first paste/share, matching, source/format/quality comparison, on-device queue, retry/cancel, validating, ready and failed states; optional AWS configuration later	Map to Document 16 job states; distinguish actual source and declared/measured quality; accessible labels; preserve transport access; not created yet

3. Sourcing and rights workflow

Use the status sequence **proposed → shortlisted → acquired/created → rights verified → integrated → verified**. A catalogue listing, subscription, search result or download link is not evidence that an asset can be embedded in a commercial app. Review fonts, bundled stock, music, model weights and demos separately. Do not redistribute commercial asset source files in an agent package without the necessary permission.

Barlow's official repository declares SIL Open Font License 1.1.[1] Lucide uses ISC, with MIT terms for listed Feather-derived icons, and requires the relevant notices.[3] These facts support shortlisting; the exact package still needs a saved license and checksum. IBM Plex also declares OFL 1.1 in its official repository; the actual selected files, notice packaging and required glyphs must be checked at acquisition.[2]

For commissioned audio, agree explicitly on app distribution, user recordings/exports, demos, promotional use, modification, stems, territory, duration and any attribution. Avoid assuming “royalty-free” means unrestricted redistribution of samples as extractable assets. Keep ownership of the compositions and the recordings clear. No procurement spend or license fee is committed by this plan.

The same review applies to video, karaoke text and ParDay examples. A music licence does not automatically cover lyric display or a particular video. Provider names and marks, if shown in source selectors, must follow the applicable permitted use rather than being redrawn as implied partnerships. Do not include downloaded Spotify/YouTube music or catalogue artwork in the shipped demo pack; Document 14 defines authorized import and discovery paths. The user's broad source ambition is preserved through adapters and eligibility states, not bundled unlicensed media.

4. Required asset manifest

Each entry records:

`asset_id`, `name`, `role`, `status`, `owner`, `source_url`, `author`, `acquisition_date`, `license_name`, `license_reference`, `permitted_uses`, `attribution`, `modifications`, `app_redistribution_permission`, `path`, `format`, `bytes`,

`dimensions_or_duration`, `checksum`, `version`, `accessible_description`, `fallback`, `review_evidence`, and `requirement_ids`.

Store editable masters separately from optimized app files. Retain provenance/license evidence with the project. Commit generated fixtures with their generator and seed. Pin source versions so an upstream replacement does not silently change visual or audio behavior. Rights-unknown assets cannot pass the release gate; use an original or appropriately licensed fallback.

5. Creation briefs for build agents

Original dial family

Create a precision instrument dial for selected Crystal Current with a pale opaque or muted titanium hub, a restrained brass chamfer, a single dark-ink high-contrast index and a readable scale. Derive its low-light variant from C's semantic tokens; do not copy A's graphite recipe as the default. Deliver parameterized vector/native geometry for normal, focused, touched, centred and disabled states at several target sizes. Text remains native and separate. Demonstrate grayscale readability, Essential-tier equivalence and nonoverlapping 48dp hit areas. The dial must reflect the engine's actual value and units; a cosmetic rotation is insufficient.

Glass material study - future image generation

“Create a material study for an original mobile DJ instrument in selected Crystal Current: three bounded pale frosted-glass panels on a mineral-light stage, brushed titanium, delicate edge highlights and restrained brass anchoring details, with stable dark-ink labels over opaque backing. Show subtle depth without refracting text. Studio lighting, front-on presentation, no logos, no tiny invented words, no music-data charts, no decorative gears or steam. This image explores visual material; it is not a functional interface.”

After concept selection, implement native recipes and fallbacks; do not ship the concept image as the control surface. AGSL requires Android 13+.[4] An equivalent static/vector tier must exist for other selected device profiles.

A/C concept comparison - completed; C selected

The completed generation brief requested two directions using Document 04's shared composition: comparable control positions, fictional track identity, BPM/key values, framing and ParDay placement. A uses graphite, smoked glass and champagne brass; C uses pale frosted glass, titanium and restrained brass details. The exact prompts, generation date, variant identity and selected traits are recorded. A raster can be reviewed for art direction but cannot pass native touch, accessibility, engine-state or performance requirements. Do not regenerate the comparison as an outstanding task or distribute it as a finished app screenshot.

The A/C concepts have now been generated and visually reviewed; the asset register records their real paths and checksums under AS-28 as `concept_generated`. Document 04 records the visible layout differences and omissions. A/C review status and production asset readiness remain separate. Do not mark the fonts, knobs, sound packs or music shown by implication as acquired simply because a concept has been generated. The owner selected C. Preserve both generated files as the comparison record, but create new production assets for Crystal Current.

ParDay identity and interface family

Create a small legible **ParDay** word treatment and optional abstract mechanical/optical mark that fits selected C and its high-contrast/low-light treatments. Keep a text-only fallback. Deliver neutral, selected, pending, local-only, needs-review and unavailable states. Semantic states should rely on text and shape as well as colour. Do not reuse provider logos as ParDay's identity or add an assistant face without a user benefit.

Create structured proposal components for crate changes, cue suggestions, transition previews and route explanations. Include compact and expanded layouts, long track names, unknown/source-restricted state, expired proposals, explicit review/apply, and accessible labels. No generated animation implies a successful command before acknowledgement. Model/provider names belong in understandable configuration/context, not decorative endorsement badges. Optional spoken feedback must remain a separate user choice and never become an unrequested master-output sample.

Video and karaoke fixtures

Produce a short original video with known frame rate, synchronized original audio and a companion timed-text track with independently authored words. Include variants with delayed captions, a missing video frame segment and an unsupported fixture only in the test suite. Keep source video, audio master, caption authoring file and synchronization ground truth. Verify seeking, loop boundaries and cue preview independently. No lyrics from commercial music are needed for this validation.

Original factory track and stems

Produce a 90-150 second original track with a verified tempo grid, clear phrases, instrumental variation and a companion deliberately drifting-tempo version. Supply full mix and independently authored stems with exact offsets, BPM/key reference and rights declaration. Use varied styles to challenge analysis, including electronic, hip-hop, soca/calypso-inspired, reggae-inspired and live drums. Do not imitate a named performer or borrow their recording. Manually verified ground truth is stored separately from automated analysis output.

Deterministic waveform/test fixtures

Generate impulse trains and layered sine/noise tracks at the supported sample rates and channel configurations. Supply sample-indexed ground truth and expected extrema at several zoom levels. Include silence, transients, clipping, variable tempo and long-duration media. Preserve generator source, version, seed and checksum. These fixtures validate correctness; they are labelled tests rather than disguised as finished music.

Optional later marketing concept

“Create an original product atmosphere for a handheld professional DJ instrument using selected Crystal Current: pale frosted liquid glass, titanium and restrained brass. The focus is tactile material and precise construction, with empty safe areas for later placement of real approved app captures. No invented brand, no fake store ratings, no text claims, no depiction of unsupported external controller connections.”

6. Packaging and budget gates

Proposed starting budgets: approximately 8 MiB of non-audio visual assets, 1-2 MiB for necessary font files, and separately downloadable or explicitly packaged demo/sample content. These numbers are planning limits to reconcile with supported languages and final distribution size. Audio fixtures used only by tests must not inflate the shipping app. No giant video, per-state 4K knob atlas or mandatory 3D scene sits behind the live surface.

For every asset, validate format, integrity, memory footprint, dimensions, fallback, accessibility, appearance at actual scale and license evidence. Audio assets additionally require channel/sample-rate metadata, headroom, loop seams, start alignment and audible review. Prompt output is only a draft; generation does not certify rights, correctness or production suitability.

Current production acquisition, native rendering, audio verification and rights status: **NOT VERIFIED / PENDING**. Concept generation/review is recorded separately when its real outputs exist. The 30-class matrix is the work order whose applicable items agents must close with evidence, not a claim that production assets already exist.

Optional community-source distribution

Document 17 defines the proposed public repository boundary. A future source release needs redistributable fonts, icons, textures, demo media, models and fixtures, with notices and reproducible source files. Personal acquired media and paid components are not automatically redistributable. Keep nonredistributable developer assets out of public history and provide a permitted build fallback where needed. The two generated concept renders remain design references; no exclusive copyright or production-asset clearance is inferred from generation. No public asset release is authorized by this planning revision.

Sources

[1] Barlow official author repository, accessed 29 October 2026. [2] IBM Plex official repository and license, accessed 29 October 2026; selected-file acquisition pending. [3] Lucide license, accessed 29 October 2026. [4] Android Developers, AGSL, accessed 29 October 2026.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 09 OF 19

Quality and Device Validation

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; implementation and validation pending **Product basis:** New product; no Mixline validation is inherited. **Platform decision:** Android first; Samsung Galaxy S26 Ultra is the user-selected first test device. **Accountable agents:** Quality/Release; Audio Engine; Android Platform; Product UI. **Target status:** Numerical performance and test thresholds below are proposed, not measured results, vendor guarantees or delivery commitments. The separate future AWS planning target of about US$35/month excluding ParDay AI charges is owner-confirmed.

Quality objective and release rule

Prove that the product can sustain a real mobile DJ session, with audible behavior matching the interface and recoverable failures. Every supported capability must have a named device, OS, output route, configuration and result. Emulator success, screenshots and a brief cold-device demo cannot establish professional reliability.

The two-deck foundation is the first integration gate. The selected complete product also includes four music decks, an independent sampler/sequencer, prepared and device-gated live stems, eligible recording, mic, automix, video, karaoke, broadcasting and third-party plugin hosting; lighting is excluded. Live stem separation is a gated capability. A failed extended profile does not justify advertising that profile as supported; it also need not invalidate an independently verified basic profile.

Release requires no open defect that can leak headphone cue to master, lose user edits without warning, start unexpected playback, corrupt source files, or repeatedly interrupt audio in a certified profile. Root-cause and retest all reproducible audible failures. A low statistical crash rate alone cannot waive a known performance-breaking defect.

Evidence progresses through G0 Decision baseline; G1 Audio feasibility; G2 Working vertical slice; G3 Core performance beta; G4 Advanced performance; G5 Source/provider and ParDay integrations; G6 Release candidate; G7 Personal installed release; and G8 Maintenance/optional community readiness. G3 applies P0; G4 applies advanced profiles; G5 adds source/entitlement and assistant failures; G6 repeats the release-relevant physical gates on the packaged build. The first version is local on the phone. AWS comparison/deployment and service-specific home-lab cases are deferred extension evidence, not prerequisites to G6/G7 local personal release. Full selected local capabilities and their applicable source/AI tests still require evidence. System-interruption expectations must follow the supported Android audio-focus rules. [S5]

Device and route matrix

The first device is **Samsung Galaxy S26 Ultra**, with software reported by the owner as “up to date.” This is a device selection, not inspected software inventory or certification. Before the first benchmark, record exact model/region variant, RAM/storage configuration, Android version/API level, One UI version, build/fingerprint, security patch, available storage, display settings and the actual audio accessory/route. Do not infer those values from the model name or the phrase “up to date.” Record app, NDK, DSP and extractor versions with each run.

Broader supported coverage remains open. If the product advertises it, add a baseline midrange Android/Samsung phone, an independent-vendor phone and tablet/foldable evidence. The S26 Ultra selection does not establish performance on those devices. Cover the proposed minimum OS, current shipping OS and meaningful intermediate behavior changes. At the date of this baseline, Android 17 background-audio behavior warrants explicit coverage. [S1]

Axis	Required coverage	Purpose
CPU/RAM	Named S26 Ultra configuration first; additional real phones only if added to the support matrix	Certify the claimed device scope; do not generalize one high-end result

Axis	Required coverage	Purpose
OS/lifecycle	Minimum, current, focus/background behavior boundaries	Service, permissions, calls and routing
Page size	Native 4 KB and 16 KB environments	Core plus every prebuilt SDK compatibility [S2]
Display/input	Small/large portrait, landscape, resize, 60/120 Hz where offered	Reachability, multitouch and frame pacing
Wired stereo	Built-in/USB-C DAC where supported	Master and splitter profiles
USB multichannel	Named ≥4-output interface and supported hub/power setup	Independent stereo master/cue
Bluetooth	Named ordinary wireless routes	Honest delay reporting and graceful limitations
iOS if included	Named iPhone/iPad and accessory combinations	Separate platform certification

For splitter tests use a dedicated DJ splitter and check channel assignment; a duplicated stereo Y splitter is not equivalent. Do not infer output channel count from a USB device marketing label. Query and verify the negotiated route and mapping. Oboe documents Bluetooth and unsupported channel configurations as reasons a low-latency request can fail. [S3]

Profiles and test conditions

P0 foundation: two music decks, validated key lock, channel EQ/filter, two selected FX per deck, eight sampler voices, waveforms, ordinary glass treatment and recording where eligible. **P1 extended:** four music decks, sixteen sampler voices, equivalent declared FX and recording. **P2 prepared stems:** the declared deck count with four aligned prepared stems per active stem deck; evaluate its mix/stretch cost. **P3 live separation:** add the exact model/runtime/inference concurrency and certify separately.

Those combinations are workload definitions to test, not promises that all phones support them. List the heaviest allowed FX chain rather than testing only cheap effects. Use high brightness, active multitouch, browsing/search and realistic charging conditions. Test both normal ambient conditions and an agreed hot-room scenario that remains within device operating guidance. Log temperature/thermal status; never claim indefinite operation at any temperature.

Google’s Thermal API supports workload adaptation because sustainable mobile performance changes with thermal state. A short benchmark cannot establish warm performance. [S4] Degradation must stop optional analysis and queued or precomputation stem jobs, reduce glass/shader cost and lower nonessential visual refresh before affecting audio. Live inference supplying audible stems belongs to the protected active workload and cannot be suspended. P3 is unavailable on a profile that cannot sustain it. Active tempo, key, stem alignment, limiter latency and route remain unchanged. If stable audio cannot be maintained, the profile fails.

Acceptance register

All targets apply to the stated corpus/profile/hardware. Change a target only through a recorded decision; do not quietly relabel a failed result.

ID	Test and proposed acceptance	Evidence owner
QA-01	P0 two-hour warm-soak, three independent runs: no app-caused audible dropout, recorded discontinuity, crash/ANR or accumulating underrun count in controlled conditions	Audio + QA
QA-02	P1/P2/P3 each tested separately under maximum declared load: same continuity rule; exact unsupported combinations listed	Audio + QA
QA-03	Physical cue/pad touch-to-output: target median ≤20 ms, p95 ≤35 ms on certified wired baseline, ≥500 trials; report tails and method	Platform + QA
QA-04	Callback workload: target p99 <50% of deadline, maximum below deadline in test; bounded queues and no callback I/O/locks/allocations	Audio
QA-05	Engine-scheduled quantized events: ≤1-frame execution error against requested frame after compensation; no accumulated click-track drift over 30 minutes	Audio
QA-06	Cue isolation: no unintended cue in master; target ≥70 dB separation on certified hardware where achievable; correct mono sum and stereo mapping	Platform + QA
QA-07	100 rapid loads/cancels plus seek/reverse/loop stress: no stale-generation playback, memory fault, deadlock, unbounded queue or unexplained spike	Audio + Data
QA-08	Blind listening at declared tempo/key ranges: no release-blocking artifact on approved corpus; panel criteria set before comparison	Audio + DJ reviewers
QA-09	Stems: aligned length/timing, acceptable blind quality, recorded preparation time/RAM/storage/heat; cancellation honored; live mode separate	Analysis + QA
QA-10	Focus/calls/lock/background/route changes: specified compliant state, safe output, no cue leakage, no accidental loud resume	Platform
QA-11	Full disk, revoked storage and process kill during recording: storage failure preserves live mix; recover to last valid chunk where feasible; incomplete status visible	Data + Audio
QA-12	Prepared local workflow: proposed p95 cold start ≤3 s and prepared track-ready ≤1 s on named ≤10-minute-track corpus and baseline device	Product + QA
QA-13	Sampler/FX: quantized triggers, documented voice stealing, choke groups, tails/bypass and parameter changes match specification	Audio
QA-14	Signal integrity: correct trim/crossfader curves, meter taps, limiter/headroom, no NaN/Inf or digital overflow under adversarial input	Audio
QA-15	Library/analysis: malformed files safely rejected; revoked URI/missing media handled; manual grid/cue edits survive analysis and migration	Data
QA-16	UI/accessibility: all primary tasks operable without color alone; semantic labels/value adjustments; touch targets and rotation preserve state	UI + QA
QA-17	Final package: reproducible signed-release candidate, native ABIs/page sizes validated, dependency versions/licences pinned	Release
QA-18	Content eligibility: source-specific recording/stem/offline restrictions respected; denied action explains reason; no unauthorized cache/export	Content + QA
QA-19	No-network local set: complete prepared set works without login/network calls; cloud retries do not interfere; account behavior as approved	Services + QA

ID	Test and proposed acceptance	Evidence owner
QA-20	Session recovery and destructive actions: source files preserved, edits durable, undo/confirmation behavior correct, restored set does not autoplay	Product + Data
QA-21	Diagnostics privacy: inject sentinel tokens, private paths, track metadata and microphone content; exported/logged/network data contains none unless the user explicitly selects that field and the source policy permits that destination/purpose; restricted provider data remains blocked or redacted despite selection; consent matches actual behavior	Security + QA
QA-22	Microphone/input: denied permission, feedback-prone route, clock mismatch, live monitoring and talkover exercise the explicit mute/gain/duck contract on supported hardware	Audio + Platform
QA-23	Automix/manual takeover: cancel, reorder, unavailable next track, tempo mismatch and intervention at transition boundaries never replace or restart the wrong deck	Audio + Product
QA-24	Backup/restore: round-trip cues, grids, duplicate entries, missing media and conflicts; reject path traversal/oversized archives; no tokens or restricted caches exported	Data + Security

ParDay, acquisition and advanced-module evidence

AI-001 through AI-024 in Document 13, SRC-01 through SRC-12 in Document 14 and AV cases in Document 15 extend the shared QA register without replacing it. First run model-free fixtures to prove permissions, source isolation, stale proposal rejection, duplicate-call handling, cancellation and manual takeover. Then run provider-specific evaluations with eligible fixture data; never use prohibited catalog payloads as convenient prompt fixtures.

Queue future fader/transport automation, then intervene manually without changing the track or route. Verify the engine-owned automation revision is invalidated and none of those already queued events can reassert old values. Inject a partial-capacity admission failure and require an all-or-nothing result; verify cancellation acknowledgements and at-most-once action behavior. Repeat under full scheduling queues, changed-payload ID reuse, master-control takeover, permission invalidation and engine restart. No affected queued event may run after the acknowledged cancellation frame.

The user selected ParDay armed assisted performance. Verify preparation assistance and bounded transition/automix execution with explicit arming and immediate takeover, while unarmed live behavior remains read/propose. Exercise late model results after deck replacement, route reopen, manual fader movement and permission change. Verify that denied or unknown provider data never enters prompts, embeddings or model-readable tool results. Test an offline set with the assistant disabled, timing out, and rate-limited. A correct recommendation is still a failure if its execution targets stale state or causes an audible surprise.

Add exact video decode/GPU/encode and plugin-chain workloads to P0-P3 when certifying combinations. Video freezes/frame drops, stream disconnects, microphone paths and plugin failures need distinct expected outcomes. Do not certify every module simultaneously by combining unrelated individual pass results. Hosting must pass with an actual supported third-party plugin, not only app-owned effects.

Measurement methods and meaningful evidence

Measure touch-to-sound with a synchronized physical trigger/observation rig and electrical output capture where possible. State exactly whether a number represents UI input delivery, engine response, output latency or round-trip latency. Buffer duration alone is not end-to-end latency. Do not estimate output latency by simply halving an unrelated round-trip figure.

Use generated impulse/click/chirp fixtures for timing, alignment and reconstruction tests; cleared musical fixtures for listening. Include steady electronic beats, swing, live drums, variable tempo, acapella, dense bass, sustained notes, percussion transients and silence. Store hashes and expected timing metadata. For proprietary SDKs use equivalent settings and identical source material. Do not compare different buffers or warmed caches as if they were controlled equivalents.

Separate engine scheduling error from perception and musical analysis error. A sample-accurate event scheduled to a wrong beatgrid is still musically wrong; inspect BPM/downbeat/grid accuracy and user correction independently. Conversely, visual waveform drift may be display interpolation rather than audio drift; capture output and correlate engine timestamps before diagnosing it.

For the personal build, the owner performs repeated concealed-reference listening and records the limits of a single-listener result. A broader quality study should include at least three experienced DJs/listeners before generalizing claims; external-panel recruitment is not a personal release gate. Use concealed backend identity, randomized order and repeated references. Proposed ordinary evaluation range is 0.8-1.2× tempo and ±6 semitones; extended ranges require their own verdict. Use a pre-agreed artifact rubric covering attacks, vocals, bass, phasing and transition glitches. These are proposed study settings, not proven universal professional thresholds.

Failure injection and release evidence

Inject storage exhaustion, permission revocation, malformed artwork/audio, missing files, cancelled imports, provider authentication expiry, network loss, busy/removed USB output, sample-rate change, foreground-service interruption and process death. Keep expected safe state separate from the normal happy path. A route change that legitimately stops audio must be labelled as interruption handling, not counted as uninterrupted performance.

Every evidence record includes build commit/hash, app/SDK/model versions, exact device/OS, route/accessories, codec and sample rate, buffer and channel configuration, profile, brightness, charging state, thermal history, duration, raw logs/captures, result and reviewer. Performance telemetry excludes music content and private filenames by default; user-shared diagnostic bundles require explicit selection.

QA must return a capability report: passed profiles, failed profiles, known limits, reproduction steps and release recommendation. Audio Engine and Platform owners sign off the relevant fixes; the Orchestrator may not turn a failing capability into a marketing claim. Re-run only affected gates plus necessary regression coverage after fixes. The complete physical suite repeats on release candidates and material audio/SDK/OS changes.

Completion and unresolved choices

Android first, the S26 Ultra first device, USB interfaces/splitters, local-first delivery and the full discussed feature intent except lighting are confirmed. The validation plan needs actual device/build/accessory evidence and a declared format/provider/workload matrix before certification, not before design. Begin with the proposed two-hour P0 soak and distinct P1-P3 profiles, recording their limits. Four-deck capability and live stems both remain selected; their simultaneous maximum workload is unverified. Design supported, unavailable and resource-denied states now. If the intended combination cannot pass, report that concrete limitation for a scope decision instead of quietly dropping either feature. Wider device support and a longer target session can be specified when needed.

Functional-to-quality traceability

QA-FR identifiers are feature-specific cases from Document 02. QA-01 through QA-24 are shared quality gates. Both must pass where applicable; a general soak result does not replace a functional case. The machine-readable requirements_register.json records every requirement, its functional case, shared gates, owner and evidence placeholders. All are currently unverified.

Requirement	Functional case	Shared quality gates
FR-001	QA-FR-001	QA-19
FR-002	QA-FR-002	QA-15
FR-003	QA-FR-003	QA-07, QA-15
FR-004	QA-FR-004	QA-15
FR-005	QA-FR-005	QA-15, QA-20
FR-006	QA-FR-006	QA-04, QA-15
FR-007	QA-FR-007	QA-05, QA-08, QA-15
FR-008	QA-FR-008	QA-05, QA-15
FR-009	QA-FR-009	QA-05, QA-16
FR-010	QA-FR-010	QA-03, QA-05, QA-07
FR-011	QA-FR-011	QA-05, QA-15
FR-012	QA-FR-012	QA-02, QA-07, QA-16
FR-013	QA-FR-013	QA-03, QA-07, QA-08
FR-014	QA-FR-014	QA-03, QA-07, QA-16
FR-015	QA-FR-015	QA-04, QA-05, QA-08
FR-016	QA-FR-016	QA-05, QA-08
FR-017	QA-FR-017	QA-05, QA-13
FR-018	QA-FR-018	QA-05, QA-07, QA-08
FR-019	QA-FR-019	QA-05, QA-07, QA-08
FR-020	QA-FR-020	QA-06, QA-14
FR-021	QA-FR-021	QA-14
FR-022	QA-FR-022	QA-06, QA-10
FR-023	QA-FR-023	QA-06, QA-10
FR-024	QA-FR-024	QA-04, QA-08, QA-13

Requirement	Functional case	Shared quality gates
FR-025	QA-FR-025	QA-13, QA-16
FR-026	QA-FR-026	QA-03, QA-05, QA-13, QA-16
FR-027	QA-FR-027	QA-02, QA-05, QA-07, QA-13
FR-028	QA-FR-028	QA-02, QA-09, QA-18
FR-029	QA-FR-029	QA-02, QA-09, QA-18
FR-030	QA-FR-030	QA-06, QA-10, QA-14, QA-22
FR-031	QA-FR-031	QA-11, QA-18
FR-032	QA-FR-032	QA-05, QA-07, QA-23
FR-033	QA-FR-033	QA-04, QA-16
FR-034	QA-FR-034	QA-20
FR-035	QA-FR-035	QA-07, QA-20
FR-036	QA-FR-036	QA-18, QA-19
FR-037	QA-FR-037	QA-10
FR-038	QA-FR-038	QA-16
FR-039	QA-FR-039	QA-10, QA-16
FR-040	QA-FR-040	QA-19
FR-041	QA-FR-041	QA-15, QA-20, QA-24
FR-042	QA-FR-042	QA-10, QA-21
FR-043	QA-FR-043	QA-17
FR-044	QA-FR-044	QA-18, QA-19
FR-045	QA-FR-045	QA-02, QA-04, QA-05, QA-10, QA-18; AV cases in Document 15
FR-046	QA-FR-046	QA-02, QA-04, QA-10, QA-18, QA-21, QA-22; AV cases in Document 15
FR-047	QA-FR-047	QA-02, QA-04, QA-05, QA-14, QA-17; AV cases in Document 15
FR-048	QA-FR-048	QA-04, QA-05, QA-07, QA-18, QA-19, QA-20, QA-21, QA-23; AI-001-024
FR-049	QA-FR-049	QA-02, QA-04, QA-07, QA-10, QA-15, QA-17, QA-18, QA-19, QA-20, QA-21; SRC-01-12 and DLA-01-20

A passing functional case requires the expected behavior specified in Document 02. Record separate cases for unsupported/error, cancellation and recovery paths. No mapping authorizes bypassing a source permission or accepting a leaking route.

Link acquisition evidence crosswalk

Document 16 and link_acquisition_acceptance_register.json add DLA-01 through DLA-20 under FR-049. They cover actual-source/version matching, quality reporting, an enabled extractor fixture, bounded transfer/validation, cancellation/publication races, recovery, batch identity, workload coexistence, worker/network isolation, source evidence, updates, ParDay egress and late completion. App-provided Cancel download must preserve unaffected mixing; Android Task Manager Stop/Force stop may terminate the app and is tested as interruption/recovery. No DLA case has been executed. Shared gates for FR-049 are QA-02, QA-04, QA-07, QA-10, QA-15, QA-17, QA-18, QA-19, QA-20, QA-21.

Comparative acquisition, AWS and home-lab evidence

The latest owner instruction selects local delivery first. Run the S26 Ultra device arm D first under Document 16 section 13 and contracts/acquisition_comparison_protocol.json; preserve its fixtures and raw observations for reuse. D-versus-AWS is a deferred expansion comparison, with no measured winner. H-LAN and H-REMOTE remain optional portability profiles. The approximately US$35/month future AWS target excludes ParDay AI charges and is not a test result or permission to deploy.

When the AWS expansion is commissioned, rerun equivalent fixtures on the phone's actual Wi-Fi/mobile networks and record cloud region, runtime and transfer path. Record exact source/format, software, caches, network and hardware. Different extractor implementations are a system comparison, not isolated proof of location effects. Deferred AWS and home-lab runs do not fail local release merely by being unexecuted; applicable DLA/SRC cases for the local source capability remain mandatory. Record results only when measured and leave selected_winner empty until a comparative decision is supported.

Cases HL-01 through HL-10 in Document 17 and contracts/home_lab_deployment_profile.json extend the existing build, runtime and independence requirements. Service-specific variants are conditional on a service profile; HL-10 applies only before optional publication. Their purpose is reproducible setup, pairing, credentials, outage independence, bounded work, upgrades and cleanup. They do not commit the product to a backend. Document 18 and contracts/aws_deployment_profile.json add AWS-01 through AWS-10 for the evaluated cloud profile, including actual cloud source feasibility, pairing, isolation, stop/recovery and resource cleanup. Those cases are unexecuted and deferred until that optional extension is evaluated; naming AWS is not evidence that an extractor works there. Reproducible local installation, backup and recovery cases remain applicable without a service. Optional source publication requires its own clean, reviewable package; public-store approval and commercial-market validation do not gate the selected personal release.

Sources

Checked 29 October 2026. Numeric targets are original proposed requirements, not claims derived from these sources.

- **S1:** Android, Background audio hardening.
- **S2:** Android, Support 16 KB page sizes.
- **S3:** Google, Oboe FAQ.
- **S4:** Android, Thermal API.
- **S5:** Android, Manage audio focus.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 10 OF 19

Delivery roadmap and agent playbook

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version 1.0 | 29 October 2026 | Design handoff baseline; implementation and deployment not started

1. Delivery strategy

Build the product through evidence gates. The objective is the full agreed professional mobile DJ product; a gate is a proof of readiness for the next investment, not permission to silently remove advanced requirements. This planning task ends with a reconciled design handoff; Document 19 provides the next agent's entry point. No further owner answer is needed to start design. No application, repository deployment, provider account, paid licence, or public release has been created.

The audience, C Crystal Current, Galaxy S26 Ultra first device, ParDay armed-assisted capability and personal-tool delivery are selected. The owner is open to a later GitHub source release for community home labs; publication and licence remain decisions for that stage. The latest decision is local first: develop the phone-native app and on-device acquisition path without a required server. The earlier on-device versus self-hosted comparison is retained for a later optional AWS expansion, with no measured winner. The future AWS planning target is approximately US$35 per month, excluding ParDay AI-provider charges; this is not a hard billing ceiling or deployment/spending authorization. Account, region, service and resources wait for a concrete expansion proposal. The critical path is exact route/build capture, audio feasibility, one real end-to-end mix, interaction validation, breadth on the same engine, then release certification. Visual components and asset exploration can run alongside audio work after they share the same interaction contracts. Source-access feasibility can run early, but an unavailable integration or home-lab service must not become a hidden dependency in the local engine.

2. Canonical delivery gates

Gate	Deliverable	Exit evidence	Dependency
G0 - Decision baseline	Recorded owner choices, local-first sequence, selected visual direction and versioned requirements	Design can start from Document 19; proposed details and later evidence gates are labelled	Current planning dossier
G1 - Audio feasibility	Two native decks, candidate stretch engines, sample-clock scheduling, real output routes	Hardware timing captures, warm-soak logs, DSP listening review, licence route, fail/revise decision	G0; capture device/route and any paid experiment envelope when needed
G2 - Working vertical slice	Import two real tracks, prepare, load, cue, transition, save and restore	Demonstration on the named real target device; UI and audio are connected; no simulated waveform or meter values	G1 engine and bridge contracts
G3 - Core performance beta	Reliable two-deck mixer, library, grids, cues, loops, FX, sampler, recording eligibility and recovery	Traceable functional tests, accessibility review, owner workflow sessions, sustained performance profile; optional independent DJ review	G2 and assets for core components
G4 - Advanced performance	Four-deck workspace, prepared stems, richer sampling/remix features, gated live stems	Per-profile sustained tests, stems quality/alignment, advanced interaction review	G3 and explicit device tiers
G5 - Source/provider and ParDay integrations	Source adapters, requested on-device link acquisition and selected assistant capability	Technical proof plus separate provider-access, content-rights, distribution, token/expiry and privacy evidence; local authority and takeover tests	Adapter-specific evidence and personal scope; cloud extension cases deferred; official API not a universal technical prerequisite
G6 - Release candidate	Complete selected personal-release matrix, final assets, installation package and operating documentation	No release-blocking failures, signed build reproducibility, licence inventory, support readiness	Required portions of G3-G5 and resolved launch scope
G7 - Personal installed release	Verified signed local build installed on the owner's S26 Ultra	Install/upgrade, backup/restore and offline/outage evidence; working local setup instructions	G6 and owner-directed installation; AWS/home-lab setup and public-store approval are not prerequisites
G8 - Maintenance and community readiness	Personal updates, regression checks and optional source/release preparation	Reproducible setup, licence/asset manifest, secret-free examples and a tested recovery path	G7; any GitHub publication remains a separate action

G4 and G5 can overlap. AWS and service-specific home-lab tests are deferred; local cases remain required. Complete personal release includes all selected advanced, source, ParDay and specialist modules. Lighting stays excluded. Run specialist feasibility in G1, integration in G4/G5 and release evidence in G6. Report blocked formats/providers explicitly; G3 is not the finished product.

3. Work ownership

Agent/role	Owns	Must not decide alone	Required handoff
Product and integration lead	Requirements, priorities, decision ledger, release assembly	Scope removals, claims of user approval	Updated traceability and unresolved decisions
Native audio/DSP engineer	Engine clock, buses, transport, DSP graph, routing adapters	Music rights, public performance promises	Versioned engine API, timing data, audio fixtures
Android platform engineer	Lifecycle, foreground playback, files, audio focus, device routes	Ignoring system interruption rules	Device/event matrix and instrumentation
UI/interaction engineer	Compose screens, touch capture, accessibility semantics, state presentation	DSP truth, unsupported device capability	Rendered screens and input-state tests
Visual/technical artist	Tokens, vector controls, materials, iconography, motion	Adding expensive effects to live path	Asset manifest, source masters, fallback variants
Library/data engineer	Track identity, crates, search, analysis jobs, migrations, backups	Overwriting user cues during reanalysis	Schema migrations and conflict/recovery examples
Analysis/ML engineer	Beat/key analysis, waveform pyramids, stem generation	Equating a model licence with all weight/data rights	Reproducible model manifest, quality/thermal results
Music integration engineer	Capability resolver, contracted provider adapters	Treating public APIs as DJ authorization	Per-provider rights/capability table and tests
Cloud/home-lab service engineer - later extension	Deferred comparative acquisition worker, pairing, optional AI gateway, retention and updates	Making a server a local-release prerequisite, selecting a winner without evidence or making live audio server-dependent	Reviewable expansion proposal, then reproducible deployment and measured cost/outage/deletion behavior
ParDay AI engineer	Provider adapters, typed proposals, source-filtered context, authority validator	Giving model output direct audio authority or unreviewed data egress	AI-001-024 fixtures, cost/retention settings, rejected-action evidence
Audio/video and plugin engineer	Video clock/composition, karaoke, broadcast encoders, real plugin host	Claiming unsupported mobile formats or bypassing distribution rules	Document 15 compatibility profile and AV evidence

Agent/role	Owns	Must not decide alone	Required handoff
QA/performance engineer	Fixtures, traces, device profiles, soak and recovery tests	Accepting visual simulation as functional proof	Evidence bundle mapped to requirement IDs
Security/release engineer	Dependency review, signing, privacy, distribution checks	Sending messages, buying licences, publishing without authorization	Release checklist and verifiable build provenance
DJ reviewer and accessibility reviewer	Actual workflow and listening acceptance	Certifying untested devices or audiences	Observations, failures, and recommended changes

One person can fill several roles, but independent review is needed for the callback, route changes, capability enforcement, and release claims. AI agents accelerate bounded implementation; their output is not evidence that a phone sounds or behaves correctly.

4. Proposed repository organization

Create these locations when implementation is authorized. They are a starting convention, not existing inspected files:

```
app-android/                Native app, lifecycle, navigation
core/domain/                Shared IDs, commands, immutable state
core/data/                  Room, migrations, import and search
audio/engine/               Platform-independent C++ mix graph
audio/android/              Oboe and platform routing adapter
audio/dsp-adapters/          Replaceable licensed DSP interfaces
analysis/                   Beat/key/waveform/stem workers
features/performance/        Deck, mixer, waveform, pads and FX
features/library/            Import, crates, preparation and search
features/routing/            Output and headphone setup
integrations/                Source adapters and capability policies
services/acquisition/        Deferred optional owner-operated worker
deploy/                      Deferred AWS/home-lab examples, no secrets
benchmarks/acquisition/      Common comparison protocol and results
docs/homelab/                Optional future service setup and recovery
assets/source/              Editable masters and licence evidence
assets/runtime/             Optimized production exports
contracts/                  Versioned public interfaces and schemas
tests/fixtures/             Rights-cleared, deterministic inputs
tests/device/               Hardware configurations and evidence
docs/                       Approved requirements and decisions
```

Keep UI feature modules dependent on domain contracts rather than each other's implementation. Only the audio owner changes the callback graph. UI and integration agents can propose API changes through the integration lead.

5. First implementation assignments

Task A01 - engine and route spike. Read Documents 05, 06, and 09. Build a bounded native audio harness with two local PCM fixtures, one sample clock, gain controls, equal-power and linear crossfades, stream restart, and independent master/cue tests where hardware permits. Output timing traces and a reproducible hardware configuration. Stop if the proposed route cannot carry the required channels; do not simulate cue separation.

Task A02 - performance interaction prototype. Start with Document 19, then read Documents 03, 04, and 08. Build representative portrait and landscape screens containing actual touch targets, deck focus,

browse sheet, and control states. Early fixtures must be visibly identified as prototype data. Validate protected seek versus scratch, playing-deck replacement, hidden-deck observability, and rotation. Integrate the real A01 engine before claiming G2.

Task A03 - durable local library. Read Documents 02 and 07. Import a permitted local folder/file set; persist identity and user annotations; handle revoked URI access and missing media; analyze asynchronously. Demonstrate that reanalysis preserves edited cues/grids and that a failed import cannot damage the previous crate.

Task A04 - DSP candidate comparison. Compare time-stretch/key-shift engines using the same files, tempo range, output format, and hardware. Include transients, vocals, sustained harmony, silence, variable tempo, and rapid cue/seek changes. Report licensing, integration time, reported latency, memory, CPU, and listening results. Do not compare marketing claims as benchmark results.

Task A05 - assets and design tokens. Produce the representative knob, fader, pad, meter, waveform markers, and glass surface at multiple states and scales. Deliver source plus optimized export and an opaque fallback. Validate those components before making every screen.

Task A06 - recovery and recording. Exercise activity recreation, route removal, focus loss, full storage, recording permission changes, and process restart. Record what is preserved and what requires user action. A force-stopped or killed process cannot be promised uninterrupted audio.

Task A07 - ParDay deterministic harness. Read Documents 07, 13 and 14. Use synthetic eligible fixtures before live API access. Prove context provenance, typed proposals, user scope, state freshness, idempotency and manual takeover. Only then attach a replaceable provider adapter. Cloud-model latency must never enter the audio deadline.

Task A08 - on-device link acquisition first. Read Documents 07, 14 and 16 plus contracts/acquisition_comparison_protocol.json. Implement the native intake/queue and validated publication contract with controlled fixtures. Prove an actual on-device candidate for the requested source path, record source/network conditions and retain the baseline data for later comparison. Preserve requested and actual-source identities; include unofficial routes as concrete candidates rather than replacing them with discovery-only handoff. Report technical, access/content and distribution evidence separately. A mocked download is not capability evidence. If an intended local adapter is infeasible or inadmissible, report the specific blocker; do not silently require AWS or mark FR-049 complete. Documents 17 and 18 guide the deferred service extension, not this task's prerequisites.

Task A09 - advanced module feasibility. Read Document 15. Verify video/audio sync under seek/scratch, karaoke microphone/output, an independent encoder failure and one real third-party Android plugin. Pin supported format/distribution choices and workload budgets. Report a plugin-hosting blocker if only built-in effects work.

Task A10 - local personal installation and recovery. Read Documents 11 and 17. Specify stable private signing, local install/upgrade, configuration/secret separation and backup/restore. Rehearse a prepared set with network and optional AI unavailable. Verify the phone-native release without requiring an AWS or home-lab service. Preserve versioned worker contracts for later expansion. A future community release needs buildable code, dependency/asset notices and truthful supported profiles; no music library, credentials or signing keys belong in it.

Task A11 - optional AWS expansion, deferred. Activate after a usable local baseline exists and the owner commissions the expansion. Read Documents 16-18 and the deployment/cost contracts. First identify the measured bottleneck or useful offload outcome, then prepare a concrete source-path trial, region/service proposal, current cost estimate within the approximately US$35/month target, pairing, retention and teardown. Keep AI charges separate. Compare D and AWS with the common protocol before selecting supported cloud behavior. No AWS account, deployment or spending is required for A01-A10. Optional LAN/remote home-lab profiles remain portability references.

These assignments define testable work; they do not authorize starting it during this planning session.

6. Required task envelope

Every coding assignment must state its objective, requirement IDs, owner, allowed paths, dependency versions, input/output schema, non-goals, failure behavior, acceptance cases, device requirements, and definition of done. Work ends with a handoff containing changed files, interface changes, tests actually run, supporting artifacts, untested cases, known limitations, and the next dependency.

Use this compact prompt pattern:

```
Implement task [ID] against dossier version [VERSION].
Read [DOCUMENTS] and the current decision ledger first.
Own only [PATHS]; coordinate any contract change with [OWNER].
Satisfy [REQUIREMENT IDS] and execute [ACCEPTANCE IDS].
Use real engine state for any live signal or transport claim.
Report evidence, device/build identity, unresolved risks, and handoff.
Do not change product scope or present provisional choices as approved.
```

Agents must distinguish proposed paths from inspected code and report missing dependencies candidly. They must not work around access controls, embed credentials, or invent successful tests. Provider outreach, paid procurement, app-store publication, and external messages need the applicable user authorization.

7. Integration and change control

Create one owner per shared contract. Give each parallel agent a bounded branch or worktree. Merge interface changes before dependent feature branches. A change to source timing, stem latency, persistent IDs, routing, or provider capability semantics requires an updated contract version and targeted migration/regression review.

Priority order for conflicting instructions: current user decisions, explicit exclusions, approved decision ledger, approved requirements, approved detailed specifications, then implementation suggestions. This baseline records user-confirmed product choices separately from proposed implementation and acceptance details. It is not authorization to build or deploy. A newer file is not automatically authoritative if its decisions were never accepted.

Require a review when a change can alter audible output, replace user data, change capability enforcement, or weaken a performance gate. For ordinary styling corrections, use the existing visual acceptance criteria without inventing another approval ceremony.

Maintain a requirement matrix with columns ID, statement, gate, owner, implementation reference, acceptance case, evidence, state, and blockers. Valid states are proposed, approved, building, verified, blocked, or explicitly deferred. A checklist tick without evidence cannot move an item to verified.

8. Schedule and cost estimation

No reliable fixed delivery date can be given before G1 and the team/SDK decisions. A useful estimation model is: effort per work package + integration/review effort + device validation + contingency, divided by effective parallel capacity and constrained by the dependency graph. Do not divide the total by the number of AI agents and call the result a schedule.

An illustrative planning allocation for a future budget is 10% discovery and experiments, 30% engine/platform integration, 20% UI and assets, 15% data/analysis, 15% QA/device certification, and 10% release/support preparation. These percentages are an assumption for estimating, not measured costs. Streaming agreements, advanced ML, iOS launch parity, or video can substantially change the allocation.

Obtain separate cost lines for DSP licences, stem-model/weight rights, music partnerships, developer accounts, test devices, audio interfaces, commissioned assets, rights-cleared test music, optional hosting, and human review. Record the currency, vendor quote date, taxes, renewal basis, and cancellation/export

terms. Do not silently spend money to satisfy a specification.

9. Demonstration and definition of done

A milestone demo should show a real track import, preparation change, deck load, cue audition where supported, a transition, a performance gesture, background/rotation behavior, and session recovery. Show the device, audio route, build identifier, and any restricted feature state. Record audio independently where possible so visible animations cannot conceal a dropout.

G6 is complete only when the agreed launch requirements pass functional, sonic, visual, accessibility, privacy, rights, and endurance gates; assets have provenance; personal installation/update/backup instructions describe real limitations; and diagnostics can explain a route or source problem. Every material failure either blocks release or has an explicit, user-accepted scope decision. No claim of "all professional features" is made until the corresponding launch matrix is verified.

10. Design readiness and later decisions

Start design from Document 19, using A02/A05 and the local import-to-mix workflow. No blocking owner question remains for that step. Use the recorded C direction, phone-native Android target, S26 Ultra first device, full professional scope, ParDay authority rules and local-first sequence. The two-deck integration view is a first slice; it does not remove four-deck capability or other selected modules.

Capture actual build/audio-route evidence during G1, select DSP and source mechanisms through measured feasibility work, and choose the first model/provider and AI spending limit before a paid integration. Naming, typed-first voice defaults, wider device support and optional publication/licence choices can be resolved when they affect concrete work. Team capacity and estimates remain unknown, so no fixed delivery date is promised.

AWS is later, with an approximately US$35/month planning target excluding AI-provider charges. Its account, region, service, current quote and operating choices belong to A11, not a questionnaire before local design. Document 12 separates confirmed decisions from these later gates; Documents 16-18 preserve the requested acquisition, portability and future cloud work without making them prerequisites to the local interface.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 11 OF 19

Release, security and personal operation

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline for local personal delivery; implementation and device validation remain unperformed **Product:** New mobile DJ application; no Mixline reuse **Platform direction:** Android first confirmed; iOS sequence unresolved **Owners:** Release lead, security lead, product owner and owner-operated service operator

1. Release promise

The user has selected Android first, serious mobile DJs as primary users, casual streaming DJs as secondary, and private practice/personal gatherings as the initial performance context. USB audio interfaces and DJ splitters are permitted. The intended feature scope includes all previously enumerated professional modules except lighting: four decks, prepared and live stems, sampler/sequencer, recording, automix, microphone, video/karaoke, broadcasting and plugin hosting. ParDay is the chosen name for the AI assistant, not a confirmed name for the whole app.

This is confirmed product ambition; technical support and source permissions still require evidence. Keep the modules in the requirements and delivery plan. If a device, license or partner prevents the intended full scope, show the gap and seek a product decision before representing a smaller release as complete. Private use does not justify reducing audio reliability, and “private gathering” must not be assumed to satisfy every provider's event or public-performance restrictions.

The product should make a defensible promise about the broader certified-device matrix, workloads, audio routes and media sources. “Professional” must correspond to measured sustained performance and recoverable workflows. Do not advertise universal latency, universal streaming compatibility or permission for commercial performance merely because a third-party catalog can technically play.

G0-G5 produce decisions, feasibility and working capabilities. G6 establishes a release candidate; G7 installs and validates the personal local release. G8 maintains it and can support separately chosen extensions. Cloud-specific G5 cases apply only when the later AWS extension is evaluated. Core beta at G3 is not a substitute for the complete intended professional product.

The owner has selected a personal tool, with openness to a later GitHub source release for others to home lab their own solutions. Public commercial distribution is not the current objective. The broader certified-device matrix, timing, exact licence/publication choice and actual source access remain unresolved. The latest owner decision is local first: build the phone-native app with no required server, including an on-device acquisition feasibility and implementation track. The on-device versus AWS comparison moves to a later optional expansion; no comparative winner is established. The owner has no local host selected and accepts AWS later with an approximate USD 35 monthly planning target excluding ParDay AI-provider charges. This target is not a hard ceiling, actual quote, selected resource configuration or deployment authorization. This document prepares personal installation and operation; optional publication, purchases and external outreach have not been executed.

2. Threat model and controls

The most important assets are uninterrupted valid audio, prepared libraries, recordings, provider credentials, private media and the release-signing chain. Untrusted boundaries include imported media/tags/artwork, archive backups, remote provider responses, external audio routes and hosted plugin code. Cloud sync, product purchase entitlements and billing callbacks are additional boundaries only if those features are selected later.

Threat	Required control	Evidence before release
Malformed media or artwork exploits a parser	Mature patched decoders; size/duration/artwork limits; defensive parsing; isolate work where practical; no parser work on callback.	Fuzz/regression corpus, crash triage and dependency review.

Threat	Required control	Evidence before release
Malicious archive overwrites files	Reject path traversal, symlinks outside permitted scope, duplicate IDs and decompression bombs. Stage and validate restore.	Adversarial backup/import cases.
Tokens or signing secrets leak	Platform secure storage for owner credentials; no shared app-owned paid secret in the APK; server secrets stay server-side if that service exists; protected signing and redacted logs.	Binary/repository/log secret scan and access review.
OAuth redirect or callback is hijacked	Use provider-approved authorization flow, PKCE where applicable, verified redirect handling and state validation.	Negative callback/replay tests per provider.
Remote response grants excessive rights	Capability decisions use validated contract/schema versions; unknown fails closed; record/export gate lives below UI.	Revoked/expired/malformed grant tests.
Sync destroys user preparation	Versioned records, explicit conflicts, tombstones, checkpoint/restore and migration fixtures.	Interrupted sync and conflicting cue/grid edits.
Telemetry exposes private music or speech	No uploaded audio, full file paths, tokens or detailed library metadata by default; opt-in diagnostics with reviewable contents.	Network inspection against actual consent/privacy text.
Unauthorized purchases or replayed entitlements, only if commercial billing is later added	Validate purchase state through supported platform flow; track idempotent transaction identity and revocation.	Test purchases, cancellation, restore, duplicate callback and offline behavior.
Plugin or model supply chain introduces code/risk	Pin exact versions/hashes, keep an SBOM and license record; prohibit downloaded executable code unless explicitly designed and policy-reviewed.	Provenance and vulnerability review; specialist module admission gate.
Route change leaks cue audio	Route identity and channel map validation; never reroute cue-only audio to main automatically.	Real-device unplug/replug and route-loss tests.
ParDay interprets an untrusted title, webpage or tool result as authority	Treat retrieved content as data; allowlist tools and parameter schemas; validate source permissions and current deck/route generations below the model.	Prompt-injection and stale-action cases cannot change playback, spend money or export restricted content.
AI gateway leaks provider data or private media	Field-level provenance, minimum prompt context, server-side secrets, model/vendor allowlist and explicit cloud consent.	Request inspection proves restricted fields, tokens and raw audio are absent from prohibited destinations.

Use least privilege. Request microphone access when the user enables microphone functionality, not on first launch. File selection should use scoped platform access. No contacts, location or broad storage access is needed merely to mix imported music. If a later feature requires additional access, add a clear purpose and update data handling documentation.

Security controls must not block the real-time thread. Queue logging, encryption, network validation and persistence elsewhere. A server outage or account login failure should not unexpectedly terminate an authorized offline local set. If product billing is added later, its app-entitlement behavior must define an offline allowance and session continuity rule while honoring separate provider expirations. Do not disguise a failed renewal as an audio-engine failure.

3. Privacy and data lifecycle

Maintain an inventory of data collected by the app and every SDK: purpose, local/remote location, retention, processor, consent, deletion and export behavior. Distinguish on-device music analysis from uploading music for cloud processing. Cloud stems or analytics require explicit product decisions and appropriate rights; they are not default infrastructure requirements.

ParDay's cloud use has its own data inventory: typed requests, optional speech, selected eligible metadata, conversation history, tool audit events, provider/model identity and retention. Core local mixing must remain functional when ParDay is off. Do not turn on ambient microphone capture to make the assistant feel present; offer an explicit push-to-talk path if voice is selected. Separate microphone permission for the mix from microphone permission/use for speech transmission. Changing between OpenAI, Anthropic or an on-device model changes processing and must be visible in settings and the data policy.

Spotify's ordinary terms prohibit AI ingestion of Spotify Content, and its policy also restricts voice-control assistants. [9,11] Review this independently of consent and cloud vendor training settings. Sending data to a “no training” endpoint is still model inference; that setting does not establish source permission. Document 07 defines provenance enforcement; Document 13 defines model routing, consent and action authority. Cached prompts, analytics and exported diagnostics require the same filter as interactive requests.

The default account-free local mode must retain a fully functional local core. Optional app-account deletion must remove associated cloud data and explain legitimately retained records. Disconnecting a music provider revokes its tokens and permitted caches without deleting unrelated user-owned media. Local preparation can remain only where allowed and chosen by the user. Export before deletion must be understandable, but deletion cannot be obstructed by unnecessary steps.

If public Google Play distribution is later selected, Google Play requires accurate Data safety information covering third-party SDK behavior and a privacy policy, including for apps that collect no user data. If the app supports in-app account creation, its account-deletion requirements include an in-app path and an external web resource. Verify exact applicability at submission. [1,2]

4. Personal installation, updates and recovery

The personal release needs an installable signed Android package, a stable package/application ID, recorded build provenance and a private signing-key backup controlled by the owner. Test fresh installation and an upgrade that preserves media access, crates, cues and settings. Keep signing material, API credentials and private configuration outside any publishable source tree. A debug or locally built prototype is not evidence of a reliable update path.

Document the installed app and optional server versions, supported S26 Ultra software/audio profile, settings and limits. Back up user preparation and configuration separately from regenerable caches and provider-controlled media; exercise restore before calling the release ready. Do not assume an older binary can read a newer database after a schema migration. Provide a tested rollback/recovery path, or clearly require restoring a compatible backup.

A later optional owner-operated service needs a paired device, bounded access, storage/retention settings, logs that omit credentials, and update/rollback instructions. AWS is the accepted expansion provider; final runtime and sizing follow the deferred comparison. None of this is a prerequisite for the first local installation. Document account ownership, region, access path, billing assumptions and resource cleanup before any deployment. Service outage must not prevent prepared local mixing. Local home-lab LAN/remote configurations remain optional portability references. Acquisition workers and ParDay cloud/gateway services have independent data and cost boundaries; see Documents 13, 16, 17 and 18.

4A. Conditional Google Play reference

Apply this checklist only if public Google Play distribution is selected later. These date-stamped research findings are not personal-release gates or a guarantee that future policy will remain unchanged. OS/native compatibility and correct installation still require testing for the selected personal package.

Gate	Deliverable / acceptance
Publisher identity	Confirm legal publisher, personal versus organization account, required verification, support contact and ownership of the package/signing process.
Target and packaging	As checked on 29 October 2026, official requirements specify Android 16/API 36 or higher for new mobile apps/updates after 31 August 2026. Recheck at release. Produce correctly signed release bundles, symbols and reproducible version metadata. [3]
Native compatibility	Verify supported ABIs and 16KB memory-page compatibility for the app and every bundled native SDK/library. Test actual packaged artifacts, not only compilation success. [4]
Foreground operation	Declare correct service types/permissions and Play Console information for user-initiated background media playback. Supply required demonstration and user-stop behavior. Microphone capture is a separate permission/lifecycle case. [5]
Data and permissions	Complete privacy policy, Data safety, permission declarations, account deletion where applicable and SDK inventory. Match real network behavior.
Content and rights	Complete content rating, audience and advertising declarations; submit truthful screenshots/descriptions; possess asset/model/codec/provider rights.
Payments	Choose a policy-compliant billing model for digital features. Google Play Billing is the default requirement unless an applicable current exception/program applies. Verify territory-specific obligations rather than assuming a universal exception. [6]
Testing access	For personal accounts created after 13 November 2023, current official guidance requires at least 12 continuously opted-in closed testers for 14 days before applying for production access. Account status determines applicability; this is not a rule triggered simply by calling an app commercial. [7]
Quality and review	Complete real-device audio certification, interruption/recovery, crash/ANR, accessibility and purchase tests. Supply reviewer access and clear instructions for optional account/provider paths.
Rollout	Use staged release, monitor supported-device failures and retain a tested rollback/forward-fix plan. Do not force an update during a live session.

If iOS is selected, create a separate Apple submission checklist from current official requirements at that milestone: App Store review, signing, privacy declarations, entitlements/background modes, purchase rules, permissions and device testing. Android approval and audio certification do not establish iOS readiness.

5. Rights and dependency review

Maintain six distinct records: software license, asset/model license, provider API/SDK agreement, end-user subscription grant, public-performance/venue context, and recording/broadcast/distribution permission. One does not substitute for another. Local file possession is not proof of permission to redistribute it. A venue's performance arrangements do not automatically change a streaming provider's contract.

For each DSP/codec/model dependency record the exact version, license, commercial terms, attribution, distribution obligations, platform/ABI support, maintenance status and exit option. A model repository's code license may not cover its weights or training-data claims. Do not infer a commercial SDK price from an old product page; request a current written quote after the required feature/workload is specified.

The user permits paid DSP licenses but prefers compatible open-source options. Evaluate open-source candidates first against quality, callback behavior, latency reporting, maintenance and distribution obligations; compare a paid candidate when those results justify it. “Open source” does not mean every license fits a closed-source commercial app, and “permitted to consider paid licenses” is not a purchase order. Record lifecycle cost and exit feasibility beside any proposed choice. Apply the same discipline to on-device AI runtimes, model weights, speech components, codec libraries and any hosted plugin standard.

Spotify's selected-DJ support explicitly excludes public venues/events/livestreams, while its ordinary Developer Policy prohibits mixing; Apple ordinary media terms also restrict commercial use. Existing named DJ partnerships do not grant this app equivalent rights. [8,9,10] Other providers require their own applicable contract review. Keep partner-blocked modules out of the public committed feature list until admitted.

The owner explicitly selected integrated link-to-file acquisition, including unofficial methods. Document 16 adds a real implementation track using replaceable extraction, direct-file or documented vendor adapters. Do not equate technical feasibility with service/content permission or store eligibility. Spotify terms restrict stream ripping and YouTube API policies restrict unauthorized download/separation; using another mechanism does not itself establish an exception to applicable service terms. [11,12] A CC/public-domain claim concerns content rights and does not by itself grant platform access permission.

The named Spotdown/Y2Mate pages demonstrate advertised consumer workflows, not verified app APIs or reusable commercial grants. The new evidence and deployment alternatives are in Document 16. Google Play specifically lists unauthorized downloading of copyrighted content and encouragement of infringement as violations; it does not prohibit every file downloader by category. [13] Public commercial distribution, a private APK and a self-hosted service have different operational implications, but none automatically supplies missing source rights. Record the four evidence dimensions independently and obtain a concrete release decision instead of deleting the requested capability or promising universal support.

Video/karaoke and broadcasting have their own rights review. A playable song does not automatically permit synchronized video, displayed lyrics, recording, remix distribution or an Internet broadcast. For user-managed local files, keep rights responsibility understandable without demanding ownership-proof uploads for routine importing. For provider-controlled sources, block only the operation that lacks permission and explain the available alternative.

6. Personal operating budget and optional commercial reference

No sales price or revenue model is required for the selected personal tool. The owner selected an approximate USD 35/month target for future AWS operation, excluding ParDay AI-provider charges, and explicitly deferred AWS until after the local version. No AWS infrastructure is needed for the first local version. Estimate development effort and recurring personal operating costs separately; local first is not a claim that optional external model calls, music services or all acquisition operations are free or offline. If commercial distribution is explicitly selected later, the historical alternatives are a one-time purchase, subscription, or paid core/modules; they are not current design requirements. Core performance remains independent of advertising or product billing.

Budget separately for native audio/DSP engineering, mobile interaction design, quality devices and audio routes, SDK/model licensing, commissioned music and assets, provider integration/certification, cloud/storage/egress if selected, privacy/dependency review, personal maintenance and ongoing OS compatibility. Store accounts, product billing and commercial support are conditional future costs. Four

decks, real-time stems, video and plugin hosting increase both development and certification effort; a cloud backend is not free simply because a service offers an initial free tier.

Add a ParDay budget with separate inputs for model requests/tokens, speech duration, retrieval/search, gateway hosting, logs, evaluations, support and optional local-model delivery. Set per-session and account spending limits, cancel stale requests, and avoid repeated full-library prompts. Consumer ChatGPT or Claude subscriptions must not be assumed to fund third-party API usage; confirm the actual selected service contract before quoting price. Do not choose a named model or estimate unit cost without a current benchmark and vendor pricing check at implementation.

Use a quote-driven model:

Development cost = role effort × agreed rates + licenses + assets + device/test equipment + external review.

Personal operating cost = model/API usage + applicable provider subscriptions + selected AWS compute/storage/network and other billed resources + maintenance.

No existing local host is selected or required. Use Document 18 and contracts/aws_cost_scenarios.json for the deferred AWS trial and approximately USD 35/month target. Keep that target distinct from an authorized trial spend, a verified bill and an enforced ceiling; do not treat estimates, free-tier eligibility or alert settings as a spending guarantee. Local hardware/electricity costs belong only to a later physical home-lab profile. If public distribution is later selected, add its store, payment and community/support costs separately.

Keep assumptions and confidence beside each input and separate one-time costs from recurring costs. Commercial break-even is outside the present personal-project acceptance criteria.

Historical commercial experiment, applicable only after a change of objective: recruit eight intended users. If four or more cannot supply 20 usable tracks for a normal set, or say missing catalog access prevents adoption, reject the claim that a local-only launch serves that audience's quick all-in-one need. Pursue a suitable partner or adjust the launch audience. This revises launch positioning without removing the offline engine or advanced final product.

7. Go/no-go record and operations

G6 requires the selected personal scope matrix, device/workload evidence, requirements traceability, no unresolved release-blocking defects, recoverable library/recording behavior, rights/asset ledger, security review, install/upgrade instructions and a tested backup/restore procedure. Store-submission materials and the deferred D-versus-AWS comparison are not gates for the first local personal release. Local acquisition, the complete selected professional scope and relevant local safety/quality evidence remain tracked; an unavailable module must be reported explicitly. Optional community publication adds Document 17's separate readiness checks. Unproven specialist modules remain visibly unavailable; they cannot be advertised as completed.

ParDay adds release evidence for task success, fabricated-track rejection, stale command rejection, prompt-injection resistance, provider-data egress, cancellation, budget enforcement and cloud outage behavior. At least one complete warm audio workload must run with active assistant requests and failures; only the device-certified combination may be advertised. AI demonstrations and concept images do not establish these results. Source integrations add Document 14's acquisition/provenance cases and Document 16's twenty DLA cases, including worker isolation, format honesty, Android stop behavior and the actual media-source distinction. No downloaded media, paid account, partner approval or new API key was acquired during this planning revision.

Assign maintenance responsibility for crash/audio regression, source or owner-operated service outage, corrupted library, credential rotation and data deletion. Commercial purchase/support workflows apply only if later selected. Incident reports record affected versions/devices, user impact, mitigation and verification.

Signed remote configuration may disable a faulty optional integration safely; it must not introduce arbitrary code or unexpectedly alter the live mix. Rollouts and migrations should occur outside active performance.

Track voluntary product metrics that answer actual questions: time to first successful mix, completion of cue/transition workflows, sustained-session success, recovery rate, repeat use and support burden. Do not collect track content merely because analytics makes it convenient. Use local, optional diagnostics for the owner's personal acceptance and maintenance. Revisit a commercial plan only if the owner changes that objective; optional GitHub interest does not itself make the app a commercial service.

Sources

1. Google Play Data safety guidance.
2. Google Play User Data policy.
3. Android target API requirements.
4. Android 16KB page-size compatibility.
5. Foreground service requirements and Android service types.
6. Google Play Payments policy.
7. Testing requirements for new personal accounts.
8. Spotify DJ support.
9. Spotify Developer Policy.
10. Apple Media Services terms, applicable territory to be confirmed.
11. Spotify Developer Terms.
12. YouTube API Developer Policies.
13. Google Play intellectual property policy.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 12 OF 19

Council decisions and owner answers

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.

DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version 1.0 | 29 October 2026 | Final planning and design handoff baseline; owner choices recorded, implementation not started

Council header

Operation: Revisit and final handoff reconciliation. Chair: Adaptive. Authority: Provisional engineering recommendations; owner decisions recorded separately. Evidence: the original answers, selected C/Galaxy S26 Ultra/ParDay C, explicit unofficial-download request, personal-tool and optional source-publication choices, AWS discussion and latest local-first instruction with budget B. Focus: make all documents agree before design while preserving the complete professional scope.

Analytical seats: First Principles, Contrarian, Expansionist, Operator, Customer Advocate, Audio/AI Architect, Product Designer, Music Access Analyst and Evidence Auditor. Three independent agents revised source/acquisition, ParDay architecture and design/assets; the chair integrated their findings with the remaining lenses. This is structured planning analysis, not interviews or nine independent experts. The previous full evaluation remains in the v0.1 archive. The seat analysis below originated in v0.2; later revisions recorded personal-tool delivery, optional source publication and acquisition comparison. Version 1.0 reconciles the latest local-first and future AWS budget decisions across the pack. This final audit is not a newly completed nine-seat exercise or evidence of app performance.

Current threshold: ready to begin design from Document 19 with no blocking owner question. Engineering remains experiment-first for device/DSP/plugin/stem feasibility, source acquisition, AI utility and workflow. Specific operations still need their applicable technical and access/content evidence; a blocked provider module is not a verdict against the whole app.

1. What the owner decided

Answer	Confirmed decision	Consequence
1 A	Android first	Native Android is the initial target. Later iOS is not a launch requirement.
2 A primary; C secondary	Serious mobile/live DJs primary, casual streaming DJs secondary	Retain professional depth and touch precision; make first-use preparation and source availability approachable.
3 A	Personal practice and private gatherings initially	Do not assume public/paid events were requested. Internet broadcasting has separate rights and audience implications.
4 Any and all	Broad source coverage is desired	Keep every relevant source in the adapter/discovery inventory; track admission rather than inventing universal access. No provider priority was supplied.
5 A	USB audio interfaces and DJ splitters allowed	Certify independent cue routes; external MIDI/HID controller integration remains excluded.
6 Everything except lighting	All discussed professional modules selected	Four decks, prepared/live stems, sampler/sequencer, recording, automix, mic, video/karaoke, broadcast and plugin hosting remain in the complete target. Lighting is excluded.
Original 7: Steampunk; renders A and C	Steampunk confirmed; comparison completed	The subsequent choice below selects C. Retain A only as comparison history.
8 A but preferred C	Paid DSP acceptable, compatible open source preferred	Evaluate an open-source route first and compare paid alternatives against the same sonic/performance/licence tests. This is not purchase authorization.
New instruction	AI assistant named ParDay	Add concrete assistant workflows, provider adapters, local authority and cost/privacy controls. The app's product name remains open.
Original acquisition suggestion	Spotify/YouTube lookup and downloading	Discovery and acquisition require distinct evidence; the subsequent explicit request below broadens the implementation study.
Follow-up 1: C	C - Crystal Current selected	Pale frosted glass, titanium and subtle brass lead. A is not an approved night theme.
Follow-up 2: S26 Ultra up to date	Samsung Galaxy S26 Ultra is the first test device	Software currency is a user report; record actual variant, installed build and route before tests. No device is yet certified.

Answer	Confirmed decision	Consequence
Follow-up 3: C	ParDay assisted performance selected	Preparation assistance plus explicitly armed transitions/automix and immediate manual takeover. Each concrete plan still needs arming.
Explicit additional request	In-app unofficial Spotify/YouTube link-to-file downloading	Evaluate own/self-hosted extraction and third-party integration candidates in Document 16. Separate technical verification, provider access, content rights and distribution review.
Previous 1 C	Personal tool only	Personal installation, upgrades and reliability define release; public commercial/store launch is not the current goal.
Previous addition	GitHub could be open sourced for community home labs	Prepare reproducible source/deployment guidance. Licence and publication remain optional decisions; no repository has been published.
Previous 2 C	Compare on-device and self-hosted acquisition before choosing	Preserve the common protocol for later expansion; latest local-first instruction now fixes the first implementation stage, with no measured comparative winner.
Prior C plus AWS	No existing local host selected; AWS can be used for server setup	Preserve an optional owner-operated AWS comparison arm; no resources or account were selected or deployed.
Latest B	Approximately US$35/month future AWS planning target, excluding ParDay AI-provider charges	Use the target in a later costed expansion proposal; it is not a guaranteed bill, hard cap or spending authorization.
Latest delivery instruction	Local version first, expand to AWS later	Build the phone-native Android version without a required server. On-device acquisition is the first path; AWS evaluation/deployment is deferred and cannot block local design or personal release.

These answers close the first eight questions and the subsequent visual, first-device, ParDay-mode, personal-delivery, local-host availability, delivery-sequence and future AWS-budget questions; they must not reappear as unanswered decisions. “Everything” is interpreted against the professional modules actually presented, not as a promise of every feature ever shipped by any desktop application. New materially different features still require a concrete requirement.

Personal practice/private gatherings describe the operating context; the latest answer separately confirms personal-tool delivery. Optional open-source publication would let others operate their own setups, but does not select a licence, authorize public posting or turn this into a hosted commercial service. Personal deployment and self-hosting do not themselves change applicable source/content permissions. AWS is the accepted later server extension while home-lab LAN/remote profiles remain optional portability references. The local version requires neither. Local first describes runtime and delivery order; online acquisition and optional provider AI still need their own network paths. This is not a finding that extraction works from an AWS network or a decision to put the live engine or model inference there.

2. Original assumptions affected

The Android proposal is confirmed as a platform choice; the Kotlin/Compose and C++/Oboe implementation remains a recommendation to verify. Interface and splitter allowance is confirmed, but no exact device/accessory combination has been certified. The previous assumption that video, karaoke, broadcast and plugin hosting could remain open is replaced by selected scope. Their delivery may be staged internally, but agents cannot call a reduced beta the finished product.

The source premise has become more demanding: broad catalogs matter to the user, particularly the secondary audience. Phone-native local delivery remains the selected first architecture. It must still preserve broad source ambition and requested in-app link acquisition; local first does not narrow the final source scope to manually imported files only. At the same time, required permissions and usable APIs are external dependencies. Track unavailable adapters honestly and raise a specific acceptance decision if concrete feasibility evidence leaves a selected source requirement unmet. A broad unanswered provider questionnaire does not block design.

The visual decision is **C - Crystal Current**. Develop pale optical glass, titanium, restrained brass and dark-ink labels on the Galaxy S26 Ultra. Low-light and accessibility treatments should derive from C; the owner did not choose the proposed A/night+C/day combination. Its exact native layout, contrast and performance remain unverified.

The owner selected ParDay preparation plus armed assisted performance. Unarmed live behavior remains read/propose, and each bounded audible plan is explicitly armed. ParDay introduces language-model data egress, cost, stale actions and prompt injection. It also creates an opportunity to make a large professional tool quicker to use. Stem separation and beat analysis remain distinct DSP/ML functions; naming an assistant does not merge these workloads into a conversational model.

3. Updated seat positions and directed challenges

Seat	Revised position and rationale	Confidence / key dependency
First Principles	Keep audio timing, identity and permissions deterministic. AI should reduce preparation work through bounded actions.	High in the boundary; utility untested.
Contrarian	A wrapper around a chat model is easy to copy and can distract from reliable mixing. Test completed DJ tasks against the manual baseline.	Moderate; no user study yet.
Expansionist	Preserve the ambitious scope and make ParDay a preparation/arrangement interface into real capabilities.	Moderate; data quality and integration determine value.
Operator	Add video, karaoke, broadcast, plugin and AI work packages now. Do not hide their cost inside “all-in-one.”	High that separate evidence is needed; effort unknown.
Customer Advocate	Finding a usable exact version matters more than merely displaying a catalog result. Explain availability with useful next actions.	Moderate; intended users need testing.
Audio/AI Architect	Use a local engine and validator, replaceable cloud models, and explicitly armed deterministic performance plans.	Moderate until timing/resource and failure tests pass.
Product Designer	Selected C establishes the material family; actual mobile layout still needs physical-size validation and simplification.	High about render limitations; C preference confirmed, native usability untested.
Music Access Analyst	Source identity and grants must travel with every field, file and operation, including AI ingestion.	High in current ordinary policy constraints; partner terms unknown.
Evidence Auditor	User preference is evidence of intent, not performance, licence clearance, market demand or model competence.	High; no app tests exist.

Acquisition update after v0.2: The owner explicitly wants unofficial link-to-file capability, and supplied Spotdown and Y2mate as examples. Preserve that capability request rather than silently replacing it with official-source-only discovery. Document 16 evaluates own/self-hosted extraction, third-party adapter options and actual-source matching. Technical feasibility, provider access, content rights and distribution review are separate statuses; neither a working public website nor the absence of an official API settles all four. Ordinary terms remain relevant to access and release decisions. [C3-C5]

Customer Advocate to Audio/AI Architect: Does asking before every assistant action destroy speed? Response: no. A direct request can authorize reversible preparation work. Live audible changes require one concrete bounded plan to be armed; deterministic execution then proceeds without repeated beat-by-beat confirmation. **Position refined:** authority follows impact and user intent, with immediate manual takeover.

Evidence Auditor to Expansionist: Can Spotify metadata be used freely for AI recommendations if no audio is uploaded? Response: ordinary published terms also restrict AI ingestion of Spotify Content, including returned metadata. **Position revised:** model context is filtered by source provenance before prompts, embeddings, tool results or logs. Consent and a model provider's no-training policy do not waive source restrictions. [C3, C4]

Operator to Product Designer: Are the two renders comparable implementation blueprints? Response: they show the same performance task and material contrast, but exact placement differs and generated

labels need cleanup. **Position unchanged:** use them to select visual identity, then build a measured component/layout prototype. Do not implement a screenshot literally or claim 48dp targets from its pixels.

Contrarian to Operator: Does “plugin hosting” really mean app-owned effects? Response: no. The user selected hosting. **Position revised:** require a real third-party Android-compatible plugin path and fail the hosting feasibility gate if only built-in effects work. Desktop binaries are not presumed compatible; supported ABI, distribution and scheduling require explicit selection. See Document 15.

4. Scenarios that change the plan

Scenario	Required response	Decision effect
A requested provider restricts an operation	Keep technical function, provider access, content rights and distribution review separate; show the specific affected operation	Research an alternate acquisition route on its own evidence; it does not inherit clearance merely because it returns a file
AI responds after the DJ replaces a deck	Reject stale generation/revision or expiry; offer a fresh proposal	Never execute on the replacement track
Phone is hot with live stems, video and broadcast	Shed optional conversation/preparation/render decoration first; test the exact active workload	If protected sound cannot be sustained, that combination is uncertified
Plugin crashes or misses its bounded budget	Follow tested bypass/isolation policy; report failure without corrupting the session	Unsafe host/format cannot ship as supported
Assistant suggests the wrong recording or downbeat	Expose confidence, exact edition and preview; preserve manual correction	Utility depends on measured metadata and user review, not fluent explanation
Source is cleared for local use but not broadcasting	Disable broadcast for that mix/source under its capability rule	Private practice is not treated as broadcast permission
ParDay API is offline or quota exhausted	Preserve all local DJ controls and valid saved preparation	Cloud assistant cannot become an audio dependency

The strongest pre-mortem is a visually impressive app that cannot admit the desired catalogs, overloads common phones and requires too many confirmations to be quick. The opportunity case is a reliable instrument whose assistant turns a large eligible local library into a prepared, understandable set faster than competing workflows. Both outcomes depend on evidence outside these renders.

5. Chair synthesis and surviving objections

Maintain the native local-engine direction with selected C, Galaxy S26 Ultra first-device validation, ParDay armed assistance, the requested link-acquisition module and the selected specialist modules. Evaluate unofficial extraction candidates rather than treating official API access as a universal technical prerequisite. Keep source eligibility and user authority enforced beneath the UI. Proceed with a replaceable model adapter rather than binding the app identity to one provider. Prefer compatible open source for DSP where it meets the actual quality and maintenance requirements, while retaining a measured paid comparison.

Strongest argument against this direction: the combined scope may exceed the available budget/team and the mobile hardware or partner ecosystem may not support every desired combination. Planning completeness cannot solve missing provider agreements or unsupported plugin formats. Before promising a release date, run G1 and obtain the actual resource constraints.

Historical minority report, updated: personal deployment and local-first implementation are selected. The earlier suggestion of a tightly reduced feature set is not. Preserve the complete professional scope and requested link acquisition on the phone, with future AWS expansion evaluated separately. Bring-your-own AI credentials and an optional gateway remain implementation options; a smaller milestone still does not complete the product.

The design choice is reversible. Vendor coupling, plugin ecosystem, source agreements, data retention and release promises are more costly to reverse. Validate those earlier than exhaustive screen polish.

6. Experiments and concrete next work

Unknown	Proposed test	Pass / failure signal	Decision impact
C native usability	Task comparison on the S26 Ultra at physical size, including reduced decoration	Readable controls and correct touch actions; fail if contrast or errors worsen	Refine selected C components; do not reopen art-direction choice without evidence
DSP route	Same corpus/device compare compatible open source and paid candidate	Meets timing/listening/route gates; failure documented per operation	Select provider or revise architecture
Catalog viability	Provider-by-provider access review plus usable-track task	Intended users can obtain exact playable versions through admitted routes	Rank launch dependencies and unavailable adapters
ParDay utility	Counterbalanced preparation study on eligible fixtures	Faster or fewer errors than manual baseline with no unauthorized actions	Keep useful jobs; change or drop weak mechanisms with scope review
ParDay authority	Adversarial fixtures before API integration	Zero unarmed, stale or duplicate audible actions	Block action tools until corrected
Video/plugin combinations	Exact device workload and one real third-party plugin	Sustained audio, truthful visual degradation, safe failure	Certify combinations or report feasibility blocker
Local resources and operation	Capture S26 Ultra build/route, effort, DSP budget if needed and AI cost limit before paid integration	A realistic local build/operation plan can be produced	Design starts now; estimate implementation after G1 evidence
On-device acquisition first	Prove D with actual eligible source fixtures, record identity, format, source/network behavior and workload	Local path passes applicable identity/audio/privacy gates or reports a concrete blocker	Preserve the requested capability; no server dependency silently substituted
Future AWS expansion	After a usable local baseline, compare D/AWS with common fixtures and a current costed proposal around US$35/month excluding AI	Source works in that actual cloud environment and justified benefit fits the reviewed operating plan	Optional extension; no measured comparative winner and no current deployment

All sample sizes and thresholds elsewhere remain proposed. The two concept images were created and visually reviewed in v0.2; the owner subsequently selected C. No device, DSP, extraction, model or performance experiment above is claimed complete by recording these answers.

7. Current decision ledger

ID	Decision	Status	Revisit trigger
D-001	New product; no Mixline reuse	Confirmed	New explicit owner instruction
D-002	No external MIDI/HID DJ controllers	Confirmed	New explicit owner instruction; DVS interpretation still open
D-003	Android first	Confirmed	New platform requirement
D-004	Kotlin/Compose + portable C++/Oboe	Proposed	Team or G1 evidence
D-005	Two-deck default, four-deck capability	Capability selected; default layout proposed	Native workflow/device tests
D-006	Prepared and live stems	Selected; live device/source/model-gated	Quality/licence/thermal results
D-007	Liquid glass + steampunk; C Crystal Current	Confirmed selection	Native usability/contrast/performance evidence or explicit owner change
D-008	USB interfaces and DJ splitters	Confirmed	Actual compatibility evidence
D-009	Offline prepared local core; no required app/server account	Local independence confirmed; detailed persistence architecture proposed	A future optional integration must not make local performance account-dependent
D-010	Video, karaoke, broadcasting, plugin hosting	Selected	Feasibility blockers reported for owner decision
D-011	Lighting	Excluded	New explicit owner instruction
D-012	Planning now; implementation later	Current task boundary	Explicit build instruction
D-013	Serious DJs primary; casual streaming secondary	Confirmed	User research or owner revision
D-014	Personal practice/private gatherings	Confirmed initial context	Public/paid usage requested
D-015	Any/all sources desired	Confirmed ambition; per-provider admission unresolved	Permissions/availability/owner priorities

ID	Decision	Status	Revisit trigger
D-016	Open source preferred; paid DSP acceptable	Confirmed preference	Comparison and budget cap
D-017	ParDay assistant name and AI exploration	Confirmed	Owner revision
D-018	ParDay preparation plus bounded armed assisted performance	Confirmed capability; unarmed live read/propose remains	Authority/takeover/usability evidence or explicit owner change
D-019	OpenAI/Anthropic replaceable API adapters	Proposed	Provider evaluation and cost/privacy choice
D-020	In-app link-to-file acquisition, including unofficial candidates	Capability requested; mechanism and four admission statuses unresolved	Document 16 technical and access/content/distribution evidence
D-021	Galaxy S26 Ultra first test device	Confirmed; up-to-date software is user report	Capture actual variant/build/route and measured workload results
D-022	Personal tool only	Confirmed	Explicit change of product objective
D-023	Possible open-source GitHub publication for community home labs	Optional future direction; licence and publication not selected/executed	Owner publication/licence decision and reproducibility/rights review
D-024	Comparative acquisition evaluation	Retained for future expansion; D first under D-026; comparative winner unmeasured	Later Document 16 comparison results; do not gate local release on AWS
D-025	No local host selected; AWS accepted for later optional extension	Confirmed future direction; account/region/resources unset and deployment unexecuted	Owner commissions expansion; Document 18 feasibility and cost proposal
D-026	Local version first, phone-native Android with no required server	Confirmed delivery order; D acquisition is first implementation/evaluation path	Explicit owner change or a concrete capability blocker requiring a decision

ID	Decision	Status	Revisit trigger
D-027	Future AWS planning target about US$35/month, excluding ParDay AI charges	Confirmed target; not a hard billing cap or spending/deployment authorization	Current regional quote and measured workload before optional expansion

8. Design closure and later decision points

No further owner discussion is required before design begins. Document 19 supplies the screen/flow inventory and first design assignment. The owner has supplied the platform, audience, visual direction, first device, assistant identity/authority, source ambition, personal delivery, local-first order and future cloud target. Do not reopen those choices or demand an AWS account before producing the local interface.

Matter	Working design basis	When it becomes a real decision or evidence gate
Product name	Use Mobile DJ as a neutral working label; ParDay remains the assistant	Final wordmark/package identity and any publication
Exact S26 Ultra build and accessory	Design responsive native layouts and explicit route states; no certified latency/channel claim	Capture actual variant/build/interface before physical benchmarks and route acceptance
Initial DSP/model providers	Preserve the proposed adapter boundaries; use labelled fixtures during design	Same-corpus DSP test; eligible-model evaluation, credentials and AI budget before paid integration
Voice	Typed/touch workflows are complete; voice is an optional proposed extension	Explicit voice scope and noisy-room/privacy tests before adding it
Genres and session length	Use the diverse cleared fixture categories and proposed two-hour P0 soak in Document 09	Refine a representative owner corpus and longer session requirement during validation
Four decks plus live stems and specialist combinations	Selected capabilities remain; show ready, unavailable and resource-denied states	Measure exact combined workloads; raise an owner scope decision if a selected combination cannot be supported
Requested source paths	Native intake/queue, actual-version preview, honest quality and failure/cancel states	Prove the local adapter and its technical/access/content/distribution statuses; no mock completes FR-049
Broader devices and iOS	S26 Ultra first; adaptive design without claiming other certification	Additional support scope and corresponding hardware evidence
Team, timing and full development budget	Gate-based work packages, no fixed completion promise	Estimate after G1 and actual capacity/paid-component choices
DVS/timecode	Outside the initial controller-free design as a proposed interpretation	Only revisit if the owner needs timecode input; do not add it by implication

Matter	Working design basis	When it becomes a real decision or evidence gate
GitHub licence/publication	Preserve dependency/asset provenance and reproducible source guidance	Before any optional public release; no publication or licence selected yet
AWS details	Deferred extension; about US$35/month target excluding AI	Account, region, service, access, current quote and controls when expansion is commissioned

These are scheduled decisions and evidence tasks, not a new pre-design questionnaire. Local first does not promise all network features work offline, erase source restrictions or certify a phone. A concrete failed gate must be reported honestly; design may represent its unavailable/recovery states without pretending it passed.

9. Updated decision record

Original direction: native local audio core, modular professional capabilities, source-specific permissions and a precision-instrument visual system. Current baseline: selected C, S26 Ultra first device, ParDay armed-assisted mode, requested unofficial acquisition, personal-tool delivery, local-first implementation, optional later AWS around US$35/month excluding AI, and possible future source publication. Outcome: **design-ready planning baseline**, with the native core maintained and cloud work deferred.

Proceed with selected C and the design-start brief in Document 19. A mockup is not evidence of implementation. The next implementation assignment should use the bounded G1/G2 tasks, actual device/route evidence and any specifically needed paid-resource decision; AWS details are not a condition for that local work. Revisit when a required provider denies access, a plugin path fails, a workload fails sustained audio, selected C fails native usability/contrast validation, or preparation studies fail to show useful assistance.

Sources checked 29 October 2026

- [C1] Android low-latency audio: https://developer.android.com/games/sdk/oboe/low-latency-audio
- [C2] Compose accessibility defaults: https://developer.android.com/develop/ui/compose/accessibility/api-defaults
- [C3] Spotify Developer Policy: https://developer.spotify.com/policy
- [C4] Spotify Developer Terms: https://developer.spotify.com/terms
- [C5] YouTube Developer Policies: https://developers.google.com/youtube/terms/developer-policies
- [C6] Spotify selected DJ integrations: https://support.spotify.com/in-en/article/dj-integration/

The owner answers are direct conversation evidence. Recommendations, experiments and analytical positions are this dossier's judgments. No provider contract, legal clearance, purchase, device benchmark or production implementation is claimed.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 13 OF 19

ParDay AI Architecture and Requirements

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; no model/provider selected, app implemented or performance result measured **Confirmed direction:** The assistant is named **ParDay**. The owner selected **C - assisted performance**: preparation assistance plus explicitly armed transitions/automix and immediate manual takeover. Android first, with Galaxy S26 Ultra as the first test device; serious mobile DJs primary, casual streaming DJs secondary; initial use is practice and private gatherings. **Accountable agents:** AI/Services; Audio Engine; Library/Data; Security; Product UI; Quality. **Related documents:** 05 Architecture; 06 Audio; 07 Sources; 09 Quality; 14 Discovery and Acquisition; 16 Link Acquisition; 17 Open Source and Home Lab; 18 AWS Deployment and Comparison.

Product decision

Build ParDay as an optional music preparation and performance assistant that can explain, propose and perform explicitly authorized application tasks. Its value comes from understanding the DJ's library, current preparation and actual supported capabilities. A generic chat window alone cannot deliver that value. A language model also cannot replace the deterministic audio engine, source permissions or physical device certification.

The leading design combines three different systems: local signal analysis for measurable musical properties; a local planner and validator for executable actions; and a replaceable language-model service for interpreting requests and explaining options. Prepared tracks, mixing, cueing, effects, recording and manual performance continue when ParDay is disabled or the network fails. Live stems remain a separate audio/ML workload governed by Document 06, regardless of whether ParDay requested their preparation.

The full product includes the owner-selected ParDay assisted-performance capability alongside the selected professional modules. ParDay remains optional to use; the selected mode is a product capability, not standing authorization for arbitrary audible changes. Introduce its features through development gates, without silently shrinking the final scope to a chatbot or advertising unverified automation. Lighting remains excluded. No consumer chat session, provider password or browser automation is required for the recommended integration.

Useful jobs and their boundaries

Job	Example request and useful result	Constraint
Library search	“Find my 118-124 BPM tracks with a long intro.” Returns local track IDs with measured fields and an explanation	Unknown BPM or intro length stays unknown; no invented analysis
Crate preparation	“Plan 45 minutes from warm-up to high energy.” Produces an editable order, estimated duration and alternatives	Respect unavailable tracks, explicit exclusions, editions and source eligibility
Transition suggestions	“Give me three ways into the next track.” Offers phrase, tempo, key and vocal-overlap choices	Use validated analysis; uncertainty and preview controls remain visible
Cue/grid assistance	“Suggest an intro cue and an eight-bar exit.” Creates a preview of proposed preparation edits	Preserve manual locks; confirm uncertain downbeats; retain prior versions
Readiness diagnosis	“Can this set run offline with my USB interface?” Checks cached eligible tracks, route, storage and device profile	Diagnose actual state; never assert compatibility from a device name alone
Rehearsal coach	“Help me practice this transition.” Offers a short exercise and optional post-attempt feedback	Opt-in, user-paced; no unsolicited spoken coaching over the master mix
Guided automix/arrangement	“Use these tracks and transition only at the marked exits.” Produces a bounded executable plan	User arms it; deterministic scheduler executes; immediate manual takeover
Discovery/acquisition help	“Find an authorized version of this track.” Opens permitted catalog results and acquisition choices	Search result, subscription and copyright licence are separate from PCM access

These are distinct experiences, not buttons that all send the same unconstrained prompt. Every result identifies which inputs were measured, supplied by the user, retrieved from an allowed source or inferred. “Energy” is a useful comparative label, not an objective fact unless the product defines a measurement. Local analysis should expose confidence and method rather than a spurious universal score.

Start with search, crate planning, readiness and transition explanations because they can improve real preparation without taking over audible controls. Cue proposals, rehearsal and bounded automix follow once the shared state and manual-takeover contracts pass. Voice input is optional and needs a separate microphone permission, noisy-room testing and an explicit listening state; typed and touch workflows must remain complete.

Recommended technical shape

The Android client owns the local library, permitted analysis, live state and action authority. The first version has no required application server. Design and test local orchestration with model-free fixtures; evaluate a personal direct-API configuration with user-supplied credentials when integrating a real provider. A later optional gateway can own provider credentials, spending controls and model configuration. Neither AWS nor a gateway is a prerequisite for the local interface, deterministic assistant harness or manual DJ features. Provider-specific response formats are translated into a common ParDay proposal format. Keep `parday-domain`, `parday-context`, `parday-policy`, `parday-provider` and `parday-ui` boundaries separate from `engine-core`.

OpenAI function calling and Anthropic client tools both support a model returning structured requests that application code handles. Their wire formats differ, so the product contract must remain independent of

either. Use provider-supported schema enforcement, then perform application validation anyway: valid JSON does not prove that a track exists, the user approved an action or the current deck still contains that track. [S1, S2]

For this bounded mobile use case, an application-owned tool loop is the leading implementation. It is justified by the requirement to enforce local state, device permissions and identical behavior across providers. An agent SDK can supply tracing or orchestration if it reduces real complexity. The OpenAI Agents API is an option for separate, longer preparation jobs; its hosted sessions and optional environments introduce state/retention decisions that must be evaluated. No cloud sandbox receives arbitrary access to the phone or live engine. The current Agents API documentation describes retained sessions and limitations on retention/residency options; do not equate a self-hosted environment with no provider retention. [S3]

The model can choose among the tools made available to its current role, but the application decides whether a requested operation is eligible. The model never receives a general shell, arbitrary filesystem access, unrestricted URL fetcher, music downloader, raw SQL tool or direct `EngineCommand` endpoint. Approved web research, if later added, uses a separate read-only broker and the source rules in Document 14.

Authority and live-session behavior

Level	Allowed behavior	How it is enabled
Read	Explain app help, inspect selected permitted state, search authorized metadata	Default within the user's requested task and current data settings
Propose	Suggest crate order, cues, transition or diagnosis	Default during live performance; no audible mutation
Apply preparation	Save explicitly requested crate/preparation edits or launch an eligible preparation job	A direct instruction or accepted diff; transactional edits and history
Armed performance	Execute a specific, bounded transition/automix plan	Explicit arm action showing affected decks, timing, limits and takeover

“Make me a crate” can authorize saving a reversible crate without repeated confirmation. “Suggest a transition” does not authorize playing it. The selected assisted-performance mode begins each unarmed live interaction in read/propose. An accepted proposal authorizes the displayed version only. Changes in its affected tracks, route or required permissions invalidate that authority; a newer model response does not inherit approval.

Arming is concrete and time-bounded. The interface names the tracks/decks, transition duration, allowed parameters, maximum gain, start condition, expiry and stop behavior. Once armed, the deterministic controller can perform that bounded plan without asking at each beat. The language model does not continually improvise new controls. Manual touch on an affected transport or parameter cancels the relevant automation according to the takeover contract; pressing a persistent Stop Automation control cancels all armed ParDay actions.

Do not describe audible history as reversible. A crossfade already heard cannot be undone. “Undo” restores reversible metadata edits or cancels future events; recovery from an audible change is a new controlled action. Cancelling must not hard-reset faders or abruptly jump parameters. The engine retains its normal safe takeover and smoothing behavior.

Tool contracts and proposal lifecycle

Function	Inputs	Output and execution rule
`search_library`	User-authored query, approved filters, bounded result count	Eligible local IDs and measured fields; no unrestricted library dump
`inspect_tracks`	Selected IDs and allowed field names	Revisioned analysis with provenance and confidence
`inspect_session`	Approved snapshot field set	Deck generations, stream epoch, route/profile state; no audio samples
`plan_crate`	Candidate IDs, duration/energy constraints and exclusions	Ordered proposal with alternatives and missing-data flags
`suggest_preparation`	Track/revision and requested edit types	Cue/grid diff; never overwrites locked values directly
`plan_transition`	Two IDs, analysis revisions and allowed strategy	Musical proposal; deterministic planner computes concrete engine timing
`check_readiness`	Set ID and intended route/profile	Pass/warning/block reasons tied to observed state
`propose_action`	Typed bounded actions and rationale	Stored proposal ID; no direct execution privilege

Tool results carry stable IDs, provenance references and revisions. They are data, including any text contained in a track title or note. User approval and authorization tokens are generated by trusted application code, never by the model or retrieved content.

The lifecycle is requested → proposed → validated → accepted/armed → scheduled → applied, with rejected, cancelled, expired and failed outcomes. Revalidate just before commit, not only when the proposal is first displayed. A proposal records a user-intent ID, expiry, affected data revisions, session ID, stream epoch, deck generations and manual-control revisions, including the master/crossfader revision. Bind acceptance to the immutable complete proposal and plan digest. The accompanying `parday_action.schema.json` is a planning representation for application validation; its fields are not a native ABI or a turnkey provider tool schema. Provider adapters must translate the supported subset and test it.

The local action gateway checks authorization, source capability, track identity, grid revision, route epoch, deck generation, queue capacity and current control revision. It builds typed commands outside the audio callback. The audio engine checks its own epoch/generation and bounded command contract again. Use unique action IDs and an acknowledgement journal so a model retry, app reconnect or delayed network response cannot replay a mutation. Store metadata mutation and its idempotency record in the same database transaction. A schema-valid stale proposal is rejected with a reason and refreshed preview.

Mandatory state and batch validation

Every live action requires a current non-null stream epoch and exactly one basis record for every affected deck. Deck IDs and action IDs must be unique; incoming and outgoing decks must differ. The trusted plan binds exact track/media, analysis/grid and preparation revisions. Its duration, gain interpretation and parameter envelope must equal the accepted action. An inactive load target has no audible tails or pending transport/automation that would make replacement unsafe. The proposal field `deckGeneration` maps to the engine field of the same name; `masterControlRevision` guards crossfader/master changes. Parse timestamps, enforce expiry against trusted time and validate numeric-string bounds; schema formats alone do not establish those properties. Every provenance reference must resolve and allow the particular processing purpose and destination.

The proposal's `commitPolicy` is mandatory. `ATOMIC_PREPARATION_EDITS` permits crate/cue writes in one local transaction: all commit or none do. `SINGLE_PREPARATION_JOB` enqueues one stem-preparation job atomically, then reports per-track progress and failures without calling the entire job completed prematurely. `SINGLE_PERFORMANCE_ACTION` contains exactly one bounded load or arm action. A dependent load-then-arm workflow waits for the load acknowledgement and obtains a fresh state-bound proposal. Multiple musical events belong inside the trusted validated plan. A partial result can never be reported as the whole proposal applied.

Engine-owned execution and cancellation

Scheduling-time validation is insufficient for future events. The trusted planner prepares one immutable, preallocated event batch and submits `ADMIT_AUTOMATION_BATCH`. The engine checks every scope binding, timing deadline and total capacity before publishing any runnable event. Admission is all-or-nothing: if a check or reservation fails, zero events become runnable and all staged siblings are discarded. Each successful acknowledgement returns an engine-issued `automationId` and `automationRevision`, attached internally to every scheduled event. These values are engine-owned register identities, not bearer credentials or user approval. The model never supplies or receives trusted execution authority.

Immediately before each event executes, the engine checks its automation identity/revision, stream epoch and affected deck/control guards. The trusted bridge also binds commands and acknowledgements to an engine-created `engineInstanceId`, so a restart cannot accidentally make old sequence/token values valid again. Control revisions track superseding external/manual changes; the active plan's own accepted parameter writes do not invalidate themselves. Manual takeover, Stop Automation, relevant deck/route changes, permission invalidation or plan expiry invalidates the revision and all pending sibling events. At the same engine control boundary, invalidation precedes remaining affected due events; direct manual values then apply through ordinary smoothing. This closes the case where a queued crossfade would otherwise resume after the DJ touched the fader without changing the track or route.

`CANCEL_AUTOMATION` uses reserved bounded control capacity or an equivalent engine-owned invalidation path, so a full normal scheduling queue cannot drop cancellation. The acknowledgement reports the effective engine frame after which the invalidated plan cannot execute further events. A command already heard cannot be reversed; cancellation does not reset current faders or hard-cut valid sound.
Admission/cancellation retries with the same immutable request return the existing outcome; reused IDs with different payloads are rejected. A failed or cancelled plan never retries its remaining events automatically. Engine restart discards tokens and queued events, and restored proposals require fresh validation/arming. Journaling, plan construction and policy resolution remain outside the callback; event-time guards are bounded numeric checks.

Worked example: a controlled transition

The user asks: “Suggest a 16-bar blend from A into B at the next marked exit; keep the vocals separate.” Both tracks are eligible local files with verified analysis. ParDay receives only selected metadata and analysis fields after the privacy/source filter, not their audio or the entire library.

The response proposes an outgoing exit marker, incoming intro marker, 16-bar duration and an explanation of the vocal overlap. The deterministic planner verifies the markers, tempo constraints and available preparation. If vocal activity is unknown, it says that overlap is unverified and offers a headphone preview instead of claiming certainty. A stem-based alternative appears only if the source and current device profile allow it.

The user selects **Arm this transition**. The app displays A→B, the intended exit, allowed tempo envelope, gain ceiling and manual-takeover control. It binds acceptance to proposal P42, stream epoch 9, the current deck generations and relevant revisions. The local planner schedules its exact frame commands from the engine clock. ParDay's network response time does not determine the transition timing.

If the DJ loads a different track into B, changes the route, moves an affected control or misses the scheduling deadline, the plan cancels or requires a fresh preview under its defined rule. It cannot silently substitute another track or execute on the next arbitrary beat. A command rejection is shown as a rejection, not “Done.” This example is a proposed behavior test, not evidence that a model has musical judgment or that timing has been implemented.

Privacy, source provenance and model boundaries

Cloud requests are optional. The setup explains which provider receives the user's request and selected metadata. No raw audio, microphone recording, full library export, artwork, lyrics, private filesystem paths or access tokens are uploaded by default. Feature-specific use of such inputs requires both a legitimate source capability and a clear user choice. Consent cannot grant rights that the app or user does not possess.

Provenance is tracked at field level. A user-entered title and a provider-returned title may look identical while having different permitted uses. Spotify-derived content is excluded by default from prompts, embeddings, model ranking and inference unless an applicable written agreement permits the specific operation. Removing a Spotify label, reformatting a response or routing it through another tool does not remove its provenance. A deterministic catalog view can remain separate from ParDay's eligible context; Document 14 owns the exact provider rules. This is a conservative design consequence of Spotify's published AI/ML restrictions, not a claim that ordinary API access permits a ParDay integration. [S8]

An AI provider's training policy does not override a music provider's restrictions. OpenAI states that API data is not used for training by default, while storage and retention depend on endpoint/features and configuration. Anthropic similarly documents retention arrangements and exceptions by API/model/feature. The app must record the selected contract and configuration rather than promise “zero retention” universally. [S4, S5]

Treat track names, comments, imported playlists, lyrics, web pages and tool-returned text as untrusted material. An instruction embedded in any of them cannot enable a tool, change system rules, approve an action or export data. Enforce this in tool availability and the gateway, not only with a prompt saying “ignore malicious instructions.” Store only concise user-approved preferences and task history; offer review/deletion and exclude source-restricted content from memory.

Offline operation, credentials and cost

When disconnected, local search/filtering, measured analysis, saved proposals, app help and deterministic readiness checks remain useful. A local language model is optional research, not a promise: model licence, memory, heat, battery, quality and supported Android hardware must be measured. It must not compete with protected audio or live-stem workloads. First reduce or stop optional conversational work. Never pause inference supplying audible stems under the label “AI degradation.”

Personal-tool delivery and local-first implementation are confirmed. The first network-model candidate is a personal direct-API configuration with user-owned credentials stored through the appropriate Android

secret-storage design; its provider and credential handling still need review before use. A gateway is a later optional deployment choice, not a hidden requirement for the first local app. Model-free fixtures allow design and authority tests to proceed before either path has credentials. Neither the source-acquisition comparison nor the AWS budget selects the model/provider. Keep credentials out of source control, examples, logs and distributable binaries; never collect consumer ChatGPT/Claude passwords. Each operator supplies their own applicable provider access and billing, and a community repository contains configuration examples rather than shared credentials.

Acquisition runtime, ParDay gateway and language-model inference are independent choices. Planning AWS for later acquisition expansion does not choose ParDay's gateway, model provider or inference location. Running a downloader at home does not keep cloud model requests local, and running extraction on the phone does not require a phone-resident language model. Document actual destinations and retention for each path. A local model remains an optional separate quality/thermal experiment; gateway or model outage leaves prepared local performance available. Document 17 defines the optional portable home-lab boundary; Document 18 defines the deferred AWS acquisition trial. The approximately US$35/month future AWS target excludes ParDay AI-provider charges and does not set an AI spending allowance. Any co-located ParDay gateway is separately selected, budgeted and isolated rather than an automatic consequence of using AWS.

Consumer chat subscriptions and developer API usage have separate access/billing systems. Current official documentation allows account-dependent API-credit benefits for some subscriptions, including specified Claude plans; those benefits still require the relevant API account/linking and do not create unlimited embedded usage. Check the user's actual eligibility at setup rather than assuming a chat subscription funds ParDay. [S6, S7]

Select the model by an app-specific evaluation set: tool correctness, music explanation quality, refusal of unsupported actions, latency distribution and total cost per completed task. Do not lock the specification to today's flagship name. Proposed operating controls are one active interactive request per session, bounded tool rounds, capped context/output, cancellation and per-user daily/monthly limits. An exact AI monetary limit remains to be chosen before paid API use; it is separate from the selected future AWS target and does not block design or model-free tests. Estimate cost from input/output tokens, cache/storage and paid tools; count retries and interrupted requests conservatively.

The UI acknowledges a request immediately and shows cancellable progress. Cloud latency is variable and cannot be advertised as sample-accurate or guaranteed. Use a proposed 20-second interactive deadline with an explicit retry for longer work; readiness checks and manual controls never wait for it. Model/provider fallback is disclosed and permitted only after the replacement passes the same action, privacy and quality tests. A cheaper model cannot inherit broader execution authority.

ParDay requirements and acceptance register

All cases below are proposed and **unverified**. AI identifiers supplement FR and QA registers; they do not replace audio certification.

ID	Requirement	Acceptance evidence
AI-001	Core DJ functions remain independent of ParDay	Complete QA-19 set with AI disabled, offline and unavailable
AI-002	Provider-neutral adapter	Contract tests include a second adapter fixture/stub; evaluate the allowed corpus on every real provider offered, with identical validation. Two production providers are required only if selected.

ID	Requirement	Acceptance evidence
AI-003	Restricted context by provenance	Sentinel provider fields never reach prompts, embeddings, logs or fallback model
AI-004	No default raw-media upload	Network capture contains no audio/mic/artwork/library dump in ordinary requests
AI-005	Selected armed assistance; read/propose until armed	Adversarial requests and model calls cause zero unarmed audible changes
AI-006	Typed tools and local validation	Reject malformed, extra-field, unknown-ID and out-of-range proposals
AI-007	Stale-state protection	Reload deck, revise grid, change route or touch a bound deck/master control before and after scheduling; old proposal/event rejected, including missing or duplicate state bindings
AI-008	Bounded arming	Expired, widened or modified plans cannot execute under prior acceptance
AI-009	Manual takeover	Physical control and Stop Automation invalidate engine-owned revision; no affected event executes after the acknowledged cancellation frame, including under full-queue load
AI-010	At-most-once mutation	Duplicate deliveries/reconnects do not repeat writes or schedule twice; changed-payload ID reuse rejected; metadata/idempotency transaction and batch all-or-nothing behavior survive injected failure
AI-011	Accurate completion state	Inject failed engine/database acknowledgement; UI reports failure, not completion
AI-012	Preparation history	Accepted edit preserves locked cues and can restore the previous revision
AI-013	Analysis honesty	Missing or low-confidence beat/key/phrase data is labelled; no fabricated certainty
AI-014	Prompt-injection containment	Malicious tags, notes, playlists and web text cannot alter tools or export policy
AI-015	Credential protection	Secret scans and network/log tests find no provider key in APK, logs or diagnostics
AI-016	Resource isolation	AI requests, timeout/retry storms and local inference cannot fail certified audio profiles

ID	Requirement	Acceptance evidence
AI-017	Cost and request bounds	Exhausted quota blocks new AI work without interrupting the mix; retry cap enforced
AI-018	Provider retention disclosure	Actual endpoint/configuration matches consent, deletion and stored-history behavior
AI-019	Approved fallback	Unknown/unqualified model disables affected action tools; no silent privilege expansion
AI-020	Rehearsal autonomy	Coach can be dismissed; no forced lessons, master speech or recording without choice
AI-021	Source permissions preserved	Prohibited download, stem, record or AI operation cannot be enabled by natural language
AI-022	Timely cancellation	Cancelled task cannot later save edits or arm playback; completion race covered
AI-023	Deterministic timing	Locally frame-scheduled plan passes QA-05/23; force failure at each batch-admission step and verify zero runnable partial events, then verify execution-time revision guards
AI-024	Useful product outcome	DJs complete agreed preparation tasks faster or with fewer errors than baseline, measured

Use a versioned evaluation set containing ordinary requests, ambiguous versions, incomplete analysis, unusual phrasing, prompt injections, stale proposals and restricted sources. For personal acceptance, compare the owner's assisted and manual preparation on the same permitted library; record time, errors and failure cases without generalizing one user's result. An optional broader study can include at least eight DJs across both audiences with counterbalanced task order. Recruitment is not a personal-release gate. No utility study has been conducted.

Delivery handoff and remaining decisions

At G1/G2, validate read-only state/context and one proposal path using fixtures; no external model is necessary to test authority boundaries. At G3, integrate useful preparation tools and failure handling. At G4, add cue proposals and bounded transition/automix execution after takeover and timing tests. G5 adds production model/source access, AI budget/account choices and eligible discovery. Its local integration does not require AWS; cloud-specific extension work remains deferred. G6 repeats AI security, privacy and audio-isolation tests on the distribution build. The full release scope remains the user's selected scope; a missing mandatory capability requires an explicit scope decision.

The owner selected ParDay option C and the Galaxy S26 Ultra as the first device. Keep read/propose before each individual plan is armed, preparation actions when requested, and immediate manual takeover. Unresolved choices are the initial model provider, acceptable per-user/month cost, how the owner supplies/stores credentials or runs a gateway, exact device software/accessory evidence, broader

supported profiles, and voice/personality. The assistant name, capability level and local-first delivery order are confirmed; do not ask those questions again. None of these implementation choices blocks design: begin with typed/touch controls, provider-neutral fixtures and the proposed authority states in Document 19, and resolve the paid-provider configuration before connecting it. Acquisition assistance, including the requested unofficial routes, must use the versioned job boundary in Document 16 rather than exposing a general downloader or shell directly to the model.

Sources

Checked 29 October 2026. Product architecture, examples, requirements and numeric limits are original proposals; vendor capabilities require rechecking at implementation.

- **S1:** OpenAI, Function calling.
- **S2:** Anthropic, Tool use with Claude.
- **S3:** OpenAI, Agents API overview.
- **S4:** OpenAI, Data controls.
- **S5:** Anthropic, API and data retention.
- **S6:** OpenAI, Managing billing for ChatGPT and the API platform.
- **S7:** Anthropic, Paid subscriptions and separate API billing.
- **S8:** Spotify, Developer Policy, section III.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 14 OF 19

Music discovery and acquisition

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; local implementation first, with no provider approval or downloader implementation **Product:** New Android-first mobile DJ application; ParDay is the user-named AI assistant **Owners:** Integrations lead, data lead, product owner and security/release lead

1. Decision and user goal

The user wants access to “any and all” music sources and explicitly requests Spotify/YouTube link-to-file acquisition within the app, including unofficial methods exemplified by Spotdown and Y2Mate. This revision retains that concrete mechanism as a required research/build track. A lookup with an external handoff alone does not complete the requested experience. The goal is to find the right recording, obtain usable media, and prepare it for professional manipulation without losing source identity or reliability.

The acquisition entry point is **Download from link**, alongside **Find a playable version** for cross-source discovery and matching. It connects resolution, exact-version selection, available formats, transfer, validation, local import and preparation. A catalog result is not itself a media file. Technical support, service-access basis, content-use basis and distribution eligibility are recorded separately. Document 16 specifies the native queue, extractor/backend alternatives and twenty additional acceptance cases.

Initial use is private practice and personal gatherings; delivery is a personal tool. Build the phone-native local version first with no required server, then consider the optional AWS extension. Optional later GitHub sharing may let others build or self-host their own solutions. Internet acquisition still requires source connectivity, and optional external AI has its own connectivity and cost requirements. That choice does not automatically settle whether a particular event is private under a provider agreement or local law. No file is being downloaded in this planning session. No music-service account has been accessed, no developer application has been approved, and no restriction has been bypassed.

2. Evidence that changes the implementation

Spotify's ordinary Developer Terms prohibit facilitating stream ripping and distinguish conditional offline syncing from permanent copies. Its Developer Policy prohibits ordinary DJ overlap, voice-control assistants and AI ingestion of Spotify Content. Selected DJ partnerships operate under different arrangements; they do not authorize this app. [1-3] A Premium subscription is not treated as a raw-file export permission. The user's requested Spotify-link workflow is now evaluated separately: a catalog identity may lead to an actual media source elsewhere, which must be labelled and assessed on its own evidence.

YouTube's API policies restrict unauthorized downloading and separation of audiovisual components. Its service terms also restrict access/use outside authorized features or the required permissions. YouTube documents official downloads of one's own uploaded videos and Takeout exports; ordinary Premium offline videos are not an MP3 export workflow. [4-7] The app retains that export/import route and now adds a separate extraction-adapter feasibility track. Using an extractor is not falsely described as an official API capability; access and release constraints remain explicit.

A Creative Commons label can grant copyright permissions subject to its terms, but it is not a blanket waiver of the hosting service's access rules. YouTube's own guidance distinguishes available licenses and says it cannot grant another creator's rights. [8] Prefer a creator-supplied original or authorized download location. Confirm that the particular recording and all incorporated material are covered; do not infer permission solely from a search filter or copied description.

2A. Unofficial acquisition evidence

Spotdown advertises Spotify-link downloads and states that it uses third-party services; its actual media origin and delivery have not been tested. Y2Mate advertises MP3/MP4 conversion, but no documented reusable app API was identified in the examined public pages. Their terms and the exact evidence are recorded in Document 16.

A maintained implementation route is technically credible: spotDL explicitly describes finding YouTube matches from Spotify metadata, while yt-dlp and NewPipeExtractor provide extraction tooling with different runtimes and licences. This does not establish Spotdown's internal implementation or authorize every use. The application will own its intake, matching, queue and media-validation contract. Validate a native on-device adapter first; retain the native versus controlled-worker comparison as later AWS expansion work, with no comparative winner assumed.

3. Source admission matrix

The table is a planning snapshot. “Candidate” means an integration to investigate, not access already obtained. Record technical behavior, applicable service/content basis and distribution status for the specific route. SDK agreements apply where that route uses an SDK; a conventional official API is not a universal technical prerequisite.

Source or route	Useful supported direction	Current admission status
User-managed local files	Android file/folder selection, validated copy/reference, local playback and supported DSP.	Core design path; user controls their files and use.
Artist or label direct delivery	Download through an authorized original-file link; preserve provenance and verify the file.	Candidate generic file-transfer adapter with explicit URL/source checks.
Bandcamp purchase	User completes the official purchase/download flow; import an available format. Bandcamp documents MP3, FLAC, WAV and other options. [9]	Candidate guided handoff; no payment automation or scraping assumed.
SoundCloud creator download	Use the uploader-enabled official web download where available; otherwise show the available provider path. [10]	Candidate handoff. Do not promise a universal in-app mobile download API.
Spotify approved DJ stream	Online DJ access through a negotiated integration, with exact operation grants.	Partnership-dependent; no approval exists for this app.
Apple Music approved DJ stream	A distinct partner integration; Apple's named partners demonstrate a route exists. [11]	Partnership-dependent; MusicKit playback alone is not evidence of custom DSP rights.
Tidal DJ	DJ stream; offline and stem capabilities vary by partner. [12]	Candidate negotiation, certification and capability matrix.
Beatport / open-format catalogs	DJ streaming and plan-dependent offline/stem options through supported integrations. [13]	Candidate negotiation; do not assume legacy provider branding or entitlement remains unchanged.
YouTube lookup/extraction	Discovery plus a separately reviewed extractor adapter returning a real media rendition when supported.	Explicit requested capability; feasibility/access/distribution not yet verified. See Document 16.
Spotify-link matching	Resolve catalog identity, then identify the actual available recording/source; preserve both provenance chains.	Explicit requested capability; do not claim matched audio is a direct Spotify master.

Source or route	Useful supported direction	Current admission status
Spotdown/Y2Mate integration	Potential vendor adapter if a usable supported interface and applicable integration arrangement are established.	Public browser workflow observed; reusable API and integration not verified.
Own eligible YouTube upload	User exports through Studio/Takeout, then imports the local video/original.	Supported acquisition concept, subject to actual rights and format support.
Other stores, record pools and providers	Evaluate each official download or integration separately.	Extensible connector interface; no “all catalogs supported” release claim.

Approved streaming caches stay in their provider domain. They are not converted into user-owned local files when a user changes a setting, loses a subscription or asks ParDay to save them. A provider integration can be excellent for practice while remaining unsuitable for the user's requested recording, broadcast or stem workflow.

4. End-to-end user experience

Search. The user types a song, artist, version, mood or existing link. Local results appear first when available. Catalog requests and extraction resolution run through enabled adapters. Return exact source identity and an intelligible result type: local file, download candidate, purchase/handoff, approved stream, discovery only, or unavailable. Do not label a discovery card “ready.”

Choose the recording. Show artist, title, mix/edit, duration, explicit/clean status where known, source and any ambiguity. A radio edit, extended mix, live performance, remaster and cover are distinct choices. A matching title does not establish matching audio. Highlight differences that affect the set; never silently substitute another recording because it is downloadable.

Choose an available route. Offer the working options: use an existing local file, select a resolved download/format, open a purchase/download page, import a file, connect a DJ provider, or save a discovery bookmark. Distinguish a temporarily broken extractor, unresolved match, unsupported format and unavailable access rather than reducing all outcomes to one error. The product should use clear language without making every user complete a legal questionnaire. A direct request for a selected download/batch can authorize that bounded preparation task without repeated per-file confirmation.

Acquire and prepare. Complete a user-approved purchase outside the audio engine where required; download only through an admitted mechanism. Validate bytes, file type, codec, size, hash and source identity. Publish the media record atomically. Schedule analysis without disturbing active decks. Report playback readiness separately from completed waveform, beat-grid or stem preparation.

Load deliberately. The ready item joins the normal protected-deck workflow. A background download completion must not replace the currently playing track. Loading uses stable deck IDs, generations and current capabilities. Keep preparation work recoverable if the user cancels, goes offline, or switches decks.

5. Matching and confidence

Build a deterministic matching service before involving a generative model. Normalize user-supplied strings for searching while preserving original values. Compare source identifiers, credited artists, version strings, duration and independently available recording identifiers. ISRC can help but does not guarantee an identical master, file timing or usable cue positions. Store the evidence behind a candidate match.

Use three proposed states: exact file identity, probable recording match and ambiguous match. Only an exact unchanged media identity can inherit sample-position preparation without validation. A probable match can be suggested; a person confirms before cues, grids or saved loops are associated. An ambiguous match remains unresolved. Do not invent confidence percentages that have not been calibrated on a

labelled fixture set.

For cross-service lookup, use only data permitted for that transfer and purpose. Do not assume that a provider's playlist-export exception permits arbitrary copying into a competing service, fingerprint service or model. A restricted result cannot be laundered by writing it to a local table first. Degrade to the user's independently provided search terms and manual selection when the connector lacks permission for automated matching.

6. Acquisition service contract

Each acquisition job has an ID, originating request, selected candidate, allowed acquisition method, user destination, source/contract version, expected media properties and cancellation token. Secrets remain in the provider adapter, not in URLs exposed to the model. The job result is a validated MediaAsset or an actionable failure, never an unverified URL labelled playable.

Supported method categories include local file selection, direct-file transfer, documented vendor conversion API, maintained extractor, external purchase/download handoff, creator export import, approved streaming session and approved offline cache. The first implementation evaluates on-device extractor execution; a controlled worker is a later optional extension; the method is not described as a direct Spotify export unless that origin is actually established. The adapter declares whether it actually returns a portable file or only a provider-controlled playback handle. These result types cannot be coerced into each other.

For an authorized direct-file transfer, constrain redirects, protocol, destination and resource size; inspect response headers and actual bytes; reject HTML login/error pages masquerading as audio. If a backend fetcher is used, prevent access to private network addresses and arbitrary credentials. Resume only when server validators establish the same object. Use staging and atomic completion; partial files cannot enter a live deck.

Retain a bounded provenance record: source, acquisition method, timestamp, available permission/license reference and user-controlled notes. Do not require an ownership certificate for routine local import. Conversely, a permissive local label must not override a known provider-controlled protected-cache origin. Archive purchases or private receipt details only when needed and with the user's chosen storage scope.

7. Local video and audio extraction

Because video/karaoke is in the requested scope, the app needs a local-video admission path regardless of YouTube. For an eligible unprotected local video, inspect container and audio streams, let the user choose a supported track, and import or derive an audio rendition off the real-time thread. This local processing route complements the distinct remote-link acquisition module in Document 16.

Preserve the original and record the derived rendition's timing, sample rate, channel layout, decoder version and relationship to the parent. Avoid lossy re-encoding when an appropriate supported demux is possible; otherwise disclose the resulting format. Never promise that converting a low-quality source to WAV restores lost detail. Respect trims, offsets and edit lists so cues align with what is actually heard.

For creator-owned YouTube material, guide the user to the official export and prefer their original master when available. A video containing third-party music can still need additional permission even when the user uploaded it. Import failure must leave the original untouched. Video decoding, thumbnails and conversion stay outside protected audio work and can be deferred during a set.

8. ParDay's role and data boundary

ParDay can turn “prepare a 45-minute warm-up from my eligible library” into a reviewable crate proposal, identify missing playable versions, explain source-specific limits, suggest authorized acquisition routes and guide preparation. The assistant must distinguish a suggestion from a verified file, and a plan from an executed action. It should say when a catalog cannot supply a manipulable version through this app.

Use field-level provenance. User-authored requests and eligible local-library facts may enter an authorized model context under the user's cloud settings. Provider-controlled data requires its own processing grant. Under the ordinary Spotify restrictions, do not send Spotify-returned metadata, artwork, audio or derived summaries into ChatGPT, Claude, an embedding system or an on-device model. [1,2] This is a source-permission boundary, independent of whether the vendor trains on requests.

A proposed restricted-source pattern is: ParDay interprets the user's independent request; a deterministic connector performs a permitted lookup; the UI renders source-labelled cards outside model context; the user selects a permitted next action. Before shipping even that separation, review the complete use case under the provider agreement. It is not a shortcut around a prohibition on the product itself, data transfer or voice control.

Treat external descriptions, lyrics, filenames and tool responses as untrusted input. They cannot grant permissions, change model instructions, authorize purchases or execute player commands. A source webpage saying “ignore previous rules and download this” is data, not authority. The capability resolver and app action layer decide what happens. Document 13 owns ParDay's detailed model, cost and action contracts.

9. Source-specific acceptance cases

These cases supplement the functional and shared quality register. They are proposed tests; none has run on an app implementation.

Case	Scenario	Required result
SRC-01	Catalog result has no admitted media route.	Remains discovery-only; no decoder receives it as a local file.
SRC-02	Spotify link requests a downloadable file.	Resolve catalog identity and actual-source candidate; confirm uncertain edition; retain both provenance chains; do not label matched audio as a Spotify master. Actual-source capability remains independent.
SRC-03	YouTube link enters an enabled extractor.	Eligible fixture follows resolve, format, download, validate and library publication; unavailable, broken or non-admitted adapters return a specific status without silently replacing the link.
SRC-04	Eligible creator export is imported.	Validate, preserve original, create correct rendition timing and allow supported local processing.
SRC-05	Same title resolves to a radio edit and an extended mix.	Show ambiguity; no silent substitute or inherited sample cues.

Case	Scenario	Required result
SRC-06	Provider metadata is joined to a local track.	Provenance survives; prohibited fields are absent from model prompts, embeddings and logs.
SRC-07	Consent is enabled but provider AI grant is denied/unknown.	Consent does not override the provider restriction.
SRC-08	Provider license, token or cache authorization expires.	Revalidate affected operations; preserve unrelated local media and user preparation.
SRC-09	Transfer is cancelled, interrupted or serves an HTML error.	No ready media artifact; clean recoverable staging and clear retry path.
SRC-10	Download finishes after deck reassignment.	Add readiness to the library; never replace the audible deck automatically.
SRC-11	Recording/broadcast is active and next source is ineligible.	Resolve before transition; do not silently capture restricted audio.
SRC-12	Retrieved text contains malicious instructions.	No secret disclosure, purchase, permission change or playback action.

10. Delivery and unresolved dependencies

Implement the local import/provenance foundation, native download queue and matching fixtures, then run Document 16's on-device extractor feasibility work. Preserve measurements and fixtures for a later D-versus-AWS comparison. Cloud deployment, service pairing and cloud-only acceptance cases do not block local design or the first local release. Add source adapters with independent technical, access/content and distribution evidence. This sequence is an engineering dependency, not a decision to remove desired sources from the final product. If a must-have provider declines access, the product owner receives a concrete gap, alternatives and schedule impact.

For API/SDK integrations, obtain the exact API/SDK access, territory coverage, account tiers, concurrency, PCM/DSP permissions, offline rules, metadata handling, recording/stems/video/broadcast limits, AI/voice restrictions, security obligations, branding and exit behavior. Test those facts in the integration, including account and track revocation. Store the evidence version with capability policy. Extractor routes instead require review of their actual interface/runtime dependencies, media origin, applicable service/content basis, transfer behavior and update/rollback path; SDK access is not a prerequisite for every method.

The first design prioritizes local import plus the explicitly requested Spotify/YouTube link journey. Source territories, actual adapter coverage, purchase handoff, optional matching services and exact runtime dependencies are implementation investigations rather than reasons to restart product discovery. The latest owner decision selects on-device local delivery first and defers the neutral D-versus-AWS comparison; no comparative winner or universal source support is established. Deliver **Download from link** and **Find a playable version** as connected capabilities with actual source, quality and operational status visible. Crystal Current, the Galaxy S26 Ultra, armed ParDay assistance and personal-tool distribution are confirmed. Optional source publication details remain open under Document 17. No local host is selected or required. AWS is accepted for later expansion with an approximate USD 35 monthly planning target excluding ParDay AI-provider charges; account, region, service, instance and an actual deployment decision remain open under Document 18. These future choices do not block local design. Hosting on AWS does not establish a working acquisition path.

Sources checked 29 October 2026

1. Spotify Developer Policy.
2. Spotify Developer Terms.
3. Spotify selected DJ integration support.
4. YouTube API Developer Policies.
5. YouTube Terms of Service, applicable territory/version to be confirmed.
6. Download one's uploaded YouTube videos.
7. YouTube offline video FAQ.
8. YouTube license types.
9. Bandcamp download formats.
10. SoundCloud uploader-enabled downloads.
11. DJ with Apple Music.
12. Tidal DJ capabilities.
13. Beatport Streaming.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 15 OF 19

Video, karaoke, broadcast and Android plugin hosting

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; selected product scope with unverified implementation contracts and certification targets **Owners:** Audio lead, Android media lead, integrations lead, security lead and device QA lead

1. Scope and completion rule

The user selected every enumerated professional module except lighting. Video mixing, karaoke, broadcasting and actual third-party plugin hosting therefore belong in the target complete personal release. They cannot be silently deferred or replaced by decorative controls. Lighting/DMX is excluded. External DJ-controller integration remains excluded; internal events used by an instrument plugin do not automatically create a controller-integration requirement.

The delivery order is now local Android first, with AWS expansion later. These selected modules are designed around the phone's engine and measured local resources, with no required AWS worker or home-lab server. Broadcasting still needs an explicitly chosen network destination; optional AI and music acquisition can need connectivity. Those independent network features do not turn cloud audio processing or a self-hosted service into a prerequisite. The future AWS planning target of about US$35/month excludes ParDay AI-provider charges and makes no promise that a cloud worker can solve an unqualified local video/plugin workload.

Android is the first platform, and Galaxy S26 Ultra is the confirmed first test device. Its software is reported up to date; the exact installed build and accessory route remain uninspected. This selection does not certify any concurrent workload. C Crystal Current supplies the visual family for all specialist workspaces. USB audio interfaces and splitters are permitted. Private practice and personal gatherings are the initial use context, but starting an Internet broadcast changes the audience and source permissions. The broadcaster must resolve those permissions separately. A playable local video or streamed song is not automatically cleared for lyrics display, recording, synchronization or broadcast; Documents 07, 11 and 14 own the source policy.

This document defines contracts before choosing exact video containers/codecs, karaoke formats, broadcast protocols or plugin formats. The targets below are proposed acceptance thresholds, not measured capabilities. A device profile can limit concurrent workloads only when that limitation is explicit. If no supported configuration can satisfy a selected feature, return the gap to the user before changing the completion promise.

2. Shared timing and resource architecture

The native audio engine's compensated sample clock remains authoritative. A media-time mapper supplies each deck's source position, transport generation, tempo map, seek epoch and presentation target. Video, timed lyrics, network timestamps and plugin automation consume that mapping. UI animation, a decoder's wall clock or a network receiver must not change musical tempo to make pictures look smoother.

Use bounded queues between demux, audio decoding, video decoding, composition, encoding and network transfer. No codec configuration, GPU wait, Binder transaction, plugin discovery, network operation or lyrics parsing occurs inside the device callback. Declare memory budgets and backpressure behavior for every queue. A slow video or network branch must not block master or cue audio.

Android's MediaCodec documentation recommends Surface-based raw video paths for efficiency, while supported codec behavior depends on platform and device capabilities. [1,2] Treat those APIs as building blocks. A successful single-video demonstration does not certify simultaneous decoding, compositing, recording and stem separation on a hot phone.

3. Video mixing contract

Begin with authorized unprotected local video and independently admitted provider video. Store audio and video track references against the same media identity with offsets, duration, rotation, aspect ratio, color characteristics and variable-frame-rate timestamps. Reject or explain unsupported streams; do not mislabel a file playable because its extension is familiar.

The proposed Android path is off-thread demux plus bounded MediaCodec decoders into Surfaces, a GPU compositor, and separate preview/program outputs. Decode only the admitted number of simultaneously needed video streams. A four-audio-deck layout does not imply four full-resolution video decoders can always run. Cache thumbnails separately and release unused decoder resources outside an active transition.

Each deck selects video-follow-audio, independent visual content, or audio-only. In linked mode, video follows the engine's source position and tempo; key changes alone do not change picture speed. A loop repeats the corresponding video interval. On a seek, invalidate old frames and resume the new epoch; never flash stale imagery from the previous track. Exact reverse/scratch video is a separate certification case because inter-frame compression may prevent immediate arbitrary-frame decoding. Until certified, freeze or replace that deck's visual according to a visible selected policy while preserving audio scratch behavior.

The program compositor supports a defined transition set, beginning with cut, dissolve and linked/unlinked video crossfade. Keep visual crossfader state separate from the audio crossfader when unlinked. Audio-only tracks use a chosen licensed still, restrained app visual or black; no invented album cover or unrelated video is silently substituted. Unsupported HDR-to-SDR conversion must be rejected or explicitly handled rather than producing incorrect washed-out output.

External display is a capability, not a guarantee for every USB-C port. Android Presentation supplies a secondary-display mechanism; actual hardware, display route and lifecycle must be tested. [3] Show a clean program surface while controls remain on the device. An external display disconnect must not expose private UI, cue controls, stream keys or the singer's private notes. A reconnect requires a valid display configuration. Wireless display latency needs its own profile and must not be advertised as equivalent to a verified wired route.

4. Karaoke and microphone workflow

Karaoke is a real performance mode with a singer queue, selected song/version, key/tempo settings, lyric/video presentation, microphone routing and an operator-controlled next singer. Keep stage names and queue notes local by default. Reordering or removing a waiting singer must not replace the active song. Include a clear pause, skip and return-to-DJ flow with recoverable queue state.

Admit licensed local video karaoke, an approved timed-text format, and a CD+G route only after parser, compatibility and rights review. WebVTT supplies a standardized timed-text basis; paired audio/CDG and archive variants exist in established open-source karaoke software, but borrowing that implementation requires its own license review. [4,5] The exact shipped subset remains a feasibility decision. Do not promise every LRC dialect, proprietary karaoke package or legacy disc format.

Store lyric timing in source-media time so the same mapper follows tempo, seeks and loops. Preserve line breaks, readable contrast, language shaping and safe display margins. Word highlighting requires word-level timing; do not fabricate precision from a line-timed file. Provide an adjustable measured display offset without changing the singer's audio. No automatic web lyric scraping or generated “official lyrics” is required. User-authored text and separately licensed lyrics are valid inputs.

Microphone features include selected input/channel, gain, meter, mute, monitoring choice, optional EQ/compression and a controlled mix/broadcast assignment. Make feedback risk understandable at routing time. An automatic ducking option uses an explicit threshold/envelope; manual talk-over remains available.

Do not route microphone audio into ParDay merely because the karaoke microphone is enabled. If a headset/USB device changes, apply the audio route-loss rules before enabling monitoring again.

5. Broadcast architecture and operation

Broadcasting consumes an explicitly selected eligible program bus, normally master plus chosen microphone/video. Cue audio and private assistant speech are excluded by default. Before Start, show endpoint/account, audience or destination, selected buses, source eligibility and selected quality profile. Starting/stopping is a deliberate user action; ParDay can prepare a proposal but cannot silently begin transmission.

Use a bounded audio/video capture queue, dedicated encoder workers, mux/protocol adapter and independent network queue. Media timestamps derive from the engine clock and declared video mapping. The encoder can report delay without feeding timing corrections into the live mix. Choose a supported protocol after target-service testing: RTMP/RTMPS or enhanced RTMP and SRT are candidates with published projects/specifications, not a blanket compatibility promise. [6,7] Container, codec, encryption, authentication and receiver support are separate decisions.

Expose connection state, encoded rate, queue delay, dropped frames, reconnect attempts and whether the destination acknowledged reception. “Connected” does not prove the audience is hearing the correct mix. Provide a test destination/receiver check before a real broadcast. Store stream keys securely and redact them from URLs, logs, screenshots and diagnostic exports. Never send them to a language model.

On congestion, use a declared bounded policy: reduce video bitrate or frames within the selected profile, reconnect, or stop transmission with an alert. Do not accumulate unlimited latency, silently send a delayed entire set, or stall local audio. Preserve temporal continuity when dropping video; request a clean keyframe/resynchronization where the format requires it. Network switching and authentication failures must be tested.

Local fallback recording is a separate user-selected, source-permitted option. It must already have storage and recording permission, and cannot be enabled merely because the stream failed. Save recoverable segments and mark broadcast gaps honestly. Stopping broadcast releases transmission promptly without muting the room mix or discarding a valid independent recording.

6. Actual Android third-party plugin hosting

Built-in effects are part of the engine, but they do not fulfill this requirement. Completion requires a documented host interface and at least one independently authored compatible plugin operating through it; the proposed validation corpus includes an effect and an instrument. Users need discovery, eligibility, load/unload, parameters, presets, state recall, automation, bypass and clear compatibility information.

Desktop VST/AU binaries must not be assumed compatible with Android CPU, operating system or packaging. The current VST3 SDK's supported-platform table does not list Android. AAP, Audio Plugins For Android, is a concrete Android-oriented candidate using separate host/plugin applications and shared-memory/IPC mechanisms; its own project warns that its APIs are immature. [8,9] Benchmark an exact pinned version rather than treating a repository's existence as production readiness.

Compare two candidates at the feasibility gate: an AAP adapter with independently packaged compatible plugin applications, and a small documented app-native plugin interface with curated third-party modules delivered through the app's approved package/update channel. The second option can support real third-party authorship, but only if independently authored modules use that interface; renaming internal effects as plugins does not count. A closed compatibility list is acceptable only when clearly stated and accepted as the supported host scope.

The plugin contract declares ID/vendor/version, ABI and CPU architecture, inputs/outputs, sample formats, supported rates/block sizes, parameter IDs, state schema, maximum memory, measured CPU budget,

latency, tail length and required capabilities. Prepare/load/state parsing happens off the callback. Processing must meet a fixed deadline; latency changes require negotiated graph reconstruction outside active use. Presets cannot contain executable code or arbitrary filesystem/network instructions.

Prefer a process boundary for untrusted third-party code. An asynchronous worker/shared-memory design must supply ready blocks at a fixed declared latency; the audio callback never waits for IPC. Measure round-trip delay, jitter, crash behavior and CPU cost. An in-process trusted plugin runs with the host's privileges and can crash it; do not label that arrangement sandboxed. State clearly which isolation model each admitted class uses.

Define fault behavior before activation: latency-compensated dry bypass for effects where valid, or a short controlled instrument mute; alert the user and preserve unaffected decks. A plugin timeout must not trigger endless repetition of a stale block. Session restore checks plugin identity/state version and offers an unavailable-plugin placeholder rather than destroying the saved setup. Missing plugins cannot silently become a different effect.

Google Play prohibits downloading executable dex/JAR/native .so code from sources outside its update mechanism. Code signing alone does not remove that restriction. [10,11] For a Play release, use packaged/delivered approved modules or separately installed compatible apps only after reviewing the actual distribution design. No arbitrary remote-native-plugin marketplace is assumed. Personal non-Play distribution is now the selected milestone. That removes public-store submission from the immediate release gates while preserving installation, signing, component-licence and isolation requirements. Any later Play release needs a new distribution review; Document 17 covers optional community-source sharing. Prefer compatible open source, while checking exact code, model, SDK, trademark and redistribution obligations independently.

7. Testable module requirements

Each row has a stable ID for traceability into the global register. All are proposed acceptance contracts within the selected feature scope.

ID	Required behavior	Acceptance evidence
AV-001	Video follows compensated audio media time.	Measured flash/click fixtures across tempo, seek and loop; no accumulating drift.
AV-002	Video sources and generations remain distinct.	Rapid load/cancel/seek never displays frames from an obsolete assignment.
AV-003	Deck video modes and program transitions work.	Linked/unlinked transitions, audio-only fallback and unsupported stream tests.
AV-004	External program output is separate and recoverable.	Wired display attach/remove/reconnect; no private UI or cue leakage.
AV-005	Karaoke queue and timed presentation remain coherent.	Singer reorder, skip, pause, transpose, tempo and session restore fixtures.
AV-006	Mic, ducking and bus assignment are deliberate.	Physical input/monitor test; route loss, clipping and consent checks.
AV-007	Broadcast starts/stops only by authorized action.	Correct endpoint/bus preview; cancellation and unauthorized action rejected.

ID	Required behavior	Acceptance evidence
AV-008	Network/encoder failures cannot block local audio.	Loss, congestion, reconnect, wrong key and worker crash during warm playback.
AV-009	Broadcast/recording enforce source restrictions.	Ineligible next track resolved before transmission; partial output labelled.
AV-010	A real independently authored plugin is hosted.	Compatible effect/instrument corpus loads, processes, automates and recalls state.
AV-011	Plugin faults and latency are bounded.	Timeout/crash/state corruption/latency-change tests preserve unaffected audio.
AV-012	Plugin distribution and isolation claims are accurate.	Package provenance, license, store-policy and process-boundary review.
AV-013	Concurrent module promises match measured profiles.	Two-hour repeated warm runs and actual receiver/display observations.
AV-014	Advanced work never enters the device callback.	Instrumented allocation/block/I/O/IPC checks and deadline traces.

8. Device workloads and release gates

Start with these candidate profiles, then publish the exact tested device, OS, audio route, codecs, resolutions, plugin builds, model and accessory configuration. Proposed video timing target: absolute picture/audio offset within 80ms for 95% of observations and no growing drift over 30 minutes; calibrate wired and wireless displays independently. This is a measurable initial target, not an established professional standard or a guarantee.

Profile	Candidate sustained workload	Required evidence
AV-P0	Two audio decks, one 720p30 video, timed lyrics, mic and external-display candidate.	Correct routing/timing; declared decoder and display limits.
AV-P1	Two simultaneously decoded 720p30 deck videos, GPU crossfade, mic and permitted recording.	Two-hour warm playback plus seek/transition stress.
AV-P2	Four audio decks, declared video-deck count, two admitted plugins, sampler/FX and broadcast.	End-to-end receiver capture, loss recovery and plugin failure isolation.
AV-P3	Selected stems profile plus video/broadcast/plugins and ParDay activity.	Combined thermal/memory/callback evidence; independently passing modules is insufficient.

Record queue ceilings, callback deadlines, memory high-water marks, video loss, thermal state, total plugin latency, broadcast delay and battery/power conditions. Before enabling a workload, check admission against the current device profile. Resource pressure can pause optional preparation and simplify decorative visuals. It must not silently alter active audio quality, timing, routing or stem alignment. A profile that cannot sustain its promised combined workload fails certification.

G1 adds focused codec/display/plugin feasibility; G2 establishes the core audio slice and clock interfaces; dedicated working slices for these selected modules must precede their G4 completion. G4 expands the full controls and workloads; G6 requires AV-001-014 evidence, asset/source rights and failure recovery. G7 complete personal-release acceptance includes all selected modules; it does not require the later optional AWS extension. If evidence supports only a smaller release, present the exact missing capabilities and obtain a product decision. No such reduction has been approved here.

Sources checked 29 October 2026

1. Android MediaCodec.
2. Android supported media formats.
3. Android Presentation.
4. W3C WebVTT.
5. OpenKJ source and format documentation, implementation license requires independent review.
6. Veovera enhanced RTMP specification project.
7. Haivision SRT project.
8. Steinberg VST3 SDK supported platforms.
9. AAP Android audio plugin framework, exact version/stability to be evaluated.
10. Google Play Device and Network Abuse policy.
11. Android dynamic-code-loading risks.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 16 OF 19

Link acquisition implementation and feasibility

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; local acquisition first, deferred AWS comparison, no implemented downloader **Selected baseline:** Crystal Current (C); Samsung Galaxy S26 Ultra on user-reported current software; ParDay with explicitly armed assisted performance **Owners:** Integrations, Android, Library/Data, Audio, Security/Release and Quality **Companions:** Documents 05, 07, 09, 11, 13 and 14; acquisition_job.schema.json; acquisition_state_machine.json; link_acquisition_acceptance_register.json

1. Product decision

The owner explicitly wants an integrated Spotify/YouTube link-to-file workflow comparable to the experience on Spotdown and Y2Mate. Record this as an acquisition requirement, including investigation of unofficial extraction. A catalog search followed only by an external browser handoff does not complete this requirement. Do not treat the absence of an official download API as proof that the mechanism is technically impossible.

The product should let a DJ share or paste a supported link, resolve its recording, see the actual available media and format, download it inside the app, validate it, and prepare it for the normal decks. The final file must be independently identifiable, playable and recoverable. A web page that advertises conversion does not establish a production API, integration permission, actual recording identity, audio fidelity or sustained reliability.

Four questions must remain separate: does the mechanism work; what source/service access basis applies; what content use is covered; and which distribution channel can accept the resulting product. Unofficial is a mechanism classification, not an automatic answer to all four questions. A permission or release concern is not a reason to erase the requested feature from the backlog. An unknown or denied operation also must not be falsely labelled approved.

No music was downloaded, no extractor executed, no account connected, and no hidden endpoint or access control probed during this planning revision. The evidence below establishes candidate approaches and open questions. It does not establish that either named website has supplied a reusable backend to this project.

2. What the named services establish

Candidate	Observed evidence	Consequence for this app
Spotdown	The supplied page advertises Spotify-link and collection downloads, including MP3. Its footer states that it uses third-party services. Actual media origin and fidelity were not verified. [A1]	Keep as a vendor candidate; do not assume direct Spotify master extraction or repeat quality claims as measured results.
Spotdown integration	The examined terms describe personal/noncommercial use. No documented developer interface or app-integration grant was identified in the examined pages. [A2]	A private API may exist, but it is not established. Obtain documentation before coupling an adapter to the service.
Y2Mate	The site advertises browser conversion to MP3/MP4. [A3] Its terms require written permission for framing, cloning, resale or commercial exploitation. [A4]	A WebView wrapper is not the recommended integration. A documented vendor arrangement would be a separate candidate.
spotDL	Its README describes matching Spotify metadata to YouTube media, then downloading that media. The project uses MIT licensing and depends on other tools. [A5]	Evidence for catalog-to-media matching, not proof that Spotdown uses the same implementation or that matching always finds the correct edit.
yt-dlp	A maintained programmatic extractor candidate. Current documentation lists Python 3.11+, FFmpeg-related tools and JavaScript-runtime dependencies for full functionality; distributed artifacts can have different licence obligations. [A6]	Evaluate the controlled-worker candidate alongside the native candidate. Pin complete dependencies; do not assume an old Android wrapper is compatible.
NewPipeExtractor	An independently usable Gradle library listing YouTube and other services, licensed GPL-3.0-or-later. [A7]	Evaluate an on-device route if its licence fits the intended distribution. Measure integration, size and runtime behavior.

The source websites' statements are vendor claims. Retrieval of their public pages was successful; conversion, delivery, API availability and audio inspection were not tested. No conclusion is drawn that either site is malicious, safe, licensed for this app, or permanently available.

3. Recommended architecture and execution location

Use an app-owned acquisition module with replaceable resolver and transfer adapters. The Kotlin interface, database job state, file validation and library publication remain ours. An adapter can use a direct download, a documented vendor API, an admitted extraction component, or an external handoff. These methods report distinct capabilities rather than pretending to be equivalent.

The latest owner decision selects the local version first. Build the native intake, library and job infrastructure and evaluate the on-device extraction path without requiring a LAN or cloud server. Preserve a replaceable adapter boundary and reusable fixture/measurement records. The previously selected on-device versus owner-operated AWS comparison moves to a later optional expansion, with an approximate USD 35 monthly AWS planning target excluding ParDay AI-provider charges. This sequence does not claim that the on-device extractor already works or wins on performance. An owner-operated worker may simplify dependency updates and isolate conversion from the phone; its benefit remains a hypothesis for section 13's deferred comparison. Testing both later does not commit the product to maintaining both, and neither cloud deployment nor paired timing results block design or the first local release.

Approach	Strength	Cost or uncertainty to resolve
On-device extraction	Local control; no app-operated media relay needed for that path	Android packaging, dependency licence, updates, battery, heat, compatibility and access conditions
Controlled backend	Independently deployable adapters; isolated conversion and easier rollback	Hosting/egress cost, retention, access reliability, endpoint security and availability
Documented vendor API	Less extraction maintenance if a suitable service exists	Contract, pricing, response semantics, media provenance, rate limits and service continuity
Browser plus import	Useful recovery option while an adapter is unavailable	Does not by itself meet the selected integrated acquisition workflow

Proposed components are LinkIntake, CatalogResolver, RecordingMatcher, SourceAdapter, FormatSelector, TransferCoordinator, MediaValidator, LibraryPublisher and JobJournal. The live audio engine consumes only published library media through its existing loader. It never invokes a website, extractor, FFmpeg process, database migration or language model.

A backend-resolved media URL may expire or be tied to its issuing session or network. Do not assume that a URL resolved on a server can be downloaded directly by the phone. The adapter declares a transfer mode: device direct, controlled relay, or server artifact followed by authenticated download. Validate the intended mode with the actual source. Relay and artifact modes need explicit storage limits, deletion and cost accounting.

4. Crystal Current acquisition workflow

Place an Add music action in the library with Paste link, Receive share, Import files and Download queue. Android share intake should resolve user-selected links without silently polling the clipboard. The selected C theme supplies pale glass, titanium and restrained brass; critical progress and error text uses stable high-contrast surfaces.

The resolution screen shows requested title/artist/version, the actual source, source duration, known edit or explicit/clean information, and a concise match explanation. A Spotify reference can resolve to a different platform's candidate. That candidate must carry its actual origin. An uncertain match offers alternatives and a preview or selection step; a matching title alone cannot approve a substitute.

Format choice uses readable options such as Keep source format, MP3 for compatibility, or a supported video/audio rendition. Display actual source codec, known bitrate or an honest unknown value, output encoding and estimated storage. A high output bitrate is not presented as proof of a high-quality original. Artwork and metadata are retained only as their source allows; generic placeholders remain usable.

After Download, show a persistent queue entry with phase, byte progress when measurable, storage destination, pause availability, Cancel and an actionable failure reason. Conversion and analysis are separate phases. An unknown total must not produce a fabricated percentage. Ready to play and fully analyzed are separate states. Finishing a download adds to the library; loading or starting a deck uses the established protected workflow.

For collections, show a bounded selection and per-item outcomes. Preserve playlist order independently from transfer completion order. A batch containing failures is partially completed, never simply Done. Avoid an accidental unbounded artist-discography job; show the enumerated count and let the user select the set before transfer. Do not implement deletion of existing local tracks as an implicit playlist-sync behavior.

5. Catalog identity, actual media and sound quality

Every job retains two identities: what the user requested and what the adapter actually obtained. Record requested provider/link ID, actual media provider/ID, candidate revision, edition evidence, duration, media hash, acquisition method and extractor version. Link the resulting MediaAsset to both provenance chains. A supplied catalog title must not overwrite contradictory file evidence.

Use deterministic comparison and user selection before involving ParDay. Title/artist normalization, version labels, durations and permitted recording identifiers can rank candidates. Confidence percentages require calibration; otherwise use exact file, probable recording or ambiguous. A content hash identifies bytes, not the quality or correctness of the performance. A different edit or encode must not inherit sample-position cues without validation.

Keep the source file where practical. Remux without re-encoding when supported and useful. Offer transcoding only when the user needs a different format or the app cannot decode the original. MP3, AAC, Opus, WAV and FLAC are format candidates subject to actual encoder/decoder and licence support; list only options the selected adapter can produce. Converting an already lossy source to WAV/FLAC preserves that decoded signal and does not recover discarded detail. A 320 kbps MP3 output can still derive from a lower-quality source.

Record source and output codec, sample rate, channels, duration, byte size and processing history. Distinguish bitrate measured from a file, bitrate reported by a service, variable bitrate and unknown values. Analyze loudness and clipping as preparation information; do not normalize or alter the preserved original silently. Re-encoding may change leading delay and padding, so cues bind to the validated actual rendition.

6. Job contract and state machine

The companion JSON schema describes one network-acquisition child-job snapshot, not an API secret store, collection aggregate or model tool. A trusted app request creates the job and intent reference. A collection has a separate persisted parent aggregate with ordered child IDs and per-item outcomes; parentJobId links to it. External handoffs and returned local files use the existing handoff/import path rather than pretending to be network transfers in this schema.

Resolution may populate candidate choices without mutating the accepted execution request. Accepting or changing a candidate/format creates a new immutable execution revision with a fresh digest and idempotency key, invalidating earlier worker leases. JobId and the originating intent can remain stable across revisions. Retries of the same unchanged revision reuse its key; an old key never authorizes the new selection.

Normal states are CREATED, RESOLVING, AWAITING_SELECTION, READY, QUEUED, TRANSFERRING, optional SOURCE_VALIDATING/CONVERTING, final VERIFYING, PUBLISHING and COMPLETED. READY means a candidate/format is ready to acquire, not that audio exists locally. PAUSED, RETRY_WAIT, CANCEL_REQUESTED, CANCELLED, FAILED and BLOCKED are explicit alternate states. The normative transition map is contracts/acquisition_state_machine.json, with guards described here; a model cannot assign states.

A preserved-source path is TRANSFERRING to VERIFYING to PUBLISHING. A conversion/remux path is TRANSFERRING to SOURCE_VALIDATING to CONVERTING to VERIFYING to PUBLISHING. Final VERIFYING always checks the exact final rendition that will be published. Validation of downloaded source bytes cannot authorize different converted bytes. PAUSED/RETRY_WAIT resume only at a documented safe checkpoint with current identity, lease and admission; cancelled and completed executions never restart themselves.

Cancellation is cooperative inside the acquisition subsystem. Increment a cancellation revision transactionally, stop future stages, close active transfers, and prevent publication by stale workers. Publication uses a database transaction/compare-and-swap fence against the current request revision, cancellation revision, eligible state and active worker lease. The fence, MediaAsset insertion, idempotency

result, publication record and terminal COMPLETED job state commit together. A cancellation transaction that wins first prevents publication; a publication transaction that wins first makes a later Cancel report already completed, even if the UI has not seen the acknowledgement. Removal after publication is a separate user action. A check before opening the transaction is insufficient.

Use a stable idempotency key for each accepted immutable execution revision. Repeating the same key/payload returns that revision's existing outcome; changed payload with the same key is rejected. Only one active worker lease owns a job revision. Parent collections have child jobs; completed children remain available if another item fails. Retry policy is bounded, distinguishes permanent from transient failures, respects service rate limits and never silently changes the chosen recording.

7. Transfer, validation and publication

Resolve and transfer in private staging. Sanitize display filenames and generate internal paths independently from titles or remote filenames. Set byte, time, duration, redirect and concurrency limits before fetching; stop oversized or unexpectedly endless responses. Validate status, headers and actual media bytes. A .mp3 extension or audio MIME type is not sufficient. HTML error pages, executable content, truncated headers, unsupported codecs and invalid duration fail with useful reasons.

Resume only against a verified identical object. Store a stable object identity, server validator where available, bytes held and the expected range. Check 206 and Content-Range for a continuation; an unexpected full response restarts safely rather than appending duplicate bytes. If an expiring URL is re-resolved, revalidate the media identity and format before attaching it to existing bytes. Session/IP binding, redirect changes and unavailable range support may require a clean retry.

Inspect media in a bounded worker; pass controlled local paths to decoding/probing tools, not an arbitrary network URL supplied by a web page. Validate complete media as appropriate to the selected container and codec. Keep failures quarantined or remove staging according to the job's retention policy. The assistant must not label a file playable based only on a successful HTTP response.

Filesystem publication and a database transaction are not one universal atomic operation. Use a staged manifest and recovery journal: validate, place the final object in managed storage, transactionally publish the MediaAsset, idempotency result and terminal job state under the revision/lease fence. Reconcile a crash between these steps, remove only proven orphan temporary objects, and never delete an existing file referenced by a deck or another job. External export through Android document access is a later, separately tracked operation where atomic rename may be unavailable. Before PUBLISHING, require passed final-rendition validation. Publication remains null until the winning completed transaction. Its hash, committed request/cancellation revisions and object identity must match the validated final object and the fenced execution; these are runtime comparisons beyond JSON Schema.

8. Android lifecycle and S26 Ultra validation

The owner selected Galaxy S26 Ultra and reported it up to date. Record the observed Android/One UI build, variant, RAM, storage, battery mode and audio route when testing begins. Do not infer them from a marketing name or treat the word latest as a reproducible build identifier. No S26 Ultra measurements have been performed in this planning session.

Android's user-initiated data transfer jobs are a candidate for explicit longer downloads on supported versions. Android documents an important distinction: its Task Manager Stop can terminate the entire app process and other foreground services, while an app-provided cancellation action can stop only our acquisition work. Persist progress incrementally and provide a clear Cancel download action. Do not promise audio survives OS-level Stop or Force stop. [A8]

For deferrable preparation, evaluate WorkManager separately from user-initiated transfers. Current Android guidance states that long-running WorkManager workers can exhaust job quota on Android 16; a

foreground service is not an unlimited-work exemption. Select scheduling from the actual task and OS behavior, then validate interruption and restart. [A9]

Download, extraction, conversion, hashing and analysis operate outside the real-time audio callback. Begin with one active media transfer and one bounded preparation task as a proposed benchmark baseline. During performance, defer expensive conversion or stem preparation when the certified budget requires it. Preserve active audio and already-running live stems; pause optional acquisition work before allowing it to compromise the set. These are proposed policies to measure, not established throughput claims.

Test Wi-Fi/mobile transitions, locked screen, low storage, low battery, thermal limits, memory pressure, loss of server response and USB route changes. The in-app Cancel test must preserve the mix. The OS Stop test verifies truthful interruption/recovery, not impossible uninterrupted playback after the application is terminated.

9. Security, privacy and ParDay

The native client accepts supported link types and passes canonical identifiers to adapters. A backend fetcher allows only its defined provider and delivery hosts, validates redirects and DNS/IP destinations on every connection, and rejects loopback, private, link-local and metadata-service targets. Prevent DNS rebinding and alternate IP encodings; use network egress controls as well as URL parsing. Local network media, if later requested, is a separate explicit source with separate scope.

Extractor workers run with least privilege, bounded CPU/memory/time, isolated temporary directories and no access to application secrets. Use structured arguments or library calls rather than interpolated shell text. The caller cannot select arbitrary command flags, executable paths, output paths, plugins or postprocessors. Restrict probe/transcode network access. Adapter configuration is versioned, reviewed and reversible; remote configuration can disable an adapter but does not become arbitrary downloadable executable code.

Do not ask users to give music-service passwords, browser cookie stores or private signed links to an unknown converter. A future authenticated adapter needs its own explicit account and data-handling design. User-selected private media links remain sensitive: redact token/query values from logs and model context. Backend artifacts need an expiry, access control and deletion schedule; do not create an indefinitely shared music cache by accident.

ParDay may help choose an eligible recording, request a bounded acquisition job, report its state and prepare a crate after validated import. The deterministic service performs source resolution, transfer and publication. The language model receives job IDs and permitted summaries, never arbitrary download credentials or execution authority. The selected armed performance mode does not authorize silent downloads, purchases, new data destinations or automatic replacement of an audible deck.

Spotify-derived metadata retains its processing restrictions even if another source supplies the media file. Do not pass such metadata to a model merely because a matching file now exists locally. Independently measured local analysis and independently user-authored text keep their own provenance; do not treat the entire library as contaminated by one restricted field. Document 13 governs model egress and action authority.

10. Service permissions and distribution

Spotify's ordinary developer terms restrict stream ripping, and YouTube's service terms restrict downloading outside permitted mechanisms or required permission. Using an unofficial intermediary does not itself grant an exception. These are service-access constraints, separate from whether an extractor can technically return bytes and from the user's rights in a particular recording. [A10, A11]

Google Play's intellectual-property policy identifies unauthorized local downloading of copyrighted content and encouragement of infringement as violations. It does not imply that all download managers or all

unofficial libraries are prohibited. A future Play release needs an accurate source/use model and distribution review. The selected milestone is now a personal tool, with optional later open-source sharing under Document 17. A private APK or source release does not automatically resolve underlying service/content rights. [A12]

Keep evidence states independently: technical status, service-access status, content-use basis and distribution status. A source adapter without a conventional official API can still be evaluated. A documented denial, access-control barrier or unestablished release right must be reported accurately rather than hidden by a generic supported badge. Ordinary user-managed local imports do not require uploading proof of ownership. Do not repeatedly ask a DJ to approve the same normal preparation action as a substitute for proper product policy.

For the named sites, the next vendor question would be whether they offer a documented integration arrangement, media-origin information, rate/availability terms, privacy handling and commercial use permission. No messages have been sent. The app-owned module remains the recommendation. On-device implementation is the selected first sequence; the later neutral comparison may justify an optional AWS path. Runtime performance, source feasibility and access status remain unverified until actual tests, irrespective of that sequencing decision.

11. Acceptance cases

These twenty cases extend FR-049 and the twelve broader SRC cases. They are specifications, not passed tests. Relevant shared gates are QA-02, QA-04, QA-07, QA-10, QA-15, QA-17, QA-18, QA-19, QA-20 and QA-21; Document 09 remains authoritative for each gate's exact definition.

ID	Scenario	Required result
DLA-01	Shared URL contains redirects, credentials, alternate host syntax or an unsupported source	Canonicalize safely; reject unsafe targets; select only an enabled adapter; no arbitrary backend fetch.
DLA-02	Spotify reference yields a probable YouTube recording with a different edit/duration	Show both identities and ambiguity; require recording choice; do not inherit old sample cues or label it a Spotify master.
DLA-03	Enabled extractor receives an eligible, known fixture	Complete resolve, format selection, transfer, validation and library publication; actual bytes/source match expected evidence.
DLA-04	Named website has a working form but no verified integration contract	Mark vendor integration unverified; do not advertise an embedded API or silently scrape the form as a production dependency.
DLA-05	Lossy source is converted to high-bitrate MP3, WAV or FLAC	Preserve original and processing history; show output format without claiming recovered fidelity or lossless source.
DLA-06	Server returns HTML, executable bytes, corrupt/truncated media or excessive size	Fail safely before playable publication; maintain bounded resource use and actionable diagnostics.
DLA-07	Transfer resumes after URL expiry or changed object	Re-resolve and validate identity/range; resume only the same object; otherwise restart or report change.

ID	Scenario	Required result
DLA-08	Cancellation races transfer, conversion or publication	Old revision cannot publish after cancellation wins; completed publication reports already completed; unrelated playback continues.
DLA-09	Process exits between file placement and database commit	Recovery reconciles the manifest once; no false-ready record, duplicate asset or deletion of referenced media.
DLA-10	Duplicate request or retry arrives	Same immutable request reuses the job; changed payload with reused key is rejected; one worker owns each revision.
DLA-11	Selected playlist contains duplicates, unavailable items and reordered completion	Preserve selected order and intentional duplicates; report per-item outcomes and partial completion honestly.
DLA-12	Acquisition and conversion compete with the selected live workload	Meet the relevant audio/thermal gate or defer acquisition work; no network/file/model work in the callback.
DLA-13	Adapter receives rate limit, login requirement, access denial or repeated failure	Use bounded retry/backoff or specific blocked state; no credential harvesting, control bypass or infinite retry.
DLA-14	Malicious URL, filename, metadata or media parser input reaches a worker	No shell/path injection, private-network fetch, secret access or unbounded decode; failures stay isolated.
DLA-15	Technical success conflicts with service, content-use or distribution evidence	Preserve independent statuses and specific reason; never present technical success as an authorization decision.
DLA-16	Extractor update breaks formats or matches	Pin and canary the update, disable/rollback the affected adapter, preserve downloaded library and live engine.
DLA-17	ParDay receives restricted metadata or sensitive delivery URL	Exclude restricted fields/secrets from prompts, embeddings and logs; permitted job state remains useful.
DLA-18	Download finishes after the user changes decks or takes manual control	Publish to library only; any subsequent load/armed plan needs current state and its own authority.
DLA-19	User cancels in-app versus presses Android Task Manager Stop	In-app cancellation targets acquisition; OS app termination is reported as interruption and recovery is tested without promising continuous audio.
DLA-20	S26 Ultra is tested across software build, storage/network and audio routes	Record exact profile and outcomes; no universal device/source certification from a model name alone.

12. Implementation work packages

DL0 - Contract and fixtures. Define adapter responses, provenance, formats, state transitions and errors. Create an original short audio/video fixture set on a controlled source, known wrong-edition pairs, corrupted responses and resumable/expiring transfer fixtures. Record hashes and permissions. No copyrighted catalog is needed to prove the job infrastructure.

DL1 - Native intake and queue. Implement paste/share, selection, source/format details, queue, cancellation and persisted states using fake and controlled adapters. Demonstrate the full C-theme journey on the S26 Ultra without touching the audio engine.

DL2 - Local extraction feasibility. Evaluate the native on-device candidate on suitable real-source fixtures and current dependencies. Verify actual media identity, source behavior, transfer mode, quality, maintenance burden, Android packaging and licensing. Record reusable fixtures and baseline measurements for the deferred controlled-worker comparison. A blocked source path remains an explicit unresolved capability; report the evidence and alternatives without silently requiring AWS or dropping the selected workflow. A useful demo is not a blanket catalog guarantee.

DL3 - Transfer and publication. Implement resumable transfer, bounded validation, optional conversion, recovery journal, deduplication and library publication. Pass adversarial and interruption cases before associating media with live decks.

DL4 - Performance and ParDay. Run acquisition alongside the selected audio workload, including manual takeover and late completion. Add bounded assistant tools only after deterministic operations are proven. Compare completion time, errors, heat, battery and required user steps against the manual baseline.

DL5 - Local source and release decision. Admit the tested on-device source paths using measured results and separately recorded source evidence. Personal local delivery is selected; optional publication details remain separate. Vendor integration, if pursued, requires a real interface and agreement. Report remaining gaps to the owner; do not delete the selected feature or promise support for every link. A blocked source capability is not completed merely because local playback works. The later AWS comparison belongs to a separate extension decision, not this local release gate.

Completion means an integrated, tested source path producing the correct usable media, with truthful quality and recovery behavior. A browser link, a mocked progress bar, or a successful metadata lookup alone is not completion.

13. Deferred comparative evaluation for AWS expansion

The earlier comparison decision remains useful for selecting a later extension, but the latest owner instruction supersedes its timing: build the local version first, then expand to AWS if justified. The initial implementation path is D, with no required server. The deferred experiment compares D and AWS without a predeclared performance winner and without committing the product to two permanent modes. Preserve baseline fixtures and results during local work; rerun comparable cases when AWS is evaluated because software/source conditions may change. Possible source sharing for independent home labs is covered in Document 17. No home host is selected. AWS is accepted for later use with an approximate USD 35/month planning target excluding ParDay AI-provider charges, not provisioning authorization. Document 18 defines its future profile. No service, device benchmark or repository publication has been performed.

Arms and comparison boundaries

Arm	Execution and network	What must be declared
D - On device	S26 Ultra resolves, acquires and performs selected processing	Extractor/runtime/encoder versions, package size, phone resource use and source access path
AWS - Owner-operated cloud	Same phone uses the evaluated AWS worker from actual Wi-Fi/mobile networks	Region, instance/runtime, limits, disk, source-to-server and server-to-phone paths, billable bytes and actual access behavior
H-LAN / H-REMOTE - Optional future home lab	Portable worker on a separately chosen local host, over LAN or configured remote access	Additional portability evidence only; no local host purchase or deployment is required for local delivery or the later D/AWS comparison

D is the first local implementation/evidence track. D versus AWS is the deferred comparison once the local version exists and a concrete AWS experiment is chosen. H-LAN and H-REMOTE remain optional network conditions for a future physical home lab. Declare whether the server only resolves a URL, relays bytes, or creates a temporary artifact. A URL bound to a server session/network may make direct phone transfer unavailable. Record this as observed capability rather than quietly switching modes inside a timing comparison.

Use the same extractor implementation/version on both sides when feasible to isolate execution location. If native and backend candidates require different extractors or encoders, report the result as a comparison of complete implementations. Do not attribute all differences to the phone versus server. Test the common source/format intersection for direct comparisons and report extra or unavailable coverage separately. Unofficial extractors remain eligible candidates; an official API is not a universal prerequisite. Source/service/content evidence stays distinct from technical results.

Fixture and condition controls

Create a versioned fixture manifest with original or appropriately permitted audio/video, exact media IDs and hashes, duration, codec/container, sample rate/channels, expected recording/edition, and applicable use/access basis. Include short and longer music, a video with audio, difficult metadata/version pairs, and controlled invalid/interrupted responses. Use the existing twenty DLA cases for failure correctness. No commercial music library needs to be published with the benchmark pack.

Run source-preservation and conversion as separate experiments. Source-preservation compares the same rendition where available; conversion holds source bytes, output codec/settings, trims and metadata behavior constant. If encoders differ, record both builds and validate final rendition timing and sound rather than demanding equal hashes for independently encoded outputs. Hash equality is appropriate for unchanged preserved files. A faster result obtained by selecting lower quality is not a valid speed advantage.

Record app build, device variant, Android/One UI build, RAM/storage, available space, output route, audio sample rate/buffer, deck/FX/stem/video workload, screen brightness/refresh, battery charge, charging state, battery-saver setting, ambient conditions and observed thermal state. Record server model/CPU/RAM/storage, AWS region and exact instance/runtime, OS/kernel, worker/container digests, CPU/memory limits, active background services and network topology. Measure each applicable transfer leg separately: verified direct source-to-phone transfer, source-to-AWS transfer, AWS-to-phone delivery and optional storage-service transfers. Do not count cloud media egress when verified direct media transfer bypasses AWS. Include start/stop and cold-start conditions; a warm service trial is not an on-demand startup result. Docker containers have no resource constraints by default; record enforced limits rather than inferring isolation from a Compose file. [B4]

Separate app/process startup, resolver startup, resolver cache, downloaded-media cache and analysis cache. A warm cache hit is not a fresh acquisition. For active-mix tests the engine stays running while only acquisition caches/workers change; do not call that an entirely cold app. Use the same foreground tasks and resource policy for each arm. Record external source responses and service outages without repeatedly probing or increasing request volume to force success.

Staged test plan and applicability

The local build applies the relevant C0 correctness, C2 audio coexistence, C3 local failure and C4 phone setup checks to D. Paired C1 and all cloud-specific variants are deferred and cannot become accidental first-local-release gates. The complete staged protocol below applies when the later AWS comparison is undertaken.

Stage C0 - Eligibility and correctness. Confirm build/licence compatibility, usable source path, job contract, identity, format reporting, validation, cancellation, safe publication and recovery using controlled fixtures. Record blocked/unavailable cells separately from technical failures. An unverified website form is not an implemented adapter. Do not compare elapsed time for an arm that cannot deliver the correct artifact. For AWS, first test an actual permitted source path from the intended region and runtime. Record unavailable, denied, rate-limited, authentication-dependent or region-dependent behavior. A cloud VM is not evidence that a source adapter works; do not escalate request volume or silently change access identity to force a pass.

Stage C1 - Paired preparation trials. Select a bounded core set covering short/long audio and video, with both preserved-source and conversion tasks. Proposed initial sampling is five paired attempts per selected case/cache condition across at least two independent sessions. Randomize or balance D/AWS order, capture pre-run conditions, and recondition the phone/server to comparable states between pairs. Keep failed and timed-out attempts in the denominator. Report paired differences, medians, ranges, counts and causes; five trials do not justify a stable tail-latency claim. Expand only the cells whose uncertainty could change the decision.

Stage C2 - Live coexistence. Compare each viable arm against the same prepared local-set baseline with acquisition disabled, then with the identical acquisition workload. Start with P0 and expand only to the intended supported P1-P3/video profiles in Document 09. Follow the same deferral policy on both arms. Queue time caused by protecting audio is part of the user-visible result. Final certification still requires the relevant shared warm-soak gates; a short comparative demo does not replace them.

Stage C3 - Failure and remote operation. Exercise server down, expired endpoint credentials, interrupted connection, Wi-Fi/mobile transition, locked screen, low storage, cancelled conversion, expired media lease, adapter update failure and process recovery. Prepared local mixing must continue without the server. AWS trials distinguish Wi-Fi and mobile service, stopped/warming workers, expired artifacts and endpoint changes. Optional remote home-lab operation is a separate condition if later evaluated. In-app cancel targets the acquisition job; OS Stop/Force stop remains the app-termination case already specified.

Stage C4 - Setup and maintenance. Using clean instructions and a permitted fixture, the owner measures setup steps, active time, failed steps, questions, update/rollback effort and backup/restore. Compare phone-only setup with a clean AWS trial deployment and verified cleanup. A future physical home-lab setup is optional portability work. Testing by another operator is conditional on community-reproducibility claims, not a prerequisite for personal release. Record cloud compute, persistent storage, public addressing, transfer, logs and remaining resources after shutdown. For a later physical host, record its operating costs separately. Account spend limits and alerts have the distinct applicability described in Document 18.

Measurements and honest interpretation

Use client monotonic time for user-visible request-to-ready duration. Separately record resolve, selection wait, queue, transfer, processing, final validation, publication and optional analysis. Human selection time is reported separately from system time. Worker spans use their own monotonic clocks and trace IDs; never subtract unsynchronized phone/server wall clocks. Parallel spans need not sum to total latency. Report first playable separately from fully prepared, and include the selected actual recording and output properties beside each result.

Measure completion/mismatch/failure counts, retries, bytes transferred on each network leg, peak staging/final storage, process CPU time, a clearly defined memory metric, audio underruns/discontinuities, callback budget, thermal observations, phone battery/energy observations, server utilization and incremental operating cost. Pair power observations with an otherwise identical local-set or idle baseline. Show phone consumption and server/network consumption separately; moving work off the phone is not proof of lower total energy.

Android's Thermal API can support observation, but availability and interpretation vary by device; unsupported or invalid readings stay unavailable. Follow the documented sampling limits. [B1] Android Studio's documented power-rail monitoring is device-specific and is not an established S26 Ultra capability. Whole-device power is not automatically app-attributed energy; battery percentage alone is a coarse observation. Record available charge/current counters or an appropriate external measurement method, its resolution and limitations, without inventing missing measurements. [B2]

Use a release-like, non-debuggable/profileable build for comparable UI measurements where supported. Android Macrobenchmark can supply repeatable startup/UI traces; it does not replace physical audio/output measurements from Document 09. [B3] Keep measurement overhead comparable and run final endurance checks under the intended production configuration.

Decision rule

First apply mandatory gates: correct recording/final format, no unarmed deck action, safe cancellation/publication/recovery, protected audio in the declared profile, source/provenance integrity, bounded worker/network behavior, and dependency compliance for personal deployment, plus compatibility with any project licence later selected for publication. A faster candidate cannot compensate for failure in those gates. An unresolved gate remains unresolved; no numerical score turns it into a pass.

Among viable candidates, present the measured tradeoffs in phone heat/battery, completion time, source coverage, storage, independence away from home, setup/maintenance, privacy and ongoing cost. Apply the established priority of local independence and the approximate USD 35/month future AWS target, then present any remaining tradeoff that could change the extension decision; no arbitrary weighted overall score is preselected. If neither candidate dominates, retain the local default unless measured benefits justify adding or changing an acquisition path. Maintain both only when their distinct benefits justify the cost and maintenance. Record the evidence, limitations, chosen configuration and revisit triggers in an architecture decision record. All results in the companion protocol are intentionally empty.

Sources checked 29 October 2026

- [A1] Spotdown public product page: https://spotdown.org/en2
- [A2] Spotdown terms: https://spotdown.org/en2/terms-of-service
- [A3] Y2Mate public product page: https://en.y2mate.st/x16/
- [A4] Y2Mate terms: https://en.y2mate.st/terms.html
- [A5] spotDL project documentation: https://github.com/spotDL/spotify-downloader
- [A6] yt-dlp project documentation and licensing: https://github.com/yt-dlp/yt-dlp
- [A7] NewPipeExtractor documentation and licence: https://github.com/TeamNewPipe/NewPipeExtractor
- [A8] Android user-initiated data transfer: https://developer.android.com/develop/background-work/background-tasks/uidt
- [A9] Android long-running workers: https://developer.android.com/develop/background-work/background-tasks/persistent/how-to/long-running
- [A10] Spotify Developer Terms: https://developer.spotify.com/terms
- [A11] YouTube Terms of Service: https://www.youtube.com/static?template=terms
- [A12] Google Play intellectual-property policy: https://support.google.com/googleplay/android-developer/answer/9888072?hl=en
- [B1] Android Thermal API and device limitations: https://developer.android.com/games/optimize/adpf/thermal
- [B2] Android Studio Power Profiler and supported measurements: https://developer.android.com/studio/profile/power-profiler
- [B3] Android Macrobenchmark setup: https://developer.android.com/topic/performance/benchmarking/macrobenchmark-overview
- [B4] Docker runtime resource constraints: https://docs.docker.com/engine/containers/resource_constraints/

Architecture, proposed contracts and acceptance cases are this project's design judgments. Source statements describe current documentation, not measured functionality of this app.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 17 OF 19

Open-source and home-lab guide

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; local delivery first, with AWS and optional publication deferred **Confirmed intent:** Personal local app first; potential GitHub open-source sharing; later optional AWS extension and D-versus-AWS comparison; no local host selected; approximately USD 35/month future AWS planning target excluding ParDay AI-provider charges **Owners:** Architecture, Integrations, Release/Security, Documentation and product owner **Companions:** Documents 05, 09, 10, 11, 13, 16 and 18; decision ledger; acquisition, AWS and portable deployment contracts

1. What this decision changes

Build the product for the owner's personal workflow first. Preserve the selected Crystal Current interface, Galaxy S26 Ultra target, explicitly armed ParDay assistance and complete professional feature scope. Personal distribution does not lower the audio, recovery, privacy or source-identity requirements.

The latest answer selects the phone-native local version first and AWS expansion later. No local host, service account or cloud worker is required to start design or deliver the first local app. Document 18 owns the deferred cloud proposal and the approximate USD 35/month planning target, excluding ParDay AI-provider charges. Account, region, instance and deployment remain open; the target is not an approved resource purchase or guaranteed bill. Local LAN/remote profiles in this guide remain optional portability references.

Plan software others can inspect, build and operate with their own equipment and credentials. This does not commit the owner to public hosting, subsidized AI or support for every phone.

Open-source publication is a planning direction. Licence, GitHub account/name and publication remain unselected. Prepare an inspectable release package for that decision. This document creates no repository, public release, container image, hosted endpoint or licence grant.

2. Personal project and open source are different decisions

Personal use describes the owner's goal and maintenance commitment. An open-source licence gives downstream users permissions defined by that licence. The Open Source Definition requires redistribution and modification rights and does not allow a restriction excluding business use. A custom personal-use-only or noncommercial clause would therefore conflict with describing that licence as open source. [H1]

This distinction need not change the owner's intention. The README can describe a personal project, supported configurations and limited maintenance without adding incompatible usage restrictions. If the owner wants a binding noncommercial restriction, record that as a different source-available licensing choice rather than silently applying it.

A public GitHub repository alone is not sufficient licensing guidance. GitHub recommends an explicit licence so others know their rights to use, modify and distribute the work; its platform viewing/forking behavior does not replace a project licence. [H2]

3. Licence candidates and recommendation

The following is a decision aid. No option is selected or applied to the code in this planning pack.

Candidate	Relevant effect	Fit to evaluate
GPL-3.0-or-later	Copyleft conditions apply when covered software is conveyed; distributed object code needs a compliant corresponding-source route. Private use does not itself require public posting. [H3]	Leading candidate for the Android application if GPL components are included and shared improvements are preferred.

Candidate	Relevant effect	Fit to evaluate
AGPL-3.0-or-later	Adds a source-offer requirement for users interacting remotely with a modified covered network program. [H4]	Consider for a server only if the owner wants that network-use obligation; self-hosting alone is not a reason to select it automatically.
Apache-2.0	Permissive reuse with explicit patent provisions and licence/notice/change obligations. [H5]	Candidate for independently licensable first-party components when broader reuse matters and the dependency combination permits it.
MIT	Short permissive licence retaining its copyright/permission notice and warranty disclaimer. [H6]	Simple alternative for independently licensable components; it does not erase obligations from included dependencies.

The provisional recommendation is to evaluate GPL-3.0-or-later for the app, then decide the service licence after selecting its components and the desired contribution model. A single coherent first-party licence may be easier to maintain than several unnecessary licence boundaries. Do not adopt AGPL, a contributor agreement or a dual-licensing programme merely because they are available.

NewPipeExtractor explicitly uses GPL version 3 or later. Its inclusion would require a compatible distribution plan for the resulting application; an MIT label on our own files is not a blanket override. [H7] Review paid DSP, plugin components and native libraries against the actual combined work before promising a redistributable build.

yt-dlp illustrates why the exact artifact matters: its project source is under the Unlicense, while some bundled executables contain GPLv3+ components and other distributions include additional licences. [H8] Record the actual package, version, build flags and dependencies that are shipped. A Docker image, process boundary or HTTP API does not automatically settle whether components form one covered work or remove redistribution obligations. Review the actual integration rather than using architecture as a licensing shortcut.

4. Repository contents people can actually use

Plan one repository with clear component boundaries initially. Split repositories only if independent release or ownership needs justify it.

Planned area	Required content
Android app	Kotlin/Compose source, resource sources, build configuration and documented device requirements
Audio engine	C++ source, adapter interfaces, test harnesses and explicit real-time invariants
Acquisition module	Native on-device adapters first; shared job/provenance contracts and a replaceable boundary for a later optional service
Deployment	Local build and install instructions first; reproducible service/container setup, health checks and rollback only when an optional service is implemented
Documentation	Architecture, local build, personal installation, home-lab setup, source capabilities and troubleshooting
Contracts	App/service compatibility version, job lifecycle, cancellation, error taxonomy and migration rules

Planned area	Required content
Fixtures	Small original or redistribution-cleared media, fixture manifests and expected outcomes; no personal library dump
Release materials	Dependency inventory, licence texts/notices, source/release mapping, checksums and known limitations
Contribution materials	Issue templates, contribution guide, security reporting route and device-result template

Configuration examples contain placeholders and explained defaults. They must not require the owner's private server, signing key, account or music collection to complete a clean build. Installation documentation should distinguish a developer build, a personally signed build and any later project release APK. A working build from a fresh checkout is the standard, not the existence of a repository screenshot.

5. Code, assets and private state

Maintain separate records for source code, fonts/icons/textures, demonstration music, sample packs, model weights and third-party libraries. The top-level code licence should state its scope and point to the asset/dependency inventory. Preserve the licence and provenance of each included item. If an asset cannot be redistributed, provide a documented replacement or user-supplied configuration rather than an unexplained missing file.

Keep personal tracks, downloaded media, recordings, playlists containing private information, ParDay transcripts, provider tokens, API keys, cookies, server credentials and signing material out of public commits and release archives. Use a small deliberately shareable demo crate. Inspect release artifacts as well as source history; a clean working tree does not establish that an earlier commit or compiled resource contains no private material.

The repository should explain which components require separate external services or credentials. Each operator supplies their own credentials. Do not ship a shared paid API key, a bundled session from a music website, or a public endpoint funded by the owner as an undocumented dependency.

6. Home-lab operating model

Deliver the phone-native app first; the phone owns its performance library, preparation, acquisition job state and live engine. If a service is added later, offer a documented single-operator deployment before considering multi-tenant hosting. That optional service may resolve/acquire/process media or provide a separately configured ParDay gateway. AWS is the accepted later server option; this section preserves physical home-lab portability. Loss of any service leaves already prepared local performance usable.

Keep acquisition and AI capabilities separable in setup. A user may want an acquisition worker but no cloud model, or ParDay with their own API account while choosing on-device acquisition. The endpoint picker must state which server receives links or permitted context. A service address is not evidence that its operator is the app's publisher.

Document host OS/architecture, measured CPU/RAM/storage needs, runtime, persistent directories, temporary-media limits, credentials, reachability and pairing. Hardware minimums require comparison evidence; generic NAS or Raspberry Pi support is not assumed.

For a local home-lab profile, default to private LAN/VPN deployment with authenticated encrypted transport, per-device pairing and revocation. Private network reachability is not authentication. Pairing should use an expiring code or equivalent reviewed mechanism and a revocable credential; health/version negotiation precedes jobs. Never disable certificate validation or use trust-all TLS.

The phone may connect to its explicitly paired private home-lab endpoint. This control-plane trust does not let the backend fetch arbitrary private, loopback, link-local or metadata-service URLs as media. Keep the source-fetch SSRF rules in Document 16 separate from the configured service address. Container workers must not require privileged mode, the Docker socket or a full-home-directory mount; provide only bounded staging and necessary data paths. Prepared local DJ playback must remain independent of this service.

The runbook must explain where configuration, job state, temporary files, completed artifacts and logs live; which items are backed up; how expiration/deletion works; and how to restore without replaying completed jobs. Use bounded concurrency and per-job limits. Backup/restore should preserve user preparation while avoiding accidental publication of private media or secrets.

7. Deferred comparison of on-device and self-hosted acquisition

The owner selected local delivery before AWS expansion, while preserving comparison evidence for the later extension choice. Build and validate D first and retain its fixture/measurement records. When AWS is evaluated, compare D and AWS over recorded real phone networks; physical LAN and remote home-lab arms are optional portability work. Both candidates use the same native intake, selection, queue and library-publishing contract. The comparison is not a first-local-release gate, and one local demonstration does not establish the cloud candidate's infeasibility or sustained performance.

Use identical suitable source fixtures and selected formats, control the phone's network conditions, and apply the same resulting-media checks. Record the different source access paths rather than pretending AWS and phone egress are the same network. Record cold/warm resolution time, transfer and processing time, time until playable, completion/failure rate, battery change, thermal state, app memory, server resources, storage use and operator steps. Run acquisition alongside the declared DJ workload and test offline server loss, cancellation and recovery.

Report exact phone software, audio route, server hardware, adapter/runtime versions and output hashes. Separate source outages from implementation faults. Record per-format results rather than declaring a universal winner. A later hybrid may be justified by evidence, but the first local release does not need two complete implementations. Document 16 owns the acquisition semantics; Document 09 owns audio/device gates.

8. Updates and community operation

Version the Android/service protocol and advertise compatible ranges. A newer server must not reinterpret an old client's action silently. Pin extractor and processing dependencies in releases, test changes on the fixture corpus, and retain the preceding working deployment for rollback. Never replace a live audio component mid-set.

Document source outages, temporary adapter disablement and upstream changes without blaming every failure on a user's setup. An unsupported device report should collect the necessary profile and logs with private values redacted. Publish measured supported combinations; community reports are useful evidence but are not automatically official certification.

Require reproducible issues, focused changes and compatible contributions; credit authors. Decide whether contributions are accepted before promising review/support. Security reports need a real private route when the repository exists; do not present a placeholder address as monitored.

9. Publication readiness

Before public release, present the licence/notices, clean-build evidence, shareable fixtures, dependency inventory, source-to-binary mapping, tested setup and limitations. The owner then reviews the exact publication package. The planning guide does not substitute for a tested deployment runbook.

The personal app remains useful before that step. Google Play submission is not a gate for the selected personal-tool milestone; public store distribution can be reconsidered only as a new decision. Applicable Android installation/signing behavior, component licences and source-service requirements still need their own current review. Open-sourcing code or self-hosting a worker does not grant rights in a music catalog or turn an unverified source adapter into a supported one.

10. Acceptance evidence

The companion home_lab_deployment_profile.json records each case's applicability for the portable profile. Document 18 and aws_deployment_profile.json add the AWS-specific acceptance cases. All are unexecuted. App build reproduction, implemented-runtime safety and local/AI independence remain required. Service-specific variants apply only when a later service profile is evaluated or selected; neither they nor the D-versus-AWS comparison are local design/release prerequisites. HL-10 applies only before optional public publication.

ID	Required demonstration
HL-01	Clean-host reproduction: build the app and selected service profile using documented prerequisites, permitted dependencies and shareable fixtures.
HL-02	Pair and revoke: a new device pairs deliberately; revocation blocks subsequent service actions without deleting its local preparation.
HL-03	Authentication and TLS: reject untrusted clients/certificates; paired private control-plane access does not permit private-target media fetching.
HL-04	Server loss: prepared local mixing continues; unavailable acquisition reports its actual state without surprise playback or lost preparation.
HL-05	Resource limits: exhausted disk, hung/crashed workers and oversized jobs stay bounded and do not compromise live audio.
HL-06	Credential isolation: logs, artifacts and one worker cannot reveal another component's secrets or private context.
HL-07	Upgrade/recovery: protocol mismatch, rollback and backup restore preserve correct jobs and do not replay completed actions.
HL-08	AI independence: acquisition can run without ParDay; the local engine remains usable with all external services unavailable.
HL-09	Artifact cleanup: successful transfer, expiry, cancellation and failure follow the documented retention/deletion policy.
HL-10	Optional publication: review the exact source/history/artifacts, licence obligations and build mapping; exclude credentials, personal music and signing keys.

No build, deployment, licence adoption or public publication has been performed in this planning revision.

Sources checked 29 October 2026

- [H1] Open Source Initiative, Open Source Definition: https://opensource.org/osd

- [H2] GitHub, Licensing a repository: https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/licensing-a-repository

- [H3] GNU GPL v3 text, Open Source Initiative copy: https://opensource.org/license/gpl-3.0 ; steward version: https://www.gnu.org/licenses/gpl-3.0.en.html

- [H4] GNU AGPL v3 text, section 13, Open Source Initiative copy: https://opensource.org/license/agpl-3.0

- [H5] Apache Software Foundation, Apache License 2.0: https://www.apache.org/licenses/LICENSE-2.0

- [H6] MIT licence text: https://opensource.org/license/mit

- [H7] NewPipeExtractor, licence and integration documentation: https://github.com/TeamNewPipe/NewPipeExtractor

- [H8] yt-dlp, licensing and distribution documentation: https://github.com/yt-dlp/yt-dlp

Repository structure, deployment defaults, comparison method and acceptance criteria are proposed project decisions. The licence discussion is conditional on the actual components and distribution; it is not a claim that any candidate combination has already been cleared.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 18 OF 19

AWS deployment and comparison plan

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.

DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline for a deferred AWS extension; no cloud resources created or source path tested **Confirmed direction:** Build the local phone-native personal version first; expand to AWS later if justified. Future AWS planning target is approximately USD 35/month excluding ParDay AI-provider charges. No existing host is selected; the later D-versus-AWS comparison has no measured winner. **Owners:** Cloud/Integrations, Android, Library/Data, Security/Release and Quality **Companions:** Documents 05, 09, 10, 11, 13, 16 and 17; acquisition comparison protocol; AWS deployment profile and cost assumptions

1. Decision and first evidence checkpoint

The owner selected a local version first and AWS expansion later. Deliver the Android app with a phone-owned library, preparation, acquisition job state and live engine, without a required LAN or cloud server. Preserve a replaceable adapter boundary and useful measurements, but do not build or deploy AWS as a dependency of that first version. This document is a deferred expansion specification. When the extension is pursued, AWS is the accepted remote service host for the personal acquisition comparison. It has different network and billing behavior from a home-LAN machine. Keep portability for a later community operator; buying a home server is not required.

The owner selected an approximate USD 35 monthly AWS planning target, excluding ParDay AI-provider charges. Account, region, service/instance, active hours, endpoint, project licence and a concrete deployment decision remain open. The target is not a guaranteed bill, hard ceiling or permission to provision resources now. The initial implementation path is local; the comparative winner for an optional expansion is unmeasured. The interface remains one native acquisition workflow with interchangeable admitted adapters.

After the local version is usable and a concrete cloud experiment is chosen, the first AWS checkpoint is a bounded source-path feasibility test. Build and validate a minimal worker against a permitted known fixture, then verify the intended media sources from the selected AWS environment. Record resolution, actual recording/edition, format, redirects, transfer mode, file integrity, source errors and extractor build. A source path that works on a residential or phone connection is not assumed to work identically from a data-center network. Authentication challenges, source restrictions and unsuccessful requests become evidence; do not reinterpret them as successful extraction or add an unreviewed evasion system.

Keep requested unofficial extractor candidates in the experiment. Technical results, provider/service access, content use and distribution status retain separate records under Document 16. The availability of AWS does not settle any of those source questions. Do not invest in a larger service architecture until at least one intended end-to-end source path works and its remaining constraints are understood.

2. Proposed later AWS trial profile

For a finite benchmark, propose one on-demand Linux x86-64 EC2 instance, one availability zone, a small encrypted gp3 EBS allocation, and the same pinned portable worker that can later run elsewhere. Use a non-burstable general-purpose instance as a candidate for repeatable conversion measurements. The m7i.large specification is 2 vCPUs and 8 GiB memory; it is a proposed test profile, not a demonstrated minimum or chosen purchase. [W1]

Start with one active acquisition job and one bounded conversion/preparation operation. Record the effective CPU, memory, process, disk, byte and time limits. These are proposed experimental controls; do not publish throughput or simultaneous-job guarantees until measured. Keep the live engine, cue/master routing, immediate manual takeover and authoritative performance state on the S26 Ultra.

A 30 GiB total gp3 allocation can be used as an initial cost-model assumption, subject to the actual OS/container footprint and bounded fixture sizes. Reserve free space for the operating system and journal; a per-job quota is smaller than the available staging capacity. Reject oversized video or offer an eligible smaller rendition before transfer; never fill the system volume.

Use one owner-authenticated HTTPS service endpoint. Administrative access should use Systems Manager Session Manager when the selected image and network setup support it; AWS documents that Session Manager does not require opening inbound administration ports. [W2] The trial does not need a load balancer, NAT gateway, Kubernetes cluster, managed relational database, GPU or multi-region deployment. Each added service needs a measured purpose and its own cost line.

The region remains open. Use a named region for every price quote and trial, measure from the owner's actual Wi-Fi and mobile networks, and record the endpoint route. A pricing reference region is not a claim about the user's location or the fastest region.

3. Why compare these AWS hosting choices

Candidate	Useful property	Consequence for this experiment
EC2 on demand	Direct control of the host, pinned containers, local journal/staging and stop/start lifecycle	Proposed finite-trial baseline. OS maintenance remains ours; attached storage and retained addresses can continue costing money while compute is stopped. [W3]
Lightsail instance	Bundled compute/storage/transfer packaging can simplify a small steady personal service	Strong alternative if always-available operation becomes preferable. Stopped instances still accrue plan charges until deletion; CPU burst behavior must be recorded during sustained conversion tests. [W4, W5]
ECS on Fargate	Managed container execution without maintaining an EC2 host	Revisit when isolated jobs or managed scheduling justify additional networking, state and deployment work. Linux platform 1.4.0+ provides at least 20 GiB ephemeral task storage, configurable up to 200 GiB; container images consume part of it. [W6]
Lambda	Useful for a small control-plane action or bounded event processing	Do not make it the default media worker merely to call the service serverless. Standard event functions have a 900-second timeout; documented Managed Instances invocation exceptions support longer execution. Packaging, storage, startup, transfer and job recovery still require a complete design. [W7]

Do not choose a burstable EC2 or Lightsail profile solely from its advertised vCPU count and then compare a fully credited cold run with a depleted sustained run. If such a profile is evaluated, record CPU credits/burst state, operating mode and related charges. Compare identical output quality and report complete-implementation differences honestly.

No long-term compute commitment or reserved capacity is proposed before the trial establishes actual duty cycle and resource needs. Exact prices and scenarios belong in the accompanying cost assumptions, with quote date, region, operating system, billable hours, storage, IPv4 and transfer stated explicitly.

4. Components and data movement

The initial service has a versioned API/controller, a persistent job journal, an isolated acquisition worker and managed staging. These are logical boundaries and may share one machine; their permissions and failure handling remain separate. The phone retains its own authoritative acquisition/publication state and prepared library.

Leg or stage	Required behavior
Phone to controller	Submit the accepted request revision and bounded options over verified TLS. Send a service-specific device credential, not AWS account credentials.
Controller to worker	Admit an immutable job specification with a lease, resource limits and permitted adapter. A model cannot choose executables, flags, filesystem paths or network policy.
Worker to actual source	Resolve and transfer through the admitted adapter. Record the actual source and edition; enforce address, redirect, response-size and timeout rules.
Validation and optional conversion	Validate source bytes before conversion and validate the exact final rendition afterward. Final format and timing evidence describe the resulting file.
Service to phone	Deliver the authenticated artifact or a separately verified direct-transfer path. Count actual source-to-phone or cloud-to-phone bytes and retries for the verified mode, alongside any source-to-server traffic.
Phone library publication	Verify received media, reconcile current revision/cancellation/lease, and publish through the existing atomic contract. Server artifact readiness is not proof of local publication.

Do not assume a source URL resolved from AWS can be downloaded directly on the phone. The existing direct, relay and server-artifact modes remain distinct. Compare the exact mode offered by the adapter and account for its storage, delay and cost.

Keep S3 optional for the first single-instance experiment. Add private object delivery only if it provides a concrete recovery or stopped-instance benefit. If added, retain Block Public Access, scoped IAM and bounded retention. Presigned URLs are reusable bearer credentials until their effective expiry, not automatically single-use tokens; protect them from logs and model context, and verify the artifact on the phone. Temporary signing credentials may expire before the requested URL expiry. [W8, W9]

All cloud-side media in this profile is temporary working data unless the owner explicitly selects another policy. The phone is the prepared library. Keep originals, analysis, cues, recordings and cloud artifacts distinguishable. A server outage, stop, teardown or expired download link must not invalidate already verified local media.

5. Access, credentials and worker isolation

Pair the phone deliberately with a known service identity and an expiring single-use enrollment mechanism. Issue a revocable per-device credential. Require TLS certificate validation and an explicit certificate/identity rotation procedure. If a private VPN/overlay replaces public application exposure, review and document its authentication, encryption, mobile connectivity and additional dependency; a private address alone is not identity verification.

For a public HTTPS trial, expose only the required application port through the security group, use bounded request/job admission, and leave administration ports closed. Endpoint naming, DNS and certificate issuance must be concrete before deployment. A changed IP address must not cause the app to accept an unverified replacement service. Do not use trust-all TLS or bundle an AWS access key in the APK.

Separate owner administration from application access. The owner initially starts/stops the trial using their AWS administrative workflow. A stopped service cannot accept a wake-up request by itself. An in-app wake-up feature would require a separately designed limited control path and is not included by implication.

Use a least-privilege instance role only for the management or explicitly admitted AWS operations the host needs. Require IMDSv2 where metadata is enabled, but do not treat it as sufficient worker isolation. AWS documents metadata tokens and hop-limit behavior separately. [W10] Enforce denial of metadata, loopback, link-local and other private targets from the acquisition worker, including IPv6 and redirect/DNS-rebinding cases. The worker must not obtain the host's AWS credentials merely because the host needs a management role.

Workers run without privileged mode, host networking, the Docker socket, broad host mounts or unnecessary capabilities. Allow only managed staging and required immutable runtime files. Keep provider cookies, API keys, device tokens and AI credentials out of images, repositories, job arguments and diagnostics. Any source-specific credential use requires an explicit adapter configuration and separate scoped storage; no owner session is distributed to community users.

Encrypt EBS volumes. AWS documents encryption of volume data, instance-volume transfers and snapshots made from encrypted volumes. [W11] This does not remove application authorization or backup-retention requirements. Default backups should contain the required configuration and consistent job state, with media staging excluded. If whole-volume snapshots are used, disclose that they may preserve temporary media and apply the corresponding retention policy.

6. Cost control and lifecycle

Treat the later cloud trial as a finite experiment with a named resource set, explicit active hours and a teardown plan. The owner has selected approximately USD 35/month as the future AWS planning target, excluding ParDay AI-provider charges. This records the desired envelope, not a current deployment/spending approval or an enforced ceiling. Cost estimates must distinguish test-hours operation from an always-on month. contracts/aws_cost_scenarios.json separates 20-hour, 50-hour and always-on examples. Its M7i compute rate is explicitly historical and requires a current regional quote before any resource is created.

EC2 stop/start differs from deletion. EBS state persists, attached storage remains billable, and retained Elastic IP addresses remain chargeable. An automatically assigned public IPv4 address can change after stop/start, so endpoint identity and reconnection need a deliberate design. [W3] Lightsail stopped-instance billing is different and must not use the EC2 savings assumption. [W4]

Set local admission limits on jobs, bytes, execution time, retry count and retained artifacts. Add an explicit test-session expiry or reviewed idle-stop mechanism; drain or checkpoint work before a scheduled stop and report interruption honestly. Do not rely only on an idle detector to enforce the spending plan. Protect the local mix even if acquisition stops immediately.

Use actual-spend notifications and inspect billed resources during the trial. AWS Budgets data and notifications can lag usage; a normal budget alert is not a synchronous hard ceiling. [W12] AWS also documents project spend limits in its new AWS experience for eligible paid-plan projects. The minimum is the greater of USD 20 and a conservative spend estimate; the limit applies before tax and excludes credits. Reaching it can pause the project, and the documentation specifies permanent deletion after 90 days of inaction while paused. This account's eligibility and resource coverage are uninspected. If selected, validate these behaviors and recovery obligations; do not assume an arbitrary smaller hard ceiling is available. [W13]

Record compute, EBS, address hours, transfer, snapshots, logs, optional object storage/requests and separately chosen DNS, VPN, model or support costs. Do not assume free-tier eligibility, remaining shared transfer allowance, credits or tax treatment. Keep AI costs separate, including on a shared host.

Teardown means reconciling jobs and required exports, deleting the intended compute/storage objects, and auditing leftovers such as volumes, snapshots, addresses, buckets, registry images and log groups. Stopping the instance is a useful operating state, not proof of a zero bill or completed teardown.

Illustrative cost scenarios in USD

The following EC2 subtotals are calculations, not current deployment quotes. They use a Linux/shared-tenancy m7i.large reference of USD 0.1008 per running hour that AWS explicitly dated 4 May 2024, 30 GiB gp3 retained for a full month at an assumed USD 0.08 per GiB-month, and one auto-assigned public IPv4 at USD 0.005 per running hour. The compute rate requires a current quote before any resource is created. Region us-east-1 is the arithmetic reference, not the owner's selected region. [W14, W15, W16]

Running hours in the reference month	Compute	Retained disk	Public IPv4	Subtotal before transfer/extras
20 hours	$2.02	$2.40	$0.10	$4.52
50 hours	$5.04	$2.40	$0.25	$7.69
730 hours	$73.58	$2.40	$3.65	$79.63

Stopping saves compute in this EC2 scenario while the retained disk remains charged. A retained Elastic IP would change the address calculation. Transfer, snapshots, logs, DNS/domain, optional services, taxes and AI usage are outside these subtotals. Any account-wide transfer allowance must be verified against its remaining availability; it is not reserved for this app. The cost contract includes a paid-egress scenario as an assumption to refresh, rather than a promised zero transfer bill.

For a simpler always-available alternative, AWS currently lists Linux Lightsail bundles with public IPv4 at USD 12/month for 2 GB RAM, 2 vCPUs, 60 GB storage and 3 TB transfer, and USD 24/month for 4 GB RAM, 2 vCPUs, 80 GB storage and 4 TB transfer. These are bundle prices and region-dependent allowances, not all-in bills or performance certifications. IPv6-only variants are cheaper but require separate end-to-end connectivity validation. [W17]

The spending-target question is closed: approximately USD 35/month, excluding ParDay AI-provider charges, for a later AWS extension. The illustrative 730-hour M7i subtotal of USD 79.63 already exceeds that target before transfer and extras, so it is not an acceptable always-on recommendation within this envelope. A limited-hours EC2 experiment remains a candidate after a current quote. The listed USD 24 Lightsail base leaves USD 11 of nominal headroom before other costs, but is neither a selected service nor proof that the complete bill or workload fits. At the expansion milestone, choose duty cycle and service from current prices, measured source behavior and actual resource needs; do not reopen AWS setup as a condition of local design.

7. Build and operating runbook required from implementation agents

Provisioning record. Before creating resources, specify the owner/account context, region, instance/architecture, AMI and software versions, storage, network exposure, endpoint identity, tags, maximum trial duration and cost assumptions. Present the infrastructure change set and cleanup list. This planning document is not an applied template or authorization to incur charges.

Reproducible build. Pin the container/runtime, extractor, encoder and dependency versions; produce a dependency/licence inventory and build hashes. Prefer one portable worker package that supports the local and AWS comparison profiles. Exact x86-64/ARM support is tested, not inferred from Docker availability.

Keep secrets outside build layers and shared examples.

Bootstrap and pairing. Verify encrypted volumes, host updates, management access, effective worker limits, source-fetch isolation, TLS identity and capability negotiation before pairing the phone. Verify that the unauthenticated endpoint cannot submit work or fetch an artifact. Document how to revoke a lost device and rotate credentials without discarding preparation.

Job recovery. Persist progress and request/cancellation revisions. A restarted service reconciles incomplete jobs under current leases instead of replaying completed actions. The phone must recognize an interrupted or expired artifact and retry or reacquire through the current request contract. A resumed download cannot overwrite a newer recording choice.

Update and rollback. Test a pinned adapter change against the fixture corpus before use. Retain a known working version and protocol compatibility information. A rollback must preserve readable journal data or perform the specified migration; swapping the container tag alone is not a recovery strategy.

Cleanup and evidence. Demonstrate a complete permitted fixture from intake to local readiness, deletion/expiry of temporary cloud media, server-loss continuity of local mixing, and accurate resource/cost records. Export redacted traces and checksums, then execute and verify the cleanup list. No public source repository or shared container release is created by running a personal trial.

8. Deferred comparison protocol and promotion conditions

The first implementation is D on the phone. The later expansion comparison is D versus AWS, and paired cloud trials are not required to complete local design or the first local release. Promote this deferred work only when the local version is usable, a concrete acquisition/update/processing limitation or operating benefit warrants cloud evaluation, and an actual region/service/costed trial is ready for an implementation decision. Reuse local fixtures and rerun comparably current D measurements; old data cannot automatically be paired with changed extractors or sources. Test the AWS path separately from the owner's actual Wi-Fi and mobile connection when both are relevant. Retain H-LAN/H-REMOTE only as optional home-hardware conditions; no home host is selected.

Preserve Document 16's paired fixture, rendition, cache, workload and timing controls. Add AWS region/availability zone, instance family, AMI/container digests, resource limits, network path, billed hours and bytes, stop/start history, burst state if applicable and service availability to each evidence record. Do not substitute cloud metrics for physical audio measurements on the S26 Ultra.

Include the minimal source-path feasibility stage before sustained comparison. Then measure request-to-local-playable and request-to-prepared time; resolution and human selection separately; completion/failure outcomes; each applicable transfer leg; phone heat, battery and audio behavior; cloud resources; operator effort; and actual/estimated costs with their coverage stated. Cold service startup is part of the relevant user workflow, not silently excluded from an on-demand result.

Prepared local mixing must survive server unavailability. The AWS arm must also demonstrate safe behavior when stopped mid-transfer, when credentials or certificates expire, and when an artifact disappears before phone delivery. A technically usable cloud path does not automatically become the chosen product runtime. Record the owner's eventual tradeoff decision and the evidence that supports it.

9. AWS acceptance cases

All cases below are proposed and unexecuted. They apply only to a later evaluated AWS profile; they are not local design or first-local-release gates. Passing them does not require choosing AWS as the final acquisition runtime. Existing FR-049, DLA and relevant QA gates remain authoritative for the implemented local path, media correctness and local audio.

ID	Required demonstration
AWS-01	Source-path feasibility: a permitted fixture and intended source path produce the correct verified local rendition from the actual AWS environment; failures and access conditions are recorded.
AWS-02	Reproducible deployment: the declared configuration can be rebuilt with pinned versions, valid endpoint identity, encrypted storage and no hidden owner-specific dependency.
AWS-03	Pairing and revocation: unauthenticated/revoked devices cannot submit work or retrieve artifacts; certificate/credential rotation is deliberate and does not erase local preparation.
AWS-04	Worker isolation: attempted private/metadata/loopback targets, hostile redirects, excessive resource use and arbitrary execution inputs fail within the documented boundaries.
AWS-05	Correct artifact publication: each applicable transfer leg, exact rendition/hash, cancellation revision, lease and idempotency are checked; server readiness cannot publish a stale or unreceived phone file.
AWS-06	Stop/start and outages: local mixing continues; interrupted cloud jobs reconcile without duplicate publication or autoplay; changed endpoint addressing preserves verified identity.
AWS-07	Workload and comparison: the declared S26 audio profile and paired acquisition experiment meet their gates, with cloud and phone measurements and failure denominators retained.
AWS-08	Cost and admission controls: finite trial hours, quotas, alerts and any applicable spend-limit feature are exercised; residual storage/address/transfer costs remain visible.
AWS-09	Update, backup and restore: protocol mismatch, adapter rollback and restoration preserve correct state while excluding or explicitly governing temporary-media backups.
AWS-10	Retention and teardown: delivered/expired/cancelled artifacts follow policy; required exports are verified and all intended chargeable resources are removed or explicitly retained.

Sources checked 29 October 2026

- [W1] AWS, Amazon EC2 M7i instances: https://aws.amazon.com/ec2/instance-types/m7i/

- [W2] AWS Systems Manager, Session Manager: https://docs.aws.amazon.com/systems-manager/latest/userguide/session-manager.html

- [W3] Amazon EC2, How instance stop and start works: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/how-ec2-instance-stop-start-works.html

- [W4] Amazon Lightsail, Billing and account management: https://docs.aws.amazon.com/lightsail/latest/userguide/amazon-lightsail-frequently-asked-questions-faq-billing-and-account-management.html

- [W5] Amazon Lightsail, CPU burst capacity accrual: https://docs.aws.amazon.com/lightsail/latest/userguide/cpu-burst-capacity-accrual.html

- [W6] Amazon ECS, Fargate task ephemeral storage: https://docs.aws.amazon.com/AmazonECS/latest/developerguide/fargate-task-storage.html

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- [W8] Amazon S3, Download and upload objects with presigned URLs: https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html

- [W9] Amazon S3, Public access and defaults: https://docs.aws.amazon.com/AmazonS3/latest/userguide/granting-public-access.html

- [W10] Amazon EC2, Configure Instance Metadata Service options: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/configuring-instance-metadata-options.html

- [W11] Amazon EBS, How encryption works: https://docs.aws.amazon.com/ebs/latest/userguide/how-ebs-encryption-works.html

- [W12] AWS Cost Management, Managing costs with AWS Budgets: https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html

- [W13] AWS Account Management, Create a spend limit in AWS Settings: https://docs.aws.amazon.com/accounts/latest/reference/create-spend-limit.html

- [W14] AWS Cloud Value Framework example, rate explicitly dated 4 May 2024: https://aws.amazon.com/blogs/migration-and-modernization/unleashing-the-power-of-the-cloud-with-the-aws-cloud-value-framework-cvf-cost-savings-2-7/

- [W15] AWS gp3 regional example (use the per-GB-month rate, not its prorated volume-retention assumption): https://docs.aws.amazon.com/solutions/latest/cloud-migration-factory-on-aws/cost.html ; current EBS pricing: https://aws.amazon.com/ebs/pricing/

- [W16] Amazon VPC, public IPv4 pricing: https://aws.amazon.com/vpc/pricing/

- [W17] Amazon Lightsail, instance bundles: https://docs.aws.amazon.com/lightsail/latest/userguide/amazon-lightsail-bundles.html

Topology, capacities, workflow, tradeoff judgments and acceptance cases are project proposals. AWS documentation establishes service behavior, not this app's source compatibility, measured performance, available account features or deployment readiness. No AWS account was inspected, no resource was created and no money was spent for this planning work.

MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 19 OF 19

Start design: local Android handoff

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.

DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline; implementation and validation pending **Owners:** Design lead, Android lead, audio lead and product lead

1. Readiness and immediate assignment

The planning baseline is sufficient to start design. No additional questionnaire is required before the first design work package. Begin a fresh native Android design for the owner's Galaxy S26 Ultra using **C - Crystal Current**. Use **Mobile DJ** as an explicitly temporary working label; **ParDay** names the assistant. Neither label chooses a public repository name or final app brand. Do not inherit Mixline.

Read this document, the root README, AGENTS.md, Document 12 and the decision ledger first. Then use Documents 03/04 for interaction and material rules, Document 02 for requirement IDs, and Document 08 for assets. Consult Documents 05/06 for control-to-engine boundaries, 13 for ParDay, 14/16 for acquisition and 15 for specialist modules. Document 10 assigns the broader build workstreams.

The first assignment is a connected portrait/landscape design and interaction specification for **local import -> prepare -> route check -> load two decks -> transition -> record/save -> resume**. Include failure and recovery states. Accompany it with the expansion map for the complete feature set. This is design work; a prototype must identify any simulated audio, source, model or hardware behavior.

2. Decisions the next agent must preserve

Confirmed direction	Design consequence
Local Android version first; AWS later	The phone owns library, preparation, session and audio. No server pairing or cloud account is required for the local journey.
Galaxy S26 Ultra first; up-to-date software reported	Design for native phone windows and measured device settings; exact build, density, insets and audio route still need capture.
Serious mobile/live DJs primary; casual streaming DJs secondary	Keep precise touch control while making preparation and source readiness understandable.
Personal practice/private gatherings; personal-tool delivery	No app billing, public-store submission or customer acquisition flow is required.
C Crystal Current	Pale frosted glass, titanium, dark-ink labels and restrained brass. A is historical comparison only.
Full enumerated professional scope, excluding lighting/external controllers	Four decks, stems, sampler/sequencer, recording, automix, mic, video/karaoke, broadcast and genuine third-party hosting remain tracked.
USB audio interfaces and DJ splitters allowed	Design truthful route/cue checks. Do not invent a supported hardware configuration.
ParDay preparation plus specifically armed assistance	Read/propose until a bounded performance plan is armed; make immediate takeover clear.
Spotify/YouTube link-to-file capability, including unofficial candidates	Design paste/share, exact-version/source review, format choice, transfer, validation and local publication. No adapter is yet proven working.
Compatible open-source DSP preferred; paid alternative acceptable if justified	Keep presentation independent of a DSP vendor; no purchase is implied.
AWS later, about US$35/month excluding ParDay AI charges	Keep expansion modular. This is a future planning target, not a quote, hard cap or deployment instruction.
Possible future open-source GitHub sharing	Keep provenance and reproducible sources. Licence, repository and publication remain later choices.

Local first does not mean every capability works without a network. Discovery/downloads, broadcasting and optional provider AI can need connectivity. A prepared local set remains usable when those services are unavailable. Establish the on-device acquisition baseline first; compare it with AWS when expansion is considered. A failed local source adapter is a specific feasibility gap, not permission to introduce a mandatory server.

3. First connected design work package

Design these scenes as one navigable journey using a small, clearly labelled fixture library. The existing fictional titles AURORA DRIVE and COPPER SKY may be reused as fixtures; their displayed waveform or meter values are illustrative until backed by real audio.

1. **Open or resume.** Empty library offers Import and clearly labelled Demo when demo media actually exists. A saved session shows missing media honestly and requires deliberate audible resume.

2. **Import and prepare.** Select local files, show per-item validation/indexing, search the result, inspect an uncertain grid, edit a cue and preserve the correction. A failed file leaves valid imports available.
3. **Check output.** Show the actual route, supported master/cue arrangement and any mono tradeoff. Without independent cue, explain the limit and retain available master playback. Never claim that headphones are isolated without route evidence.
4. **Perform.** Load A/B deliberately, play, cue, nudge/sync, operate EQ/faders and complete a transition. Browse the next track while the transport, crossfader, route and meters remain reachable. Protect audible-deck replacement.
5. **Record and finish.** Start eligible recording, show storage/finalization, recover writer failure without interrupting unrelated playback, save useful preparation/session history and return to a recent set.
6. **Recover.** Show route loss, pointer cancellation/rotation, missing next track, low storage and interrupted-session recovery. Specify the actual action and the music that remains available.

Map these scenes to FR-001/002/004/006/008/010/020/022/023/031/033-042. Document 02 remains the authoritative requirement set; this work package does not replace it.

4. Screen inventory and expansion map

Two visible decks are a proposed default composition, not the product's final deck limit. Keep four stable deck identities and an accessible A-D focus path. At least eight hot cues per track and the selected sampler/sequencer remain in the control-access map, even when only one bank is visible.

Family	First design coverage	Essential states
Launch, library and crates	Complete connected journey	Empty, missing file, partial import, duplicate, no results, offline, analysis pending
Track preparation	Grid/cue edit and saved correction	Uncertain, edited/locked, undo, pending analysis, save failure
Readiness and routing	Master/cue review and route change	No isolated cue, split mono, unsupported route, disconnected
Performance/mixer	Portrait and landscape A/B, A-D focus map	Empty/loading/playing/paused, held touch, pending quantized action, clipping, protected load
Pads, loops, FX and stems	Workspace architecture and representative components	Bank/mode, edit, pending/active latch, unavailable stems, resource refusal
Sampler/sequencer	Sixteen-slot/bank and pattern access map	Empty slot, sounding pattern, queued next-bar revision, immediate/quantized stop
Queue/automix	Next readiness and takeover sequence	Missing next track, committed transition, expired edit, manual takeover
Recording/history	Complete save/resume journey	Restricted source, full storage, interrupted writer, recoverable segment

Family	First design coverage	Essential states
Sources/downloads	Local execution flow and state specification	Matching, ambiguous version, unavailable adapter, format/quality, progress unknown, cancel, validate, ready
ParDay	Preparation proposal and armed-transition states	Local only, thinking, stale proposal, needs review, applied, cost limit, unavailable
Video/karaoke/broadcast /mic	First-class workspace map and critical controls	Preview/program distinction, singer queue, mic assignment, disconnected receiver, restricted source
Plugin rack/settings/recovery	Compatibility and configuration structure	Missing/failed plugin, bypass, dirty/save/discard settings, restore mismatch

Complete this map before treating a polished two-deck screen as the whole design. Selected specialist controls may use focused destinations; they must not become tiny permanent controls squeezed into one phone view. AWS service setup stays outside the first local onboarding journey.

5. Material, geometry and input rules

Use `assets/design_tokens_proposed.json` as a proposed token source. C is selected; its exact colors, fonts and geometry still need verification. Prioritize track identity, time/beat position, transport, levels and output state. Keep text on stable opaque backing. Express steampunk through precise rims, grips and index marks; do not add moving gears, steam or a heavy live 3D scene.

Use Android dp/sp and inspect actual safe areas and text scaling. Document 03 records the Android 48dp target guidance; frequent transport/pads have a proposed 56dp target. Draw hit bounds separately from visible artwork and check overlap. The concept render's aspect ratio and pixel count are not device measurements. Design portrait and landscape deliberately; broader tablet/foldable layouts are later qualification work, not a reason to postpone the phone composition.

Each custom control needs a stable deck/parameter identity, value/unit/range, gesture, accessible alternative, pending/acknowledged state and cancellation behavior. Hold gestures retain their original identity through focus changes. A compact library or ParDay panel preserves essential live controls. Performance values apply immediately; configuration changes use explicit Save and honest persistence status.

Deliver C daylight, low-light and opaque high-contrast treatments, plus reduced transparency/motion. The Essential rendering tier retains every qualified musical action. Simplify optical decoration before reducing legibility or changing audio behavior. Visual acceptance and audio/functional acceptance are separate.

6. Assets to create first

Priority	Work order	Required output
First	AS-04/05, AS-06/07/08	Coherent icon/DJ symbol subset and editable native/vector controls with state and hit-bound specifications
First	AS-09/10/12/13	C material recipes, markers and calibrated-meter design with opaque fallback
First	AS-02/03/15	Verified font shortlist or system fallback; original deterministic artwork placeholders
First	AS-27/30	ParDay proposal/status family and truthful local acquisition state components
Before functional prototypes	AS-11/21/22/23	Authorized local demo/fixture media, actual waveform data and exact route diagrams
Module slices	AS-14/19/20/24/29	Stem/FX symbols, original samples/loops, rehearsal content and video/karaoke fixtures

AS-28's two concept PNGs already exist and record the selected material direction. They are not production control atlases. The other production work orders remain proposed or to create. AS-01 branding can wait for the app name; optional promotional/3D assets do not block the personal app. Every acquired/created item needs the provenance, licence, version/hash, editable master and fallback fields in Document 08. No asset procurement is authorized by the AWS target.

7. Design handoff acceptance and evidence

Deliver editable screen/component sources, exported previews, the connected interaction flow, a screen/state index, token changes, control-to-requirement mapping, asset manifest updates and a concise verification report. Separate states actually exercised from merely drawn. Provide native-layout measurements where available and explicitly label unmeasured device assumptions.

The design passes its handoff review when the owner can follow the local first-mix journey; portrait/landscape retain the same musical meaning; essential controls remain reachable during browsing/ParDay use; four-deck and specialist navigation are accounted for; and empty/loading/error/offline/recovery states have useful next actions. Check long titles, large text, color-independent identities, reduced motion, route ambiguity, stale proposals and unknown progress. The proposed usability guardrail is zero wrong-deck or destructive-seek actions in scripted trials; no such result is claimed yet.

Screens alone cannot pass engine, source-extractor, latency, plugin, video or thermal gates. Report **functional status**, **visual status**, **physical-device evidence** and **asset rights status** separately. Current application status is not implemented/not verified; existing concept generation and this planning audit are separate completed activities.

8. Questions assigned to the phase that needs them

The app name can wait for branding. Exact S26 Ultra build, dimensions/settings and audio accessories must be captured before physical-device claims. DSP/plugin/source feasibility needs experiments before support promises. ParDay provider, credentials and a separate spending limit are needed before paid model use; optional voice can remain absent from the first prototype. AWS account/region/service/current quote are needed only when expansion starts. Project licence and public repository contents are needed before any source publication. None requires another planning round before design begins.

Next agent task: design the connected local journey in section 3, preserving the confirmed choices and the complete navigation map. Use Document 10's bounded work packages for implementation after the owner requests it. Log any new decision that changes scope; resolve ordinary reversible layout choices within the selected C direction.